

COSE471: Data Science

Clustering

Department of Computer Science and Engineering

Korea University

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April 3, 2026

Course Schedule

Week	Day	Content	HW/Project	Note
1	Mar 06	Introduction		
2	Mar 13	Data		
3	Mar 20	Pattern mining		
4	Mar 27	Tutorial 1	HW1	<i>Latex, Preprocessing/Pattern mining</i>
5	Apr 03	Clustering		
6	Apr 10	Outlier Detection & Invited Talk		<i>An invited talk from Meta</i>
7	Apr 17	Tutorial 2	HW2	<i>Clustering, Outlier detection</i>
8	Apr 24	Classification		
9	May 01	Classification		
10	May 08	Tutorial 3		<i>Classification, Feature engineering</i>
11	May 15	Mid-term exam		<i>Offline, Closed-book, No electronic devices</i>
12	May 22	Review & Project Notice	Project	
13	May 29	Project discussion		
14	Jun 05	Project discussion		
15	Jun 12	Final presentation		
16	Jun 19	Wrap-up and Final report		

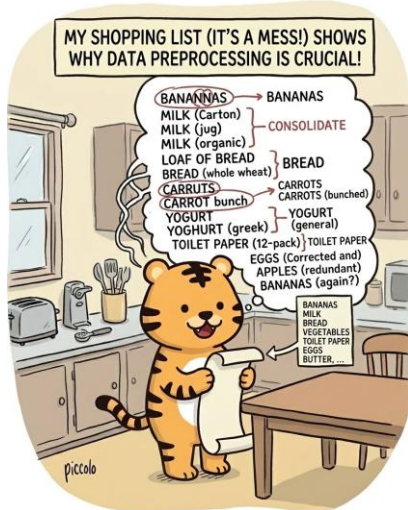
Merged to

RECAP

- ❑ Basic concepts of pattern mining
 - Pattern discovery
 - Frequent itemsets
 - Association rule
 - Closed and maximal patterns
- ❑ Pattern mining methodologies
 - Association rule discovery
 - Apriori
 - FP-Growth
 - Rule generation
- ❑ Pattern mining evaluation
 - Motivation
 - Lift



RECAP



Outline

☐ **Basic concepts of clustering**

- Overview
- Applications
- Considerations
- Multi-aspect categorization

☐ Distance-based methods

☐ Density-based methods

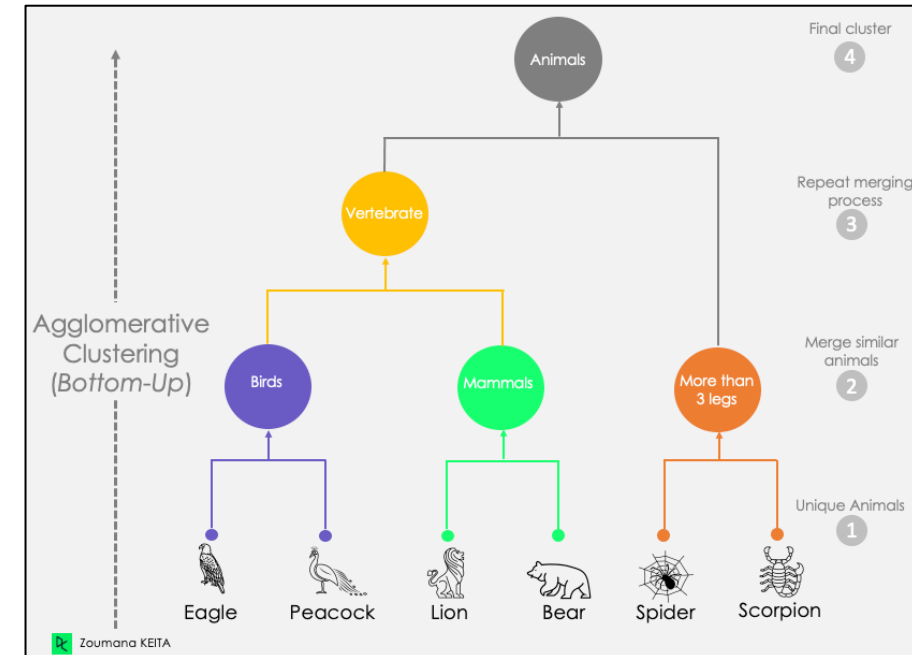
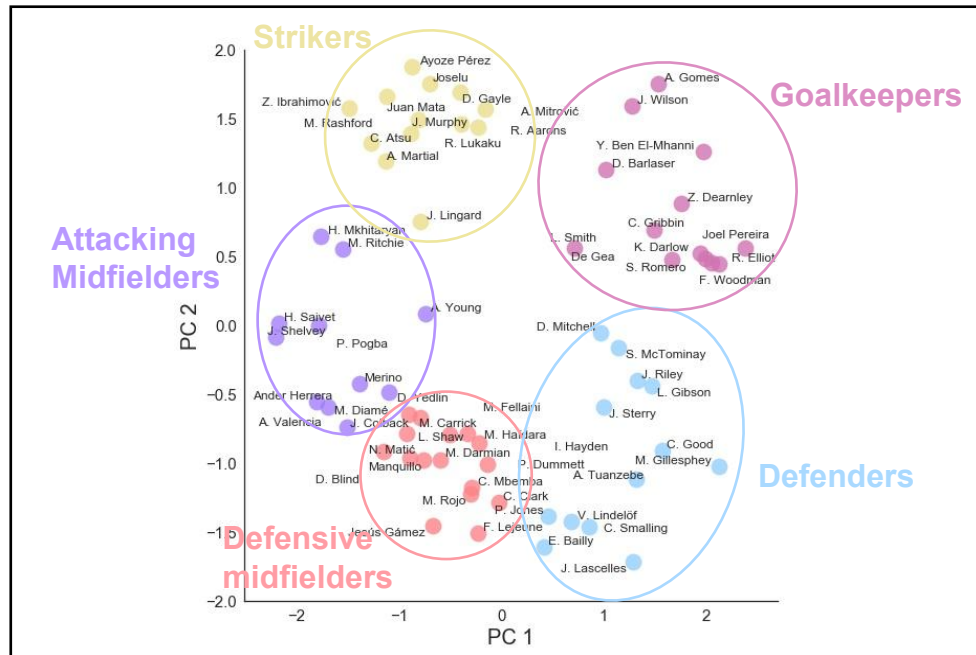
☐ Probabilistic methods

☐ Clustering evaluation

Clustering > Overview

❑ What is a cluster?

- A cluster is a collection of data objects
 - Objects are similar (or related) to one another within the same group
 - Objects are dissimilar (or unrelated) to the objects in other groups
- Examples



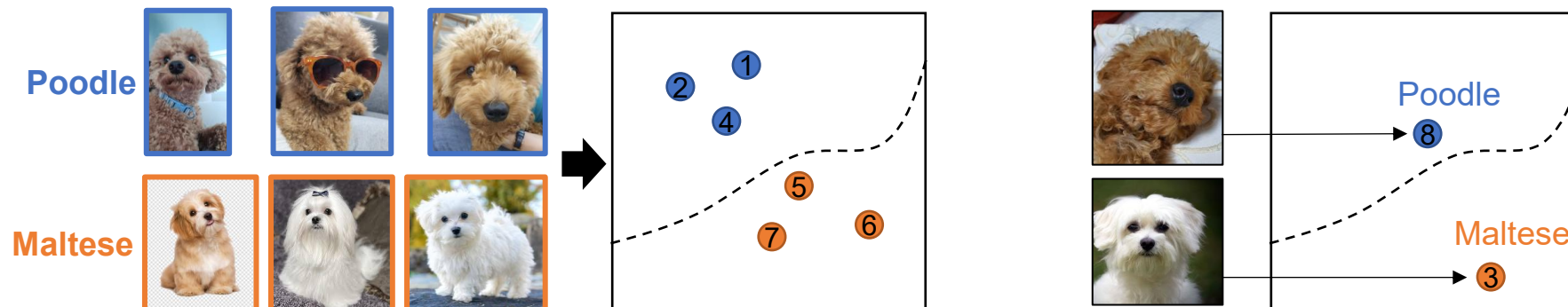
Clustering > Overview

□ Cluster analysis (or clustering, data segmentation, ...)

- Given a set of data points, partition them into a set of groups (i.e., clusters)
- Learning philosophy
 - Unsupervised learning with unknown cluster labels



- cf. classification is supervised learning with known class labels



Clustering > Overview

□ Cluster analysis (or clustering, data segmentation, ...)

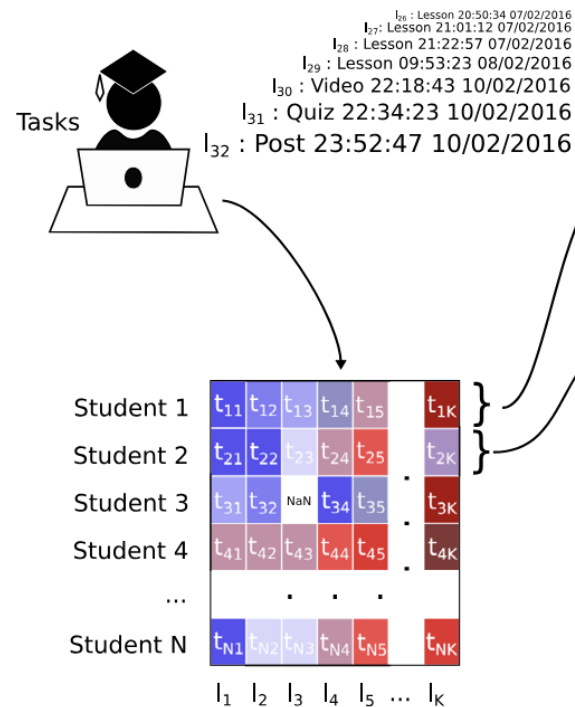
- Given a set of data points, partition them into a set of groups (i.e., clusters)
- Learning philosophy
 - Unsupervised learning with unknown cluster labels
 - cf. classification is supervised learning with known class labels
- Quality
 - A good clustering method will produce high-quality clusters
 - High intra-class similarity → cohesive within clusters
 - Low inter-class similarity → distinctive between clusters
 - There are usually separate quality measures for the “goodness” of a cluster for different applications
 - It is hard to define “similar enough” or “good enough” → The answer is typically highly subjective

Clustering > Applications

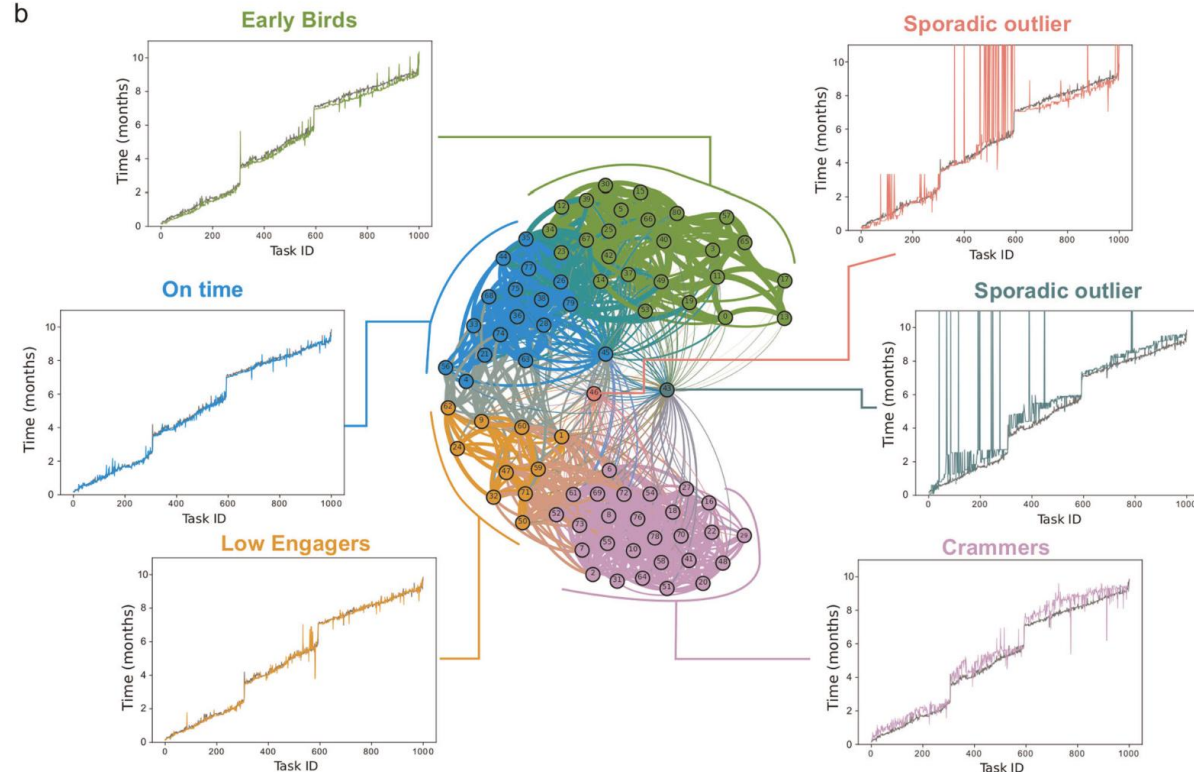
□ A stand-alone tool to get insight into data distribution

- Visual insights - one picture is worth a thousand words
- Example: types of students in an online course

a Time Series



b

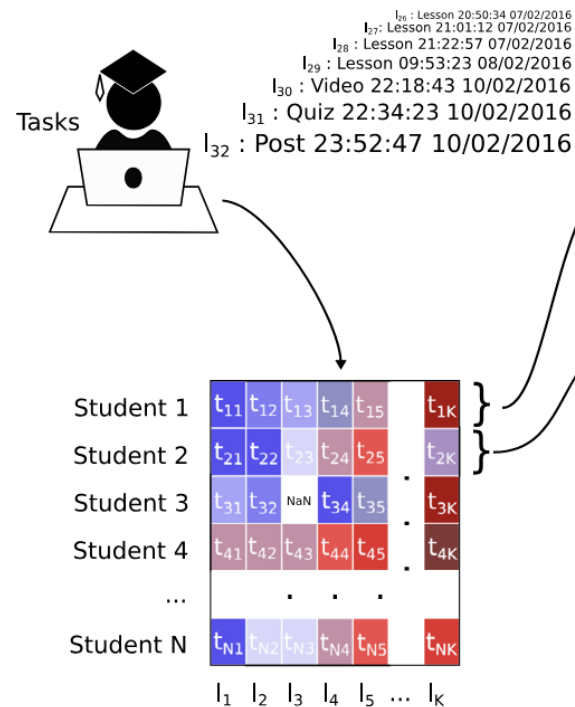


Clustering > Applications

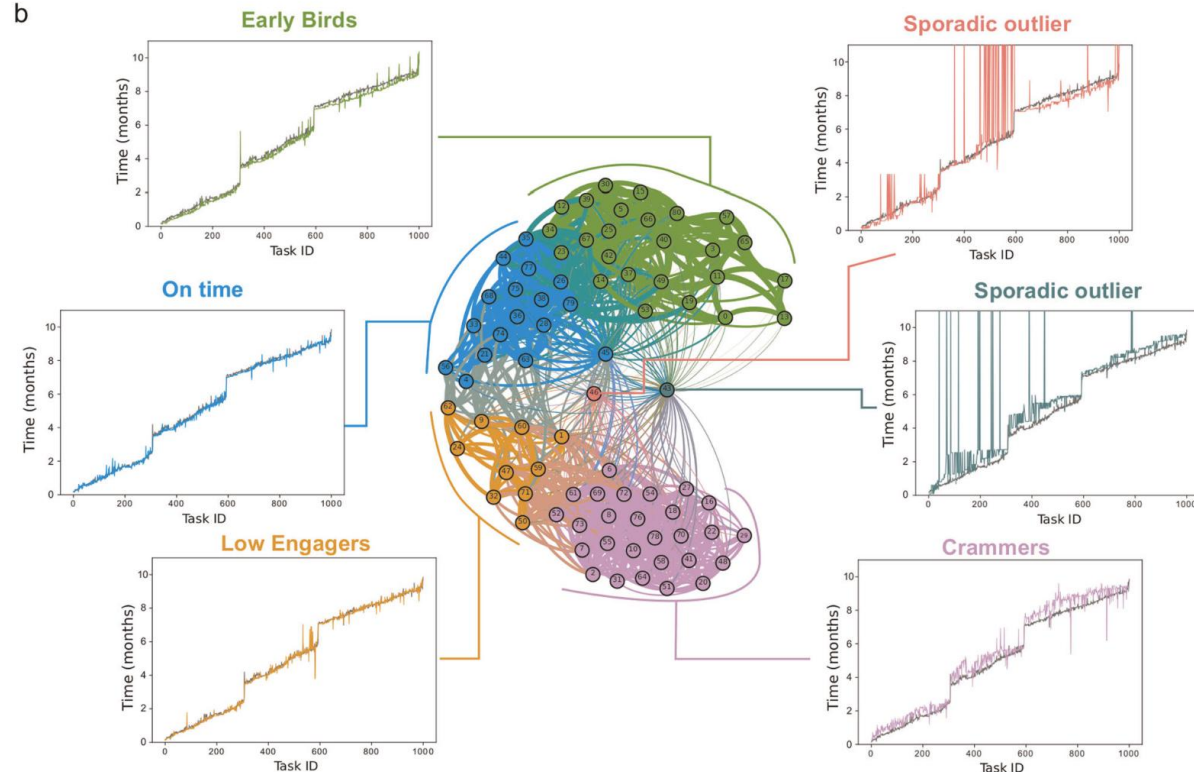
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Clustering > Applications

- ❑ A stand-alone tool to get insight into data distribution
 - Visual insights - one picture is worth a thousand words
- ❑ A key intermediate step for other data science tasks
 - A compact summary of data for downstream tasks
 - e.g., the basis for classification, pattern discovery, outlier detection
 - Data summarization, compression, and reduction
 - e.g. binning, discretization, sampling
 - Recommendation systems, customer segmentation
 - e.g. Finding like-minded users or similar products
 - Multimedia data, biological data, and social data analysis
 - e.g.. grouping images/video/audio, gene/protein sequences, trend detection and tracking

Clustering > Considerations

❑ Partitioning criteria

- Single level **vs.** hierarchical partitioning
- e.g., flat concepts (gender) **vs.** multi-level concepts (county – city – country – continent)

❑ Cluster separation

- Exclusive **vs.** non-exclusive
- e.g., a customer belongs to only one region **vs.** a news article may belong to multiple topics

❑ Similarity measure

- Distance-based **vs.** connectivity-based
- e.g., Euclidean, road network, vector **vs.** density or contiguity

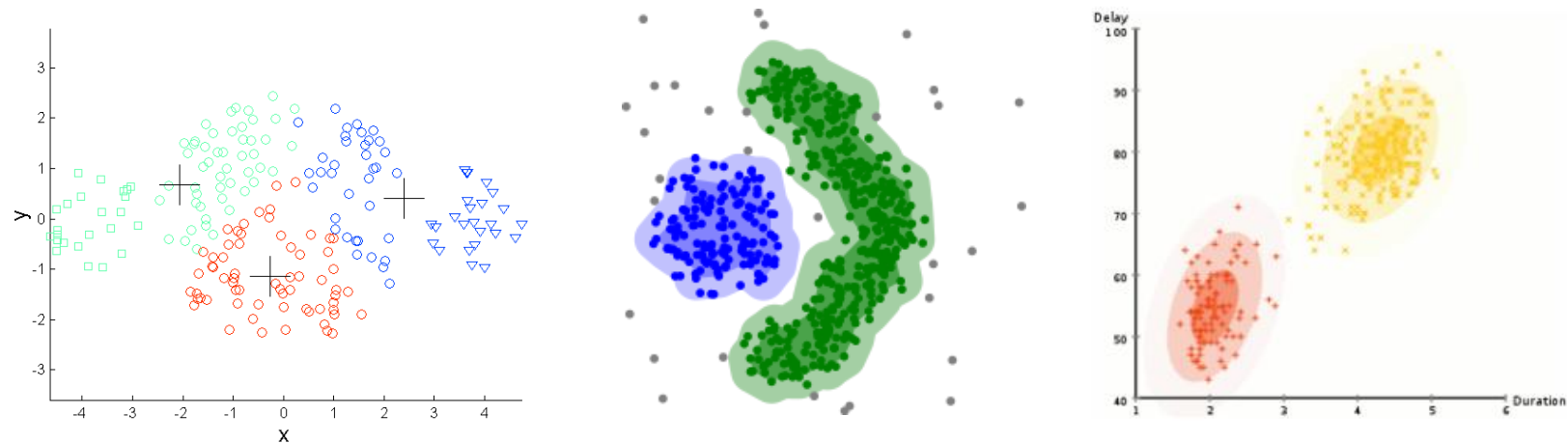
❑ Clustering space

- Full space **vs.** subspaces
- e.g., Low dimensional **vs.** high-dimensional

Clustering > Multi-Aspect Categorization

□ Technique

- **Distance-based**
- **Density-based**
- **Probabilistic**
- High-dimensional
- Scalable



□ Data type

- **Numerical data**, categorical data, text data, multimedia data, time-series data, sequence data, streaming data, network data

□ Additional insight

- Visual insights, semi-supervised, ensemble-based, validation-based

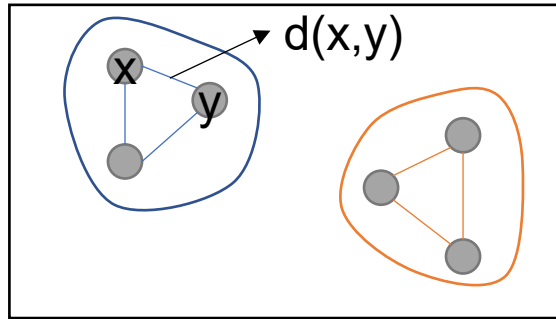
Outline

- ❑ Basic concepts of clustering
- ❑ **Distance-based methods**
 - Partitioning methods
 - Hierarchical methods
- ❑ Density-based methods
- ❑ Probabilistic methods
- ❑ Clustering evaluation
- ❑ Summary

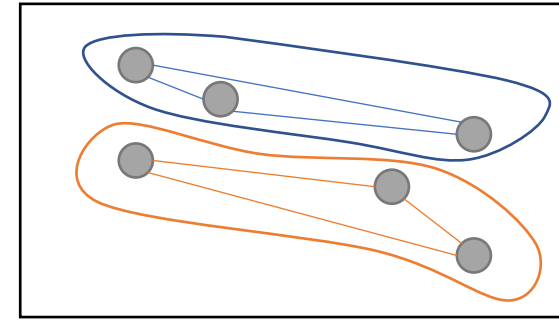
Clustering > Distance-based > Partitioning

□ Partitioning method

- Discovering the groupings in a dataset based on the distances of data objects



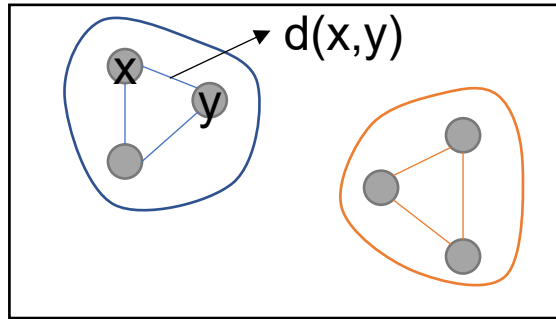
Partitioning objects
by examining $d(x,y)$
for every pairs



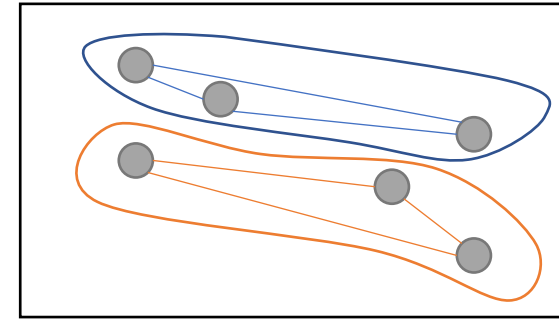
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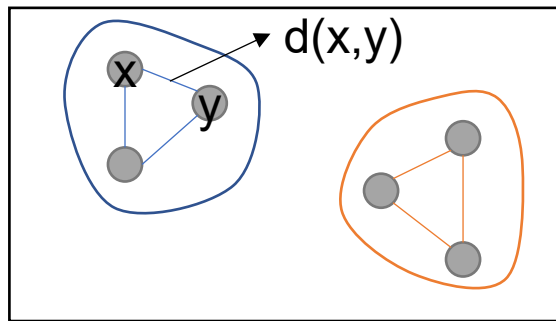


- Problem definition
 - Given k , find a partition of k clusters that optimizes the chosen distance-based objective function
 - e.g., minimize the sum of squared errors between objects and centers in clusters, $\sum_{C_k \in C} \sum_{x_i \in C_k} d(x_i, c_k)^2$

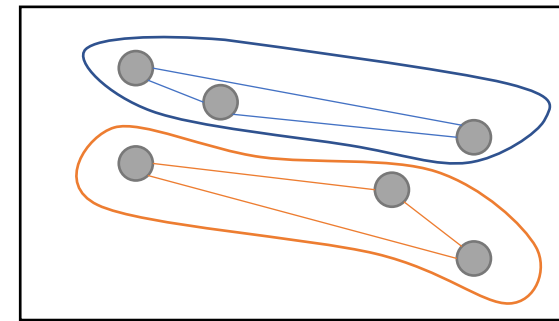
Clustering > Distance-based > Partitioning

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- Approaches
 - Global optimal needs to exhaustively enumerate all possible partitions
 - Heuristic methods (i.e., greedy algorithms): k-Means, k-Medoids, k-Medians, etc.

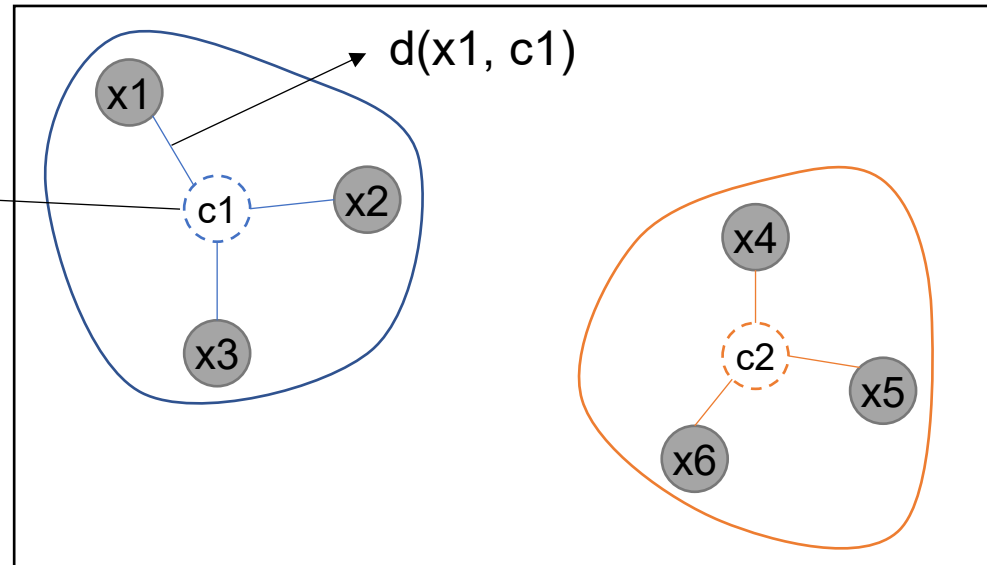
Clustering > Distance-based > Partitioning > k-Means

□ k-Means clustering

- Each cluster is represented by the cluster center and the sum of $d(\text{object}, \text{center})^2$ is minimized
 - i.e., minimize the sum of squared errors (SSE) in clusters

An average of x_1, x_2, x_3
(center or centroid)

$$c_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

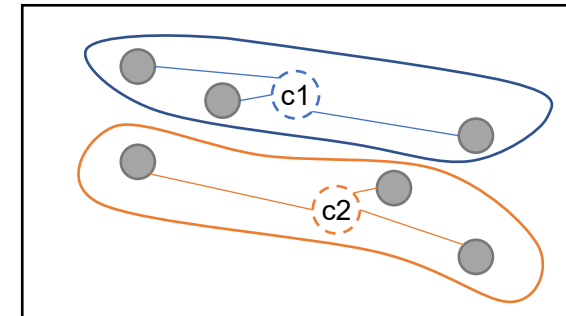


$$\begin{aligned} & d(x_1, c_1)^2 \\ & + d(x_2, c_1)^2 \\ & + d(x_3, c_1)^2 \end{aligned}$$

$$\begin{aligned} & d(x_4, c_2)^2 \\ & + d(x_5, c_2)^2 \\ & + d(x_6, c_2)^2 \end{aligned}$$

$$\sum_{C_k \in C} \sum_{x_i \in C_k} d(x_i, c_k)^2 = \text{SSE in cluster 1} + \text{SSE in cluster 2} \ll \sum_{C_k \in C} \sum_{x_i \in C_k} d(x_i, c_k)^2$$

vs.



Clustering > Distance-based > Partitioning > k-Means

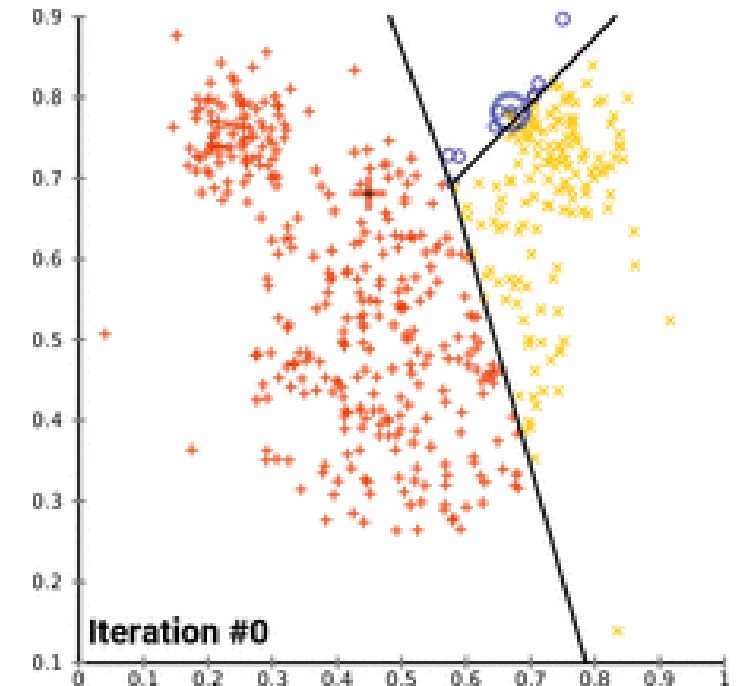
□ k-Means clustering

- Each cluster is represented by the cluster center and the sum of $d(\text{object}, \text{center})^2$ is minimized

□ Procedures with a given number of clusters k

- Select k objects as initial centroids (centers)
- **Repeat**
 - Form k clusters by assigning each object to its closest centroid
 - Re-compute the centroids of each cluster
 - i.e., the average value of objects in each cluster
- **Until** the convergence criterion is satisfied
 - e.g., the centroids do not change

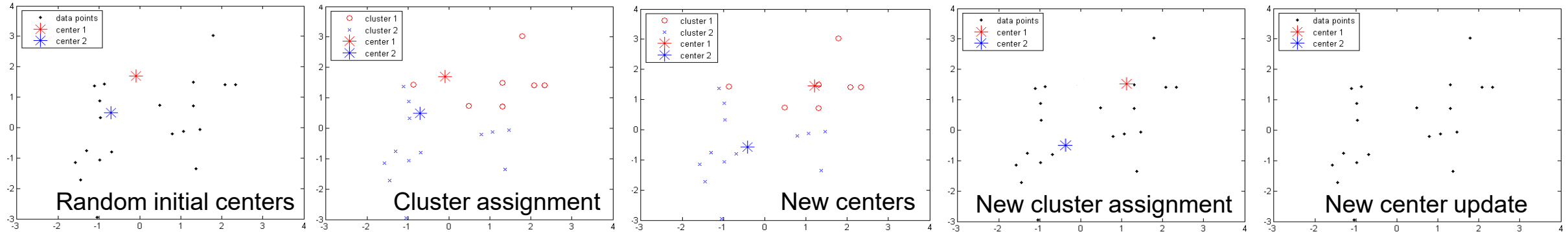
□ [Interactive demo](#)



Clustering > Distance-based > Partitioning > k-Means

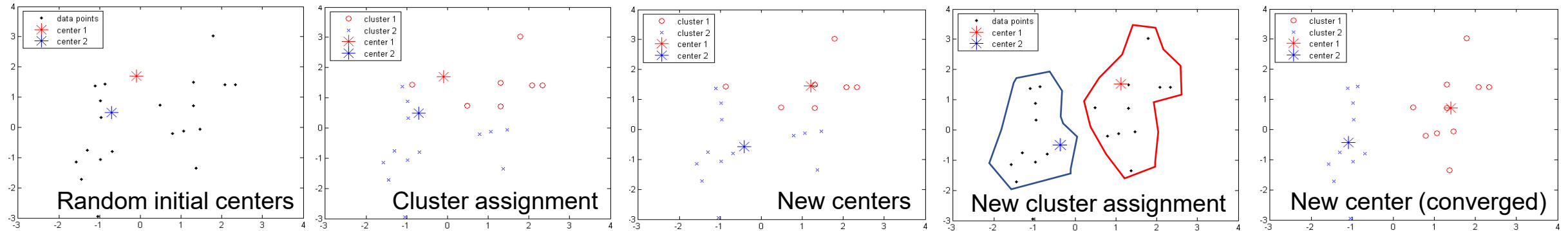
□ Example

Do it yourself!

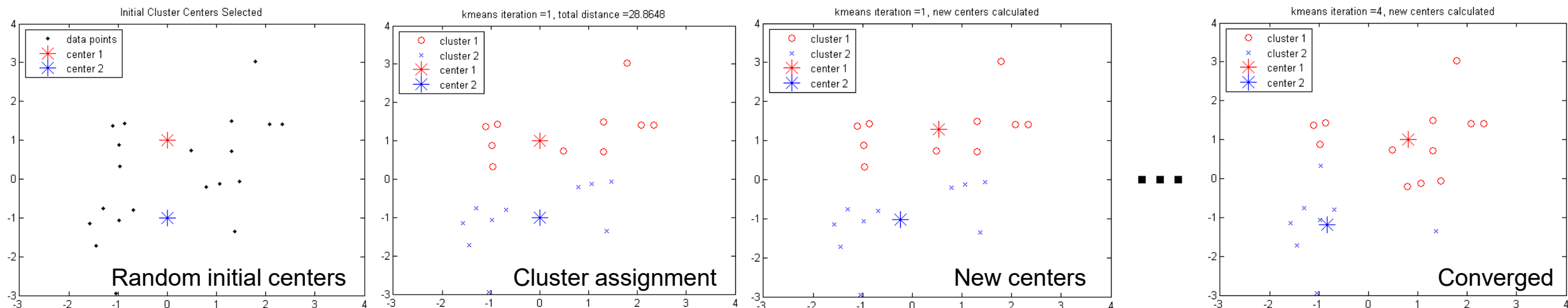


Clustering > Distance-based > Partitioning > k-Means

Example



Another example



Clustering > Distance-based > Partitioning > k-Means

❑ Complexity: $O(tkn)$

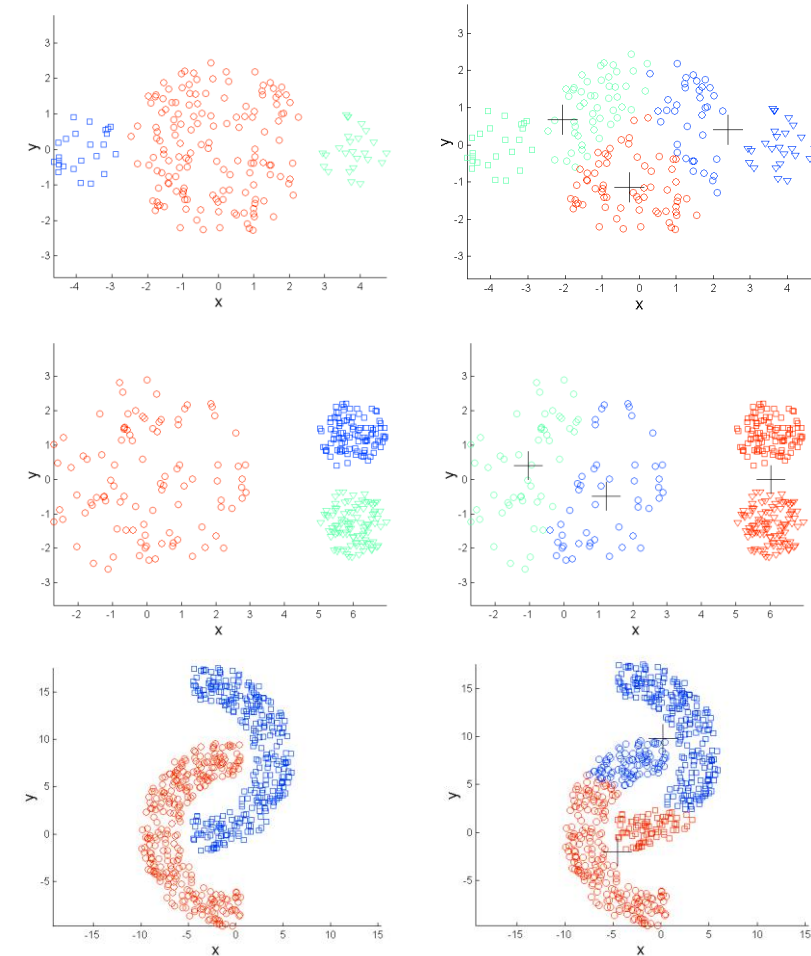
- n : # of objects, k : # of clusters, t : # of iterations (usually, $k, t \ll n$)

❑ Limitations

- Applicable only to objects in a continuous space
- Need to specify k , the number of clusters, in advance
- Often terminates at a local optimal
- Sensitive to noisy data and outliers
- Not suitable to discover clusters with non-convex shapes or clusters of very different sizes or density

❑ Variants

- Better initial centroid estimates (e.g., k-means++)
- Different representative cluster prototypes (e.g., k-Medoids)
- Feature transformation (e.g., Kernel method or embedding)

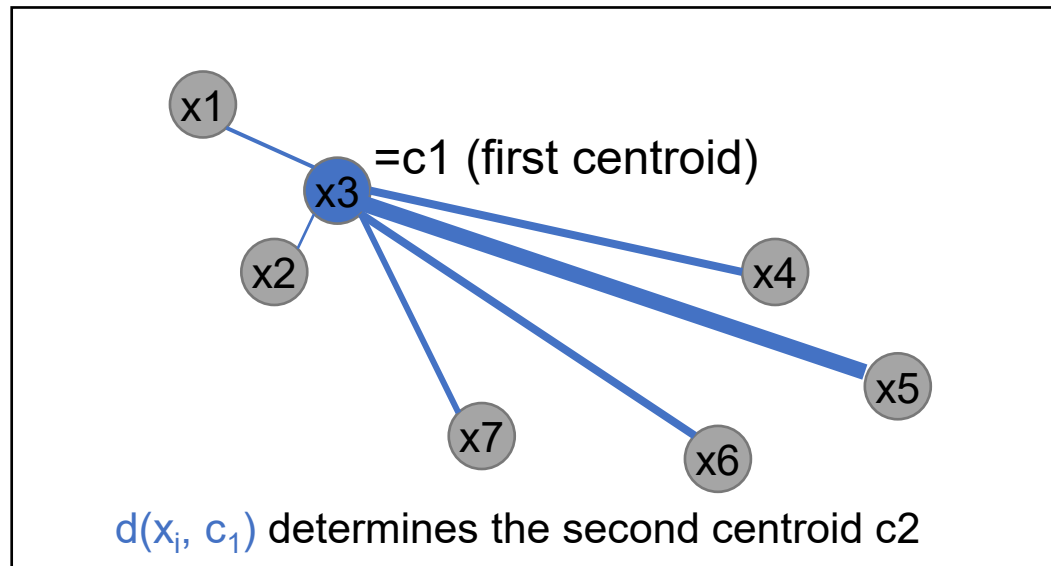


Clustering > Distance-based > Partitioning > k-Means++

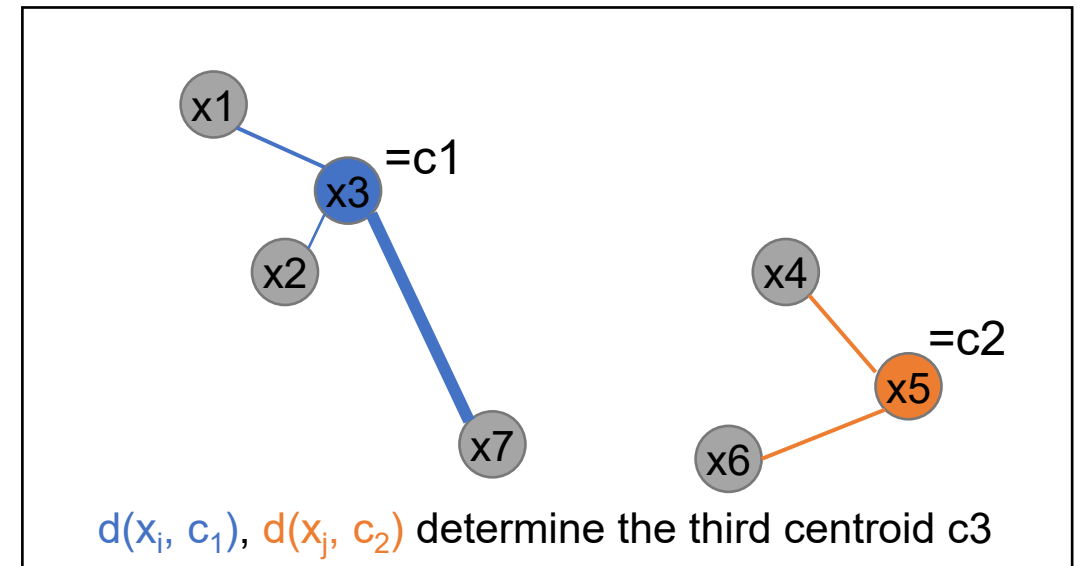
□ k-Means++ clustering (better initial centroid estimation)

- Pick the initial centroids sequentially with a weighted probability score and conduct k-Means

- Probability of choosing x_i as the next center = $\frac{\min_j d(x_i, c_j)^2}{\sum_i \min_j d(x_i, c_j)^2}$



e.g., x_5 has the highest probability to be chosen while x_2 has the lowest probability to be chosen



e.g., x_7 has the highest probability to be chosen while x_2 has the lowest probability to be chosen

■ Clustering > Distance-based > Partitioning > k-Means++

□ k-Means++ clustering

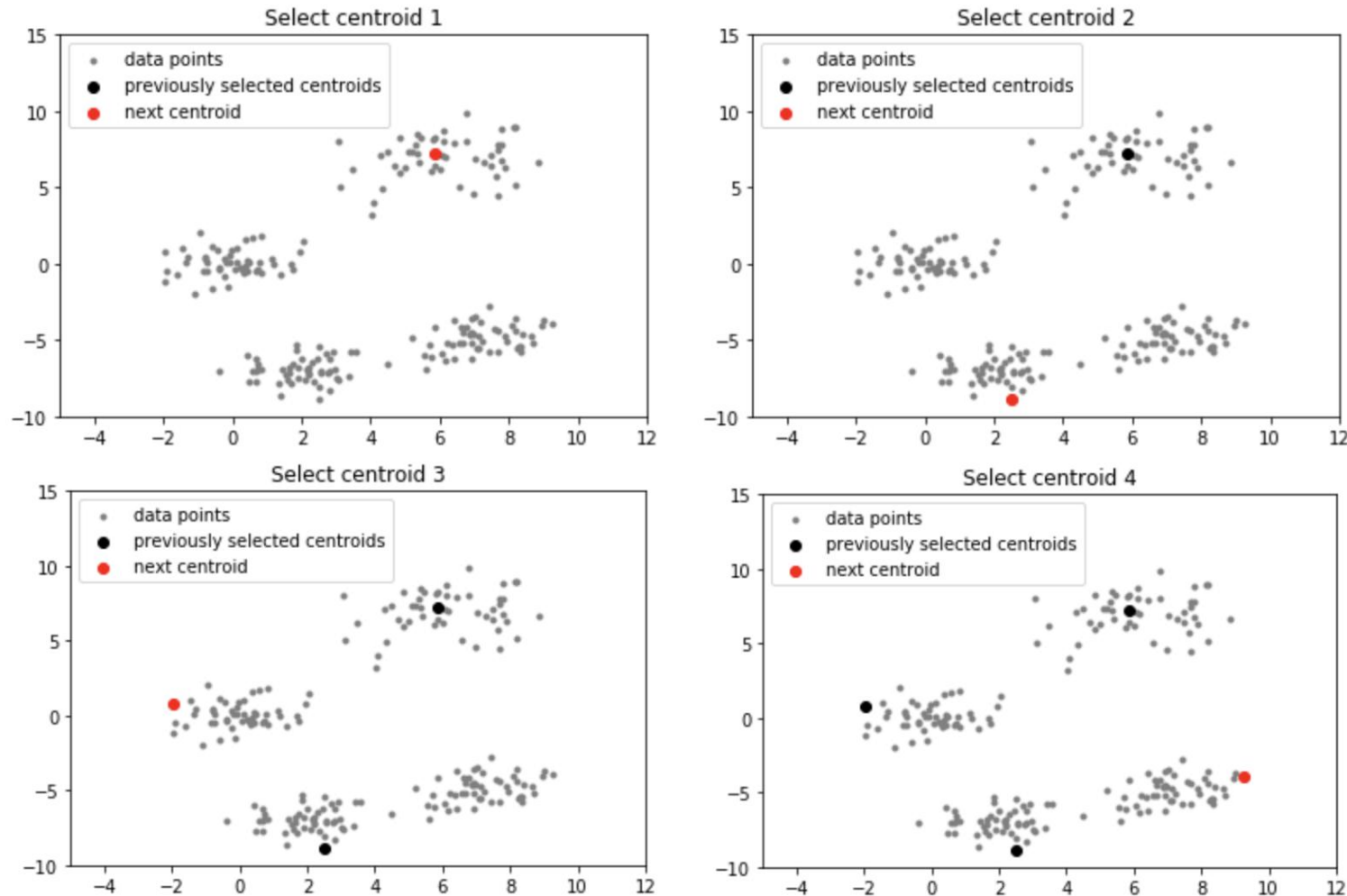
- Pick the initial centroids sequentially with a weighted probability score and conduct k-Means

□ Procedures with a given number clusters k

- Select an initial object at random to be the first centroid
- **For** k – 1 steps
 - For remaining objects, find the squared distance $\min_j d(x_i, c_j)^2$ to the nearest centroid
 - Randomly select a new centroid by choosing an object with probability proportional to $\frac{\min_j d(x_i, c_j)^2}{\sum_i \min_j d(x_i, c_j)^2}$
 - i.e., further from existing centroids, higher probability to be picked
- **End For**
- Conduct standard k-Means clustering with the initial centroids

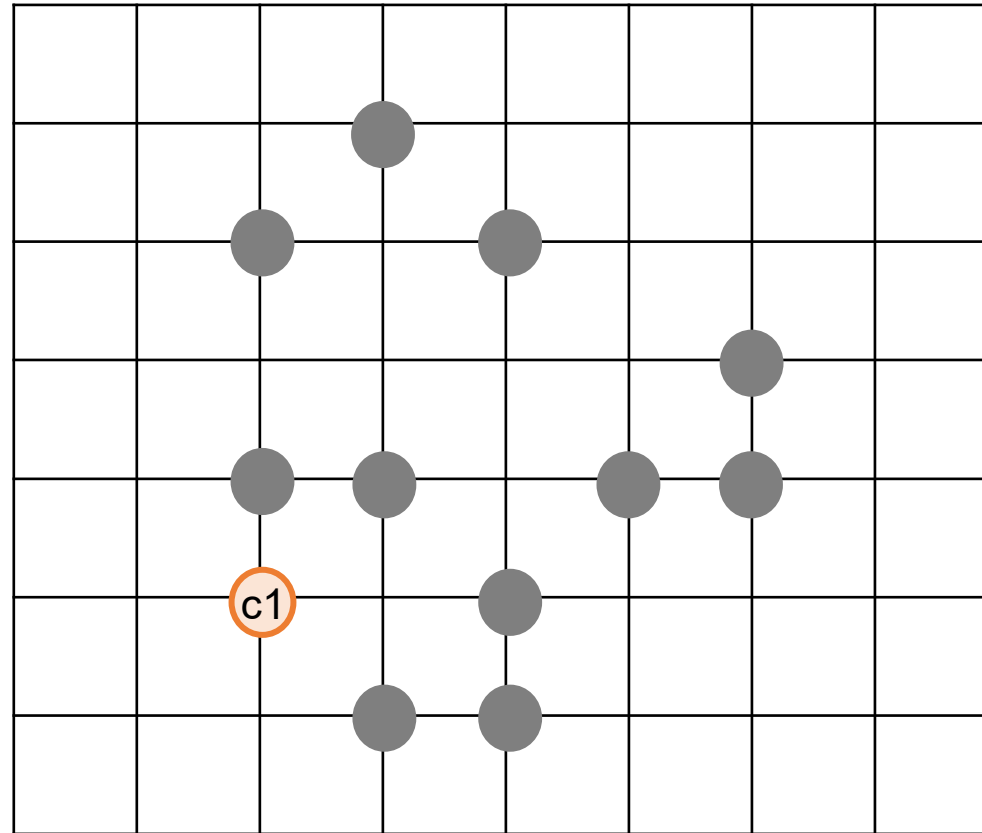
Clustering > Distance-based > Partitioning > k-Means++

□ Example of centroids initialization



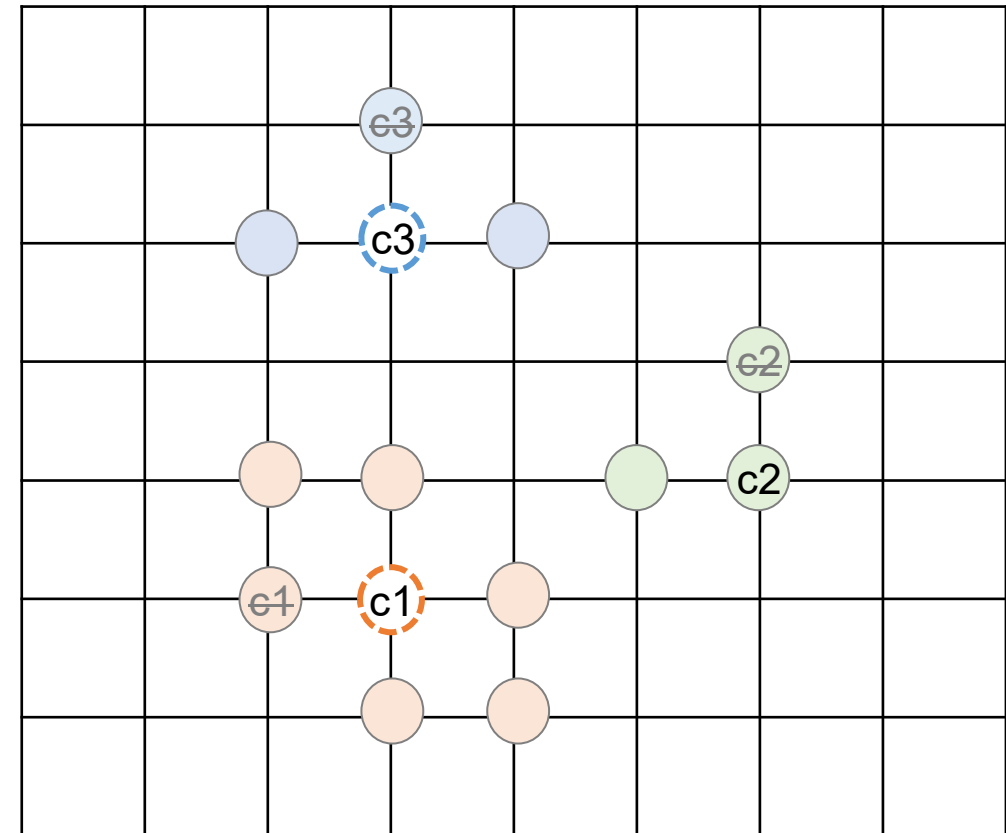
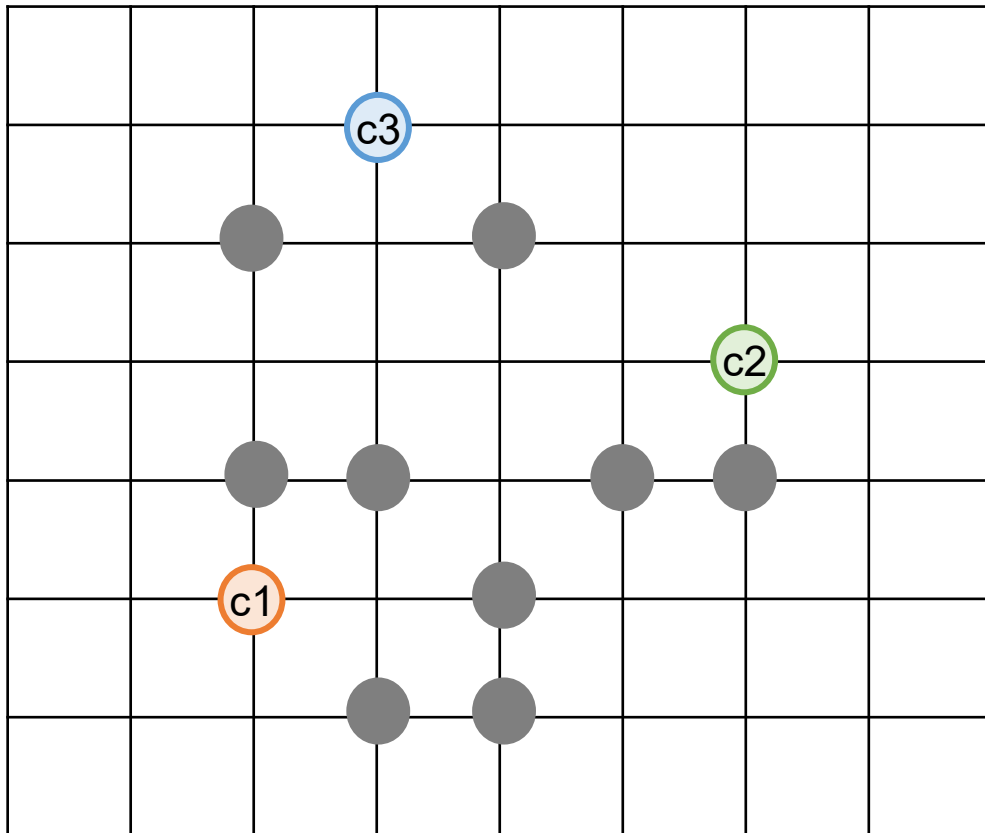
Clustering > Distance-based > Partitioning > k-Means++

- Quiz: conduct k-Means++ with the number of clusters $k = 3$
 - The first initial centroid is given



Clustering > Distance-based > Partitioning > k-Means++

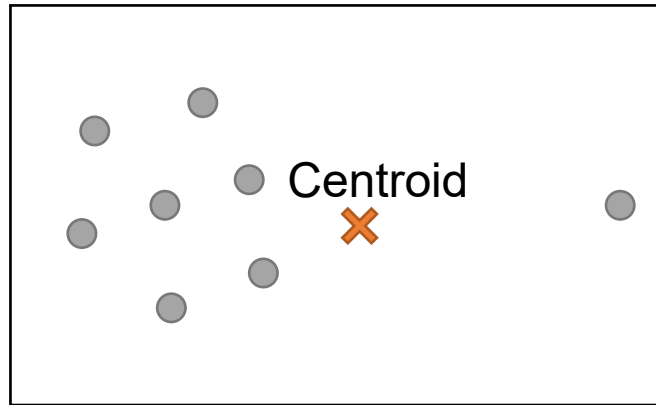
□ Quiz: conduct k-Means++ with the number of clusters $k = 3$



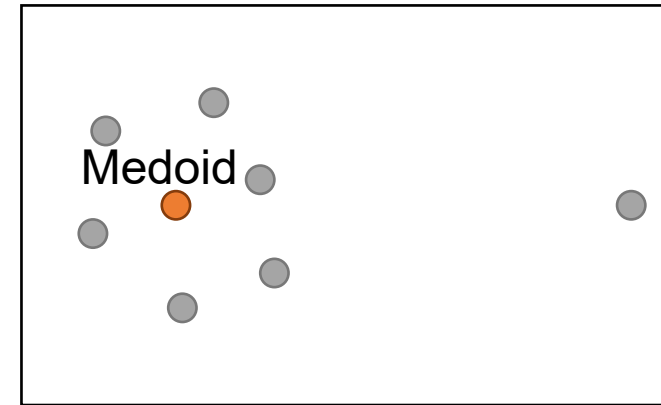
Clustering > Distance-based > Partitioning > k-Medoids

□ k-Medoids clustering (better cluster prototypes)

- Each cluster is represented by the cluster medoid and the sum of $d(\text{object}, \text{medoid})^2$ is minimized
 - Medoids: the most centrally located **object** in a cluster
- More robust to outliers than k-Means



$$c_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

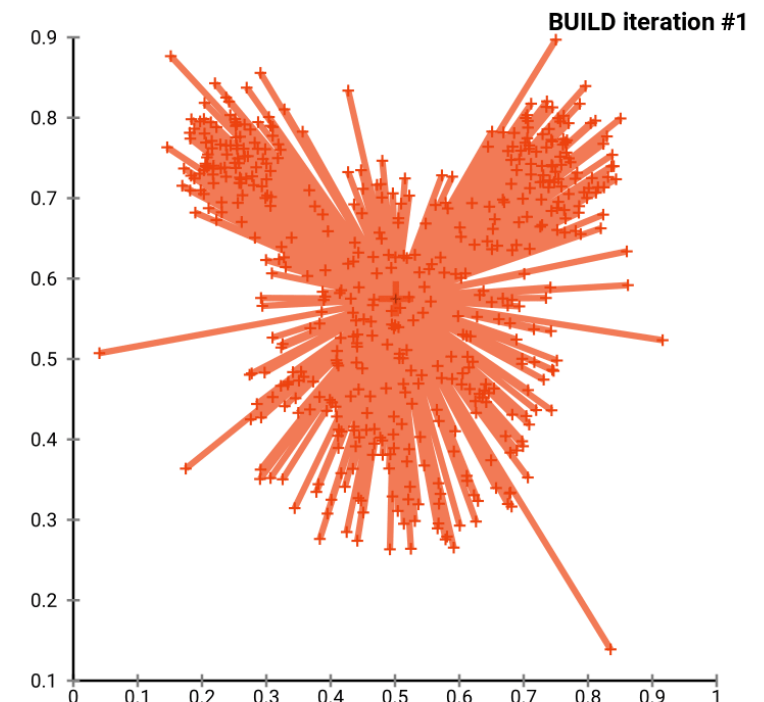


$$m_k = \operatorname{argmin}_{x_i \in C_k} \sum_{x_j \in C_k} d(x_i, x_j)$$

Clustering > Distance-based > Partitioning > k-Medoids

□ k-Medoids clustering

- Each cluster is represented by the cluster medoid and the sum of $d(\text{object}, \text{medoid})^2$ is minimized
 - Medoid: the most centrally located **object** in a cluster
- Procedures of PAM (Partitioning Around Medoids)
 - Select k points as the initial medoids
 - **Repeat**
 - Assigning each point to the cluster with the closest medoid
 - Get the total cost (e.g., SSE) of swapping the medoid with a non-medoid
 - If the total cost decreases, swap the medoid with the chosen object
 - **Until** convergence criterion is satisfied
- Issue
 - Not scalable with large data due to high computational complexity
 - i.e., $O(tk(n-k)^2) > O(tkn)$ where t is the number of iterations



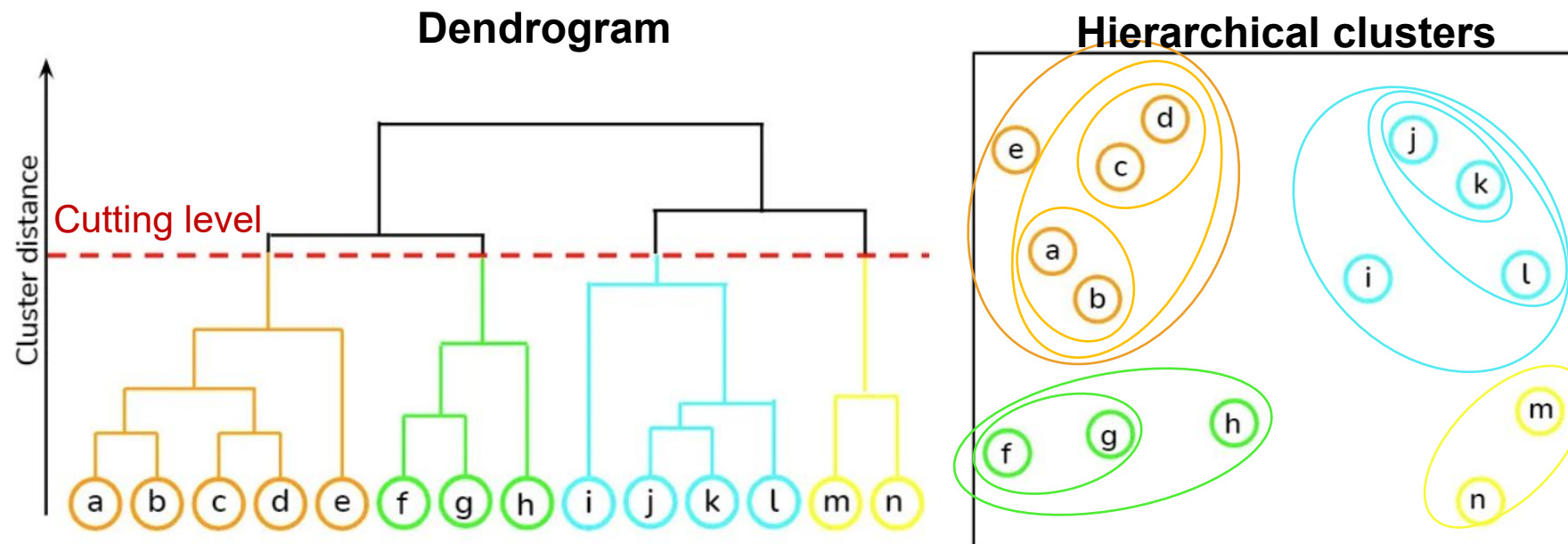
Outline

- ❑ Basic concepts of clustering
- ❑ **Distance-based methods**
 - Partitioning methods
 - **Hierarchical methods**
- ❑ Density-based methods
- ❑ Probabilistic methods
- ❑ Clustering evaluation

Clustering > Distance-based > Hierarchical

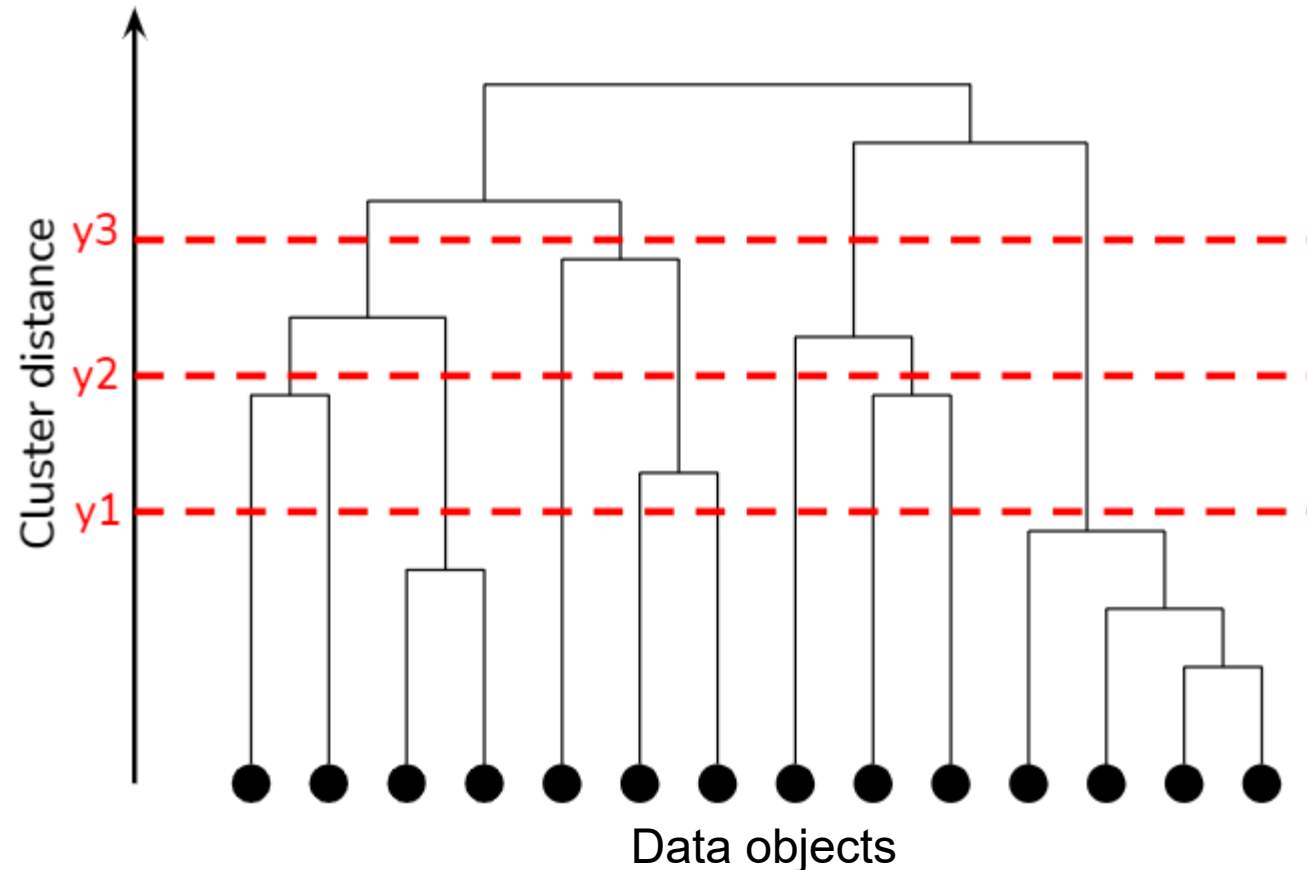
□ Hierarchical clustering

- Generate a clustering hierarchy, visualized as a dendrogram
- Not required to specify the number of clusters (k)
 - The desired number of clusters can be obtained by cutting the dendrogram at the proper level
- More deterministic than partitioning approaches
 - No iterative refinement required



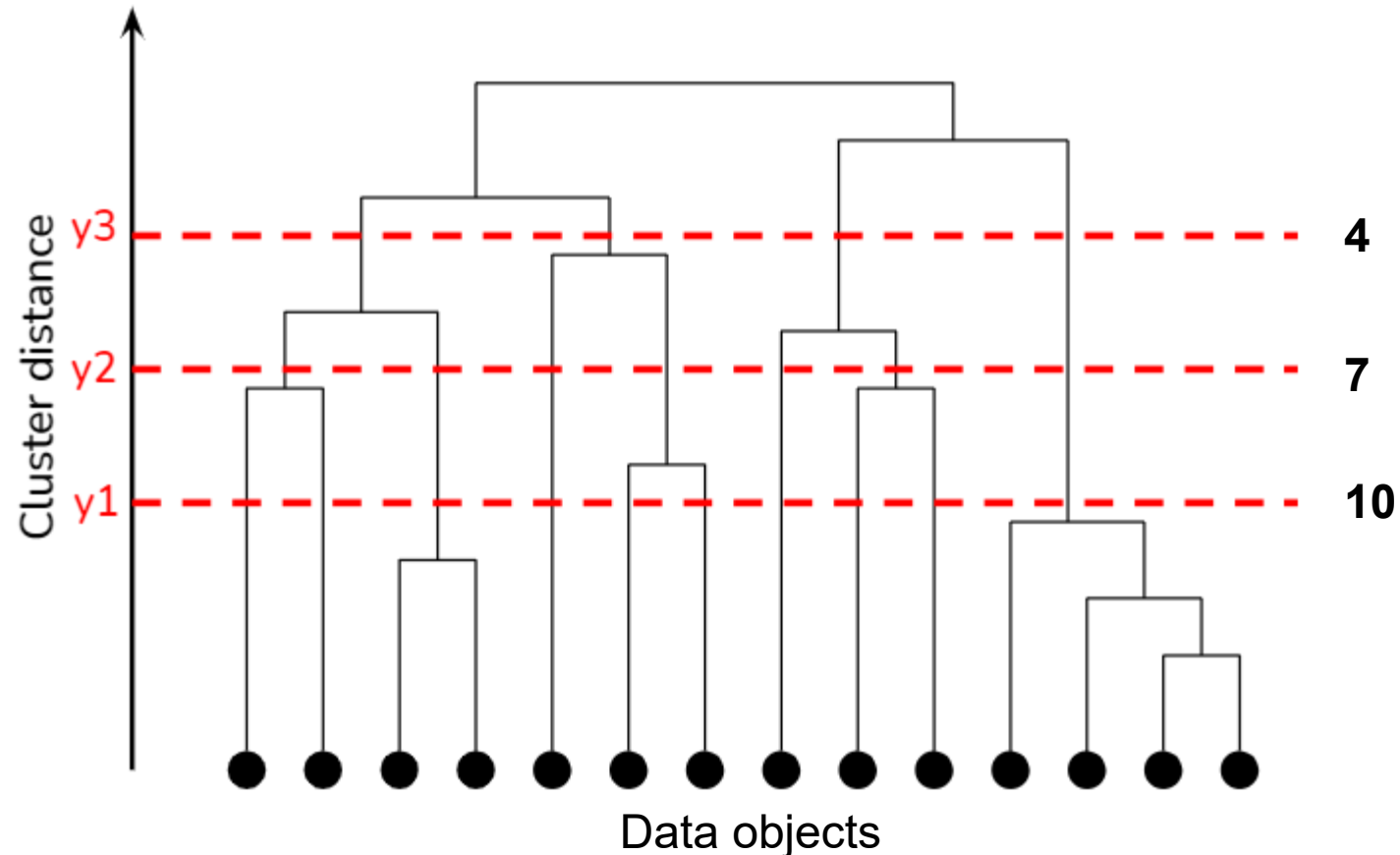
Clustering > Distance-based > Hierarchical

- Quiz: How many clusters will be returned with each of three cutting levels?
- Note: a single object can be also considered as a cluster



Clustering > Distance-based > Hierarchical

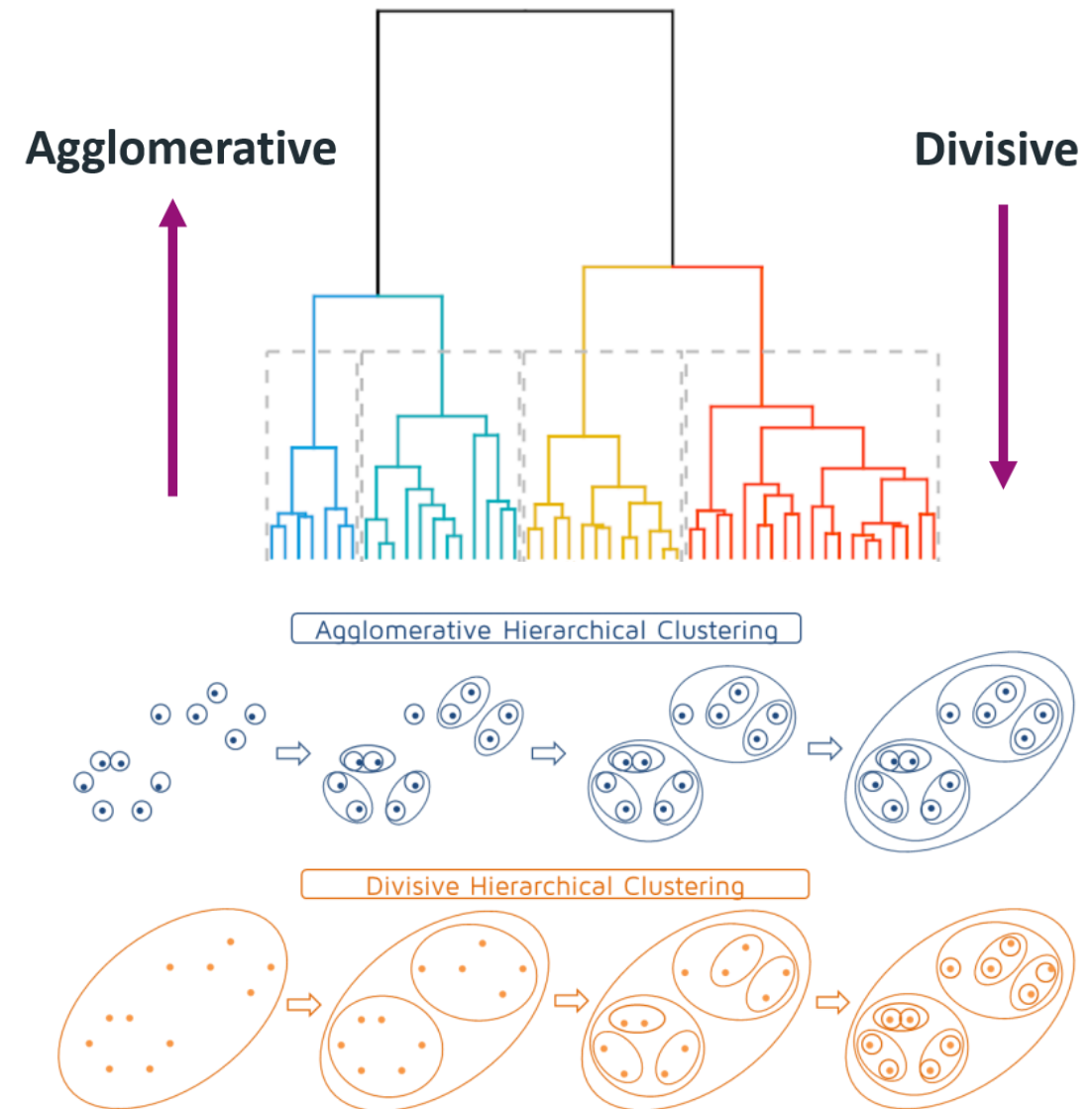
- Quiz: How many clusters will be returned with each of three cutting levels?
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Clustering > Distance-based > Hierarchical

□ Two approaches

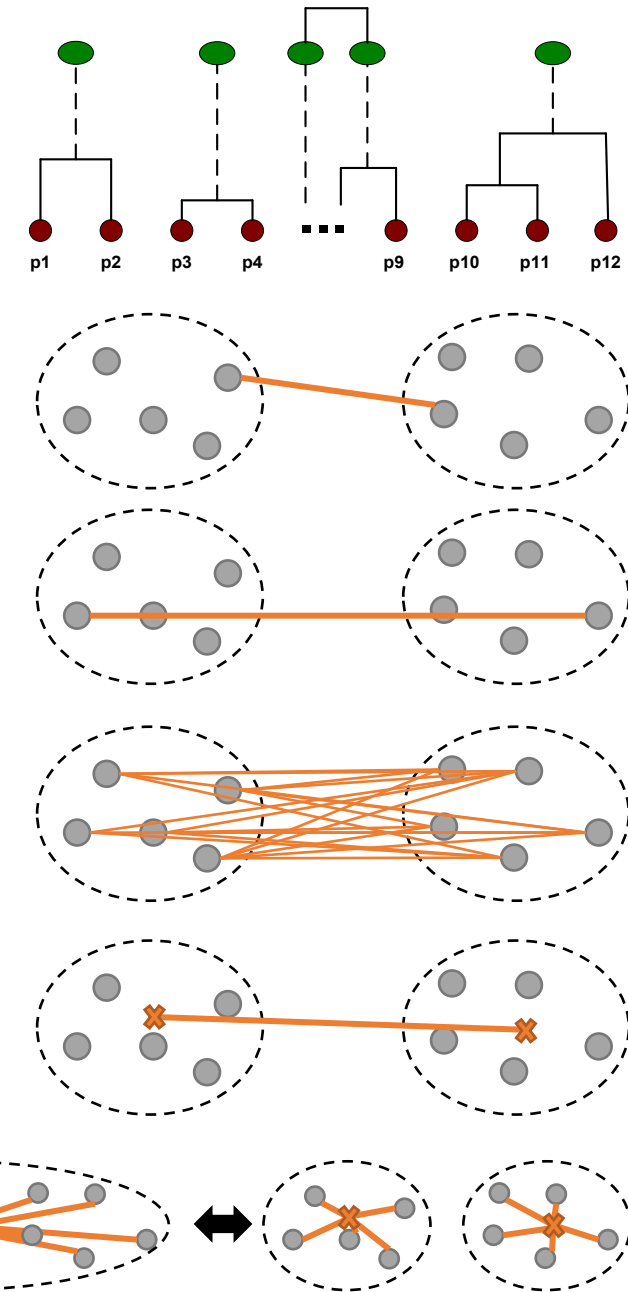
- Agglomerative (i.e., bottom-up)
 - Start with the points as individual clusters
 - Merge the closest pair of clusters continuously
- Divisive (i.e., top-down)
 - Start with one, all-inclusive cluster
 - Split it continuously into two clusters



Clustering > Distance-based > Hierarchical

Cluster distance measures for split/merge

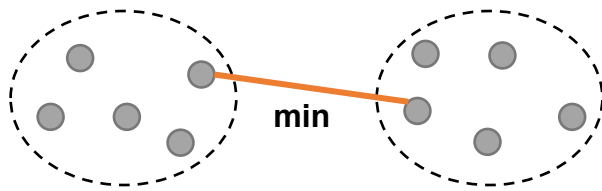
- Single link (min)
 - Smallest distance between an object in one cluster and an object in the other
 - i.e., $d(C_i, C_j) = \min(d(x_{ip}, x_{jq}))$
- Complete link (max)
 - Largest distance between an object in one cluster and an object in the other
 - i.e., $d(C_i, C_j) = \max(d(x_{ip}, x_{jq}))$
- Average link
 - Average distance between an object in one cluster and an object in the other
 - i.e., $d(C_i, C_j) = \text{avg}(d(x_{ip}, x_{jq}))$
- Centroid link
 - Distance between the centroids of two clusters
 - i.e., $d(C_i, C_j) = d(c_i, c_j)$
- Ward link
 - The change in the SSE of the clusters (i.e., similar to the partitioning approach)
 - i.e., $\Delta d(C_i, C_j) = \sum_{x_k \in C_i \cup C_j} d(x_k, c_{new})^2 - (\sum_{x_k \in C_i} d(x_k, c_i)^2 + \sum_{x_k \in C_j} d(x_k, c_j)^2) = \frac{N_i N_j}{N_i + N_j} d(c_i, c_j)^2$



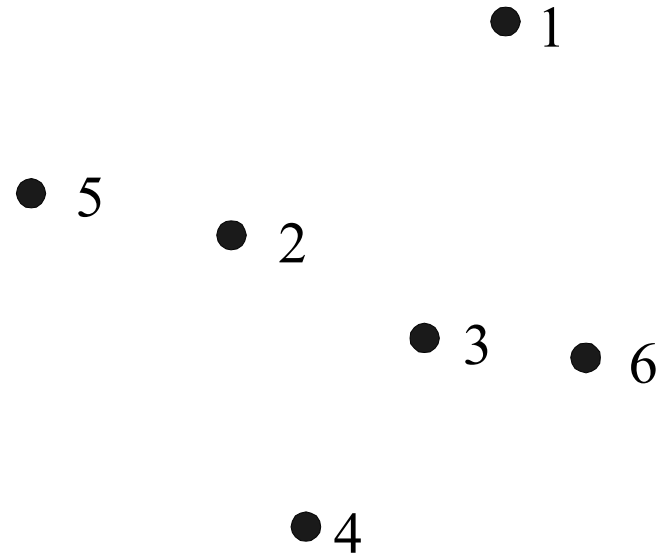
Clustering > Distance-based > Hierarchical

Cluster distance example

- Single link (min)



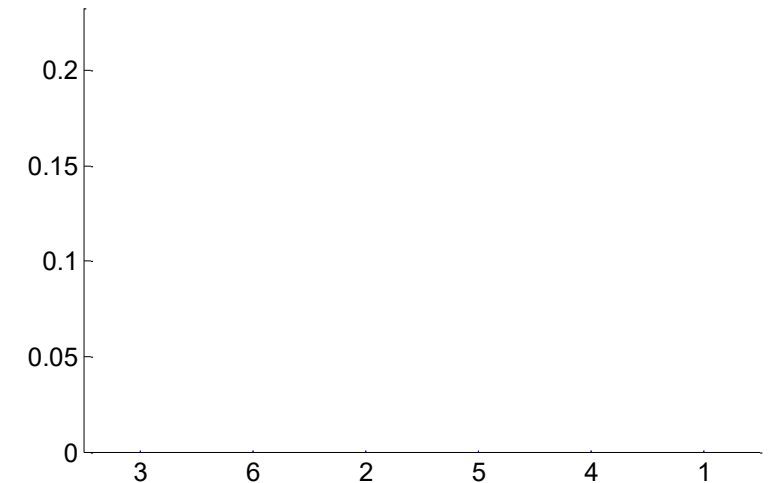
Agglomerative hierarchical clusters



Distance matrix

	1	2	3	4	5	6
1	0	.24	.22	.37	.34	.23
2	.24	0	.15	.20	.14	.25
3	.22	.15	0	.16	.28	.11
4	.37	.20	.16	0	.29	.22
5	.34	.14	.28	.29	0	.39
6	.23	.25	.11	.22	.39	0

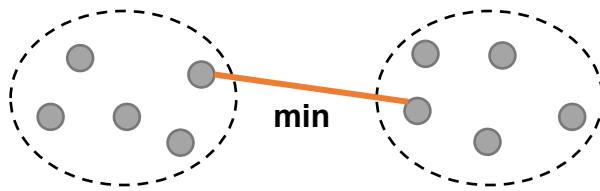
Dendrogram



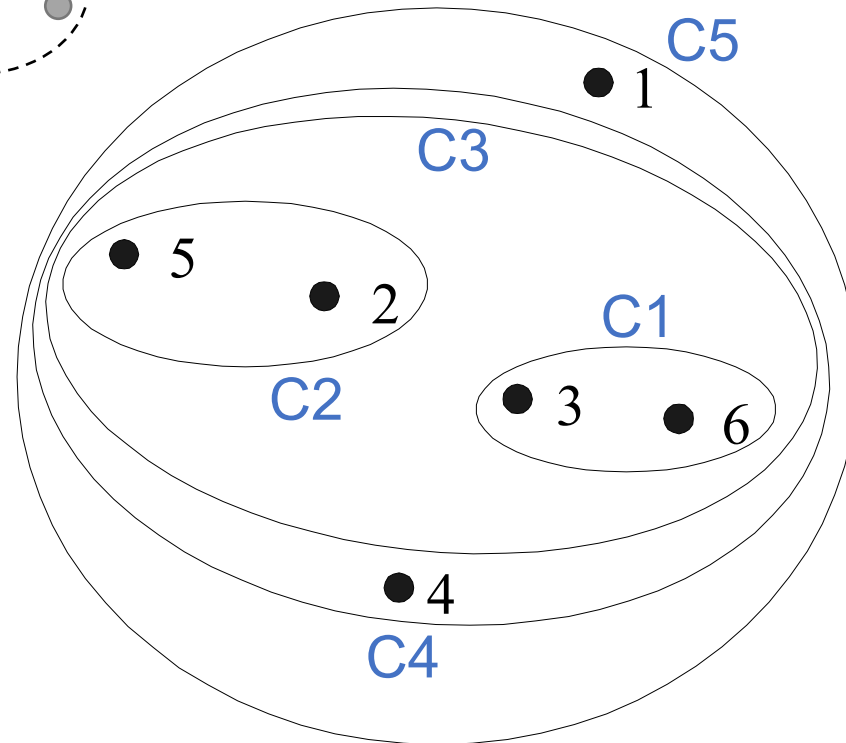
Clustering > Distance-based > Hierarchical

Cluster distance example

- Single link (min)



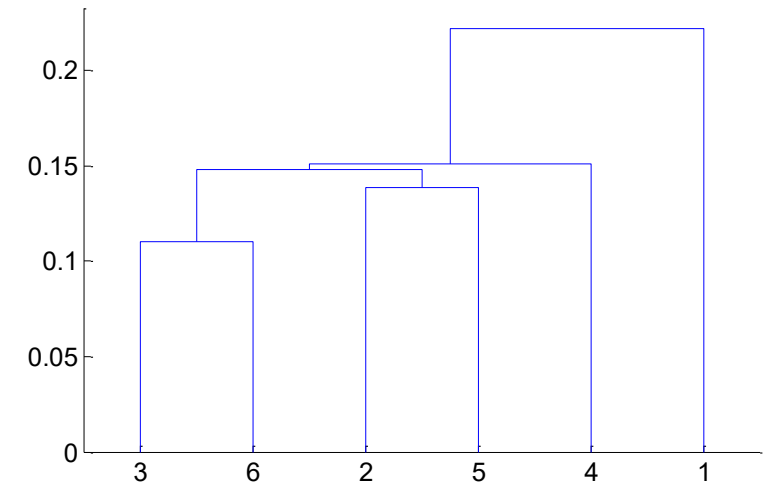
Agglomerative hierarchical clusters



Distance matrix

	1	2	3	4	5	6
1	0	.24	.22	.37	.34	.23
2	.24	0	.15	.20	.14	.25
3	.22	.15	0	.16	.28	.11
4	.37	.20	.16	0	.29	.22
5	.34	.14	.28	.29	0	.39
6	.23	.25	.11	.22	.39	0

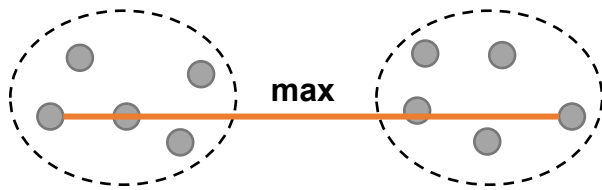
Dendrogram



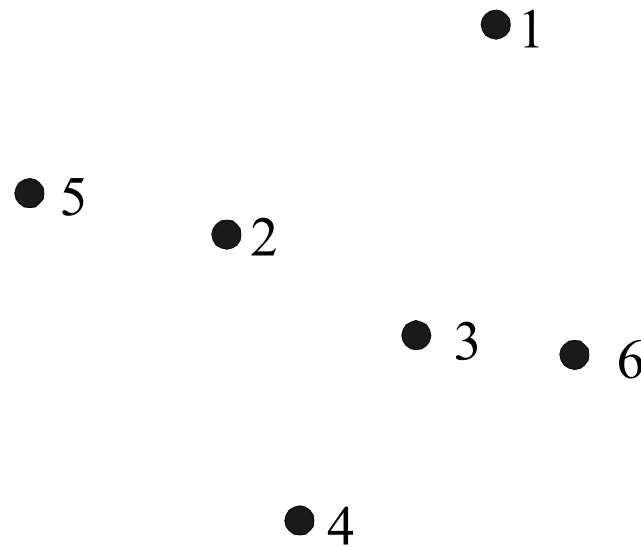
Clustering > Distance-based > Hierarchical

Cluster distance example

- Quiz: try hierarchical clustering with complete link (max)!



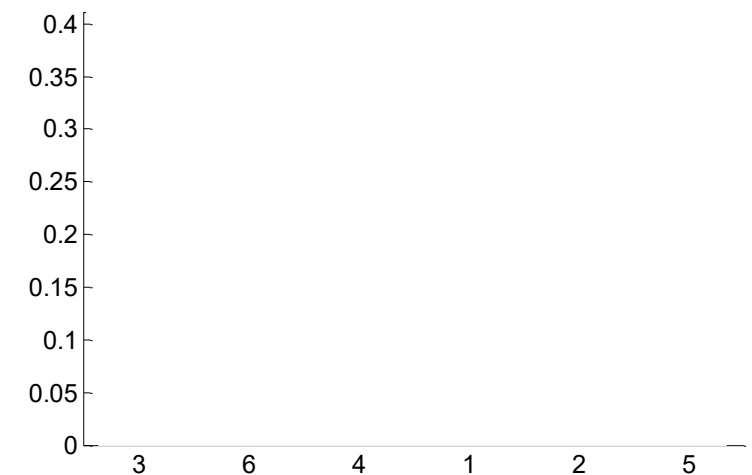
Agglomerative hierarchical clusters



Distance matrix

	1	2	3	4	5	6
1	0	.24	.22	.37	.34	.23
2	.24	0	.15	.20	.14	.25
3	.22	.15	0	.15	.28	.11
4	.37	.20	.15	0	.29	.22
5	.34	.14	.28	.29	0	.39
6	.23	.25	.11	.22	.39	0

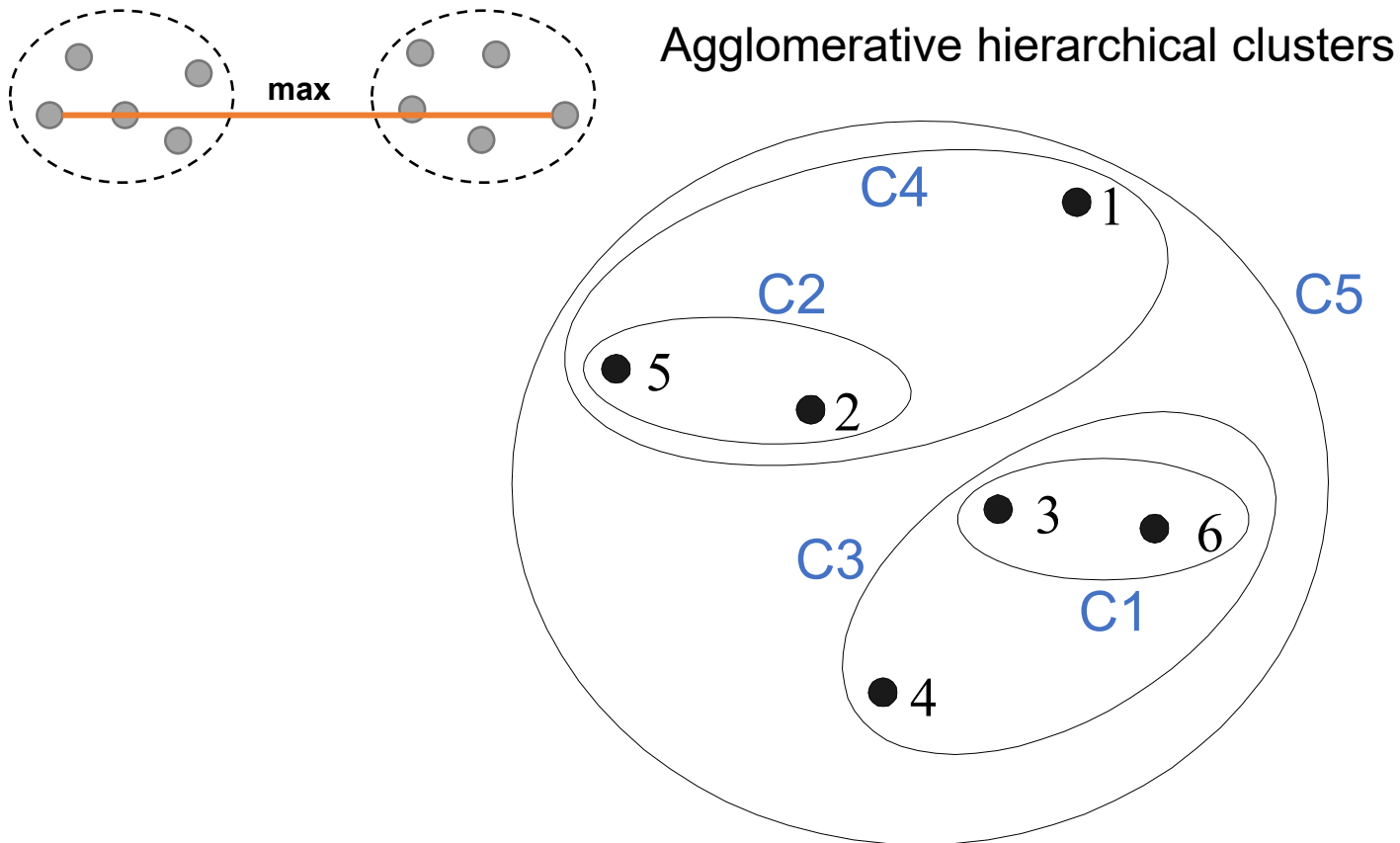
Dendrogram



Clustering > Distance-based > Hierarchical

Cluster distance example

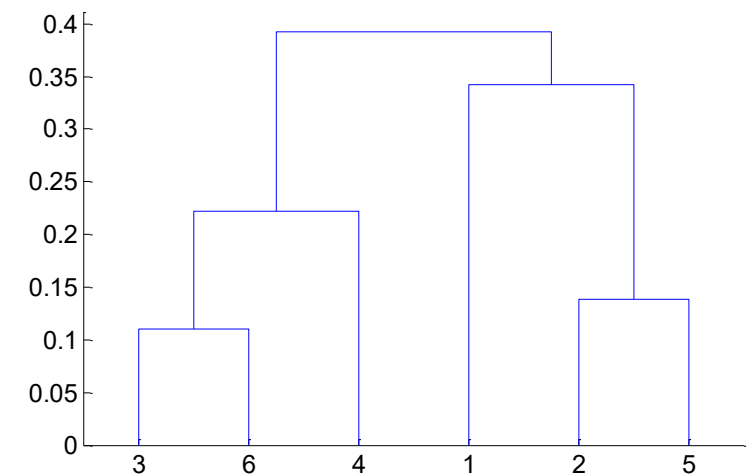
- Complete link (max)



Distance matrix

	1	2	3	4	5	6
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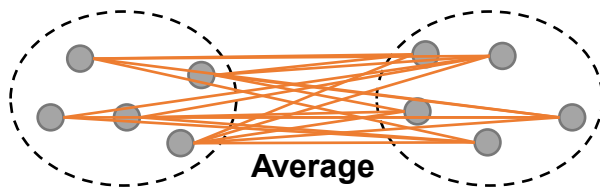
Dendrogram



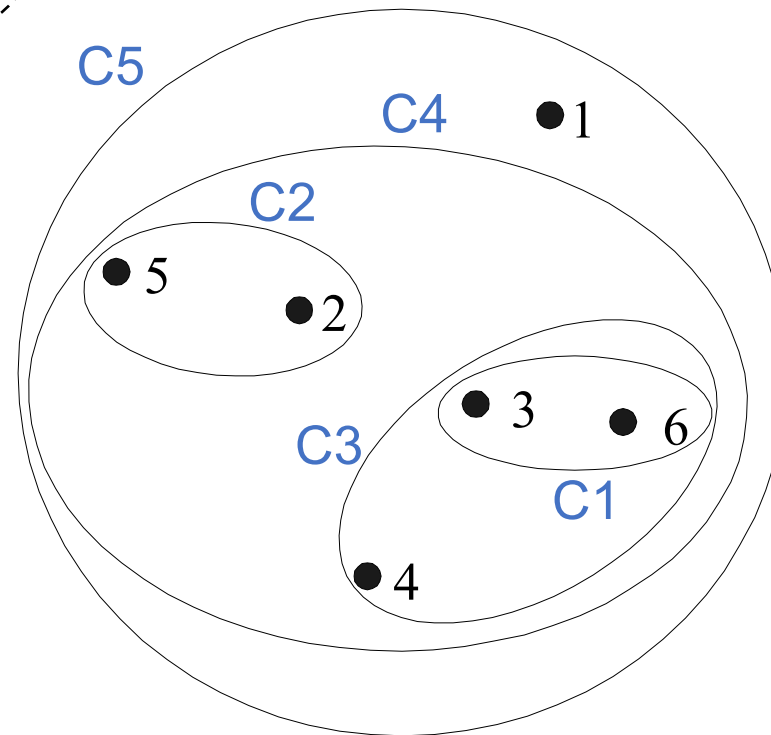
Clustering > Distance-based > Hierarchical

Cluster distance example

Average link



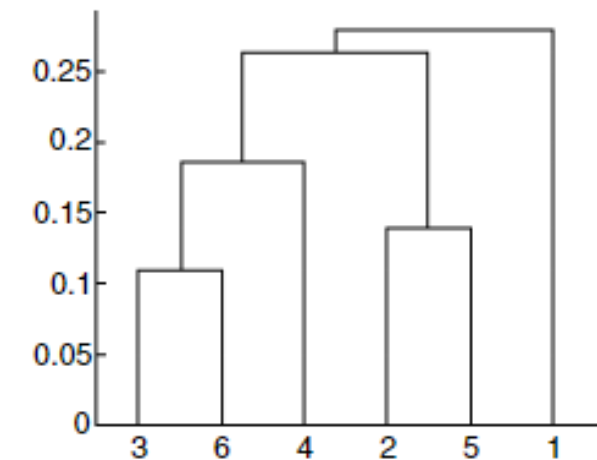
Agglomerative hierarchical clusters



Distance matrix

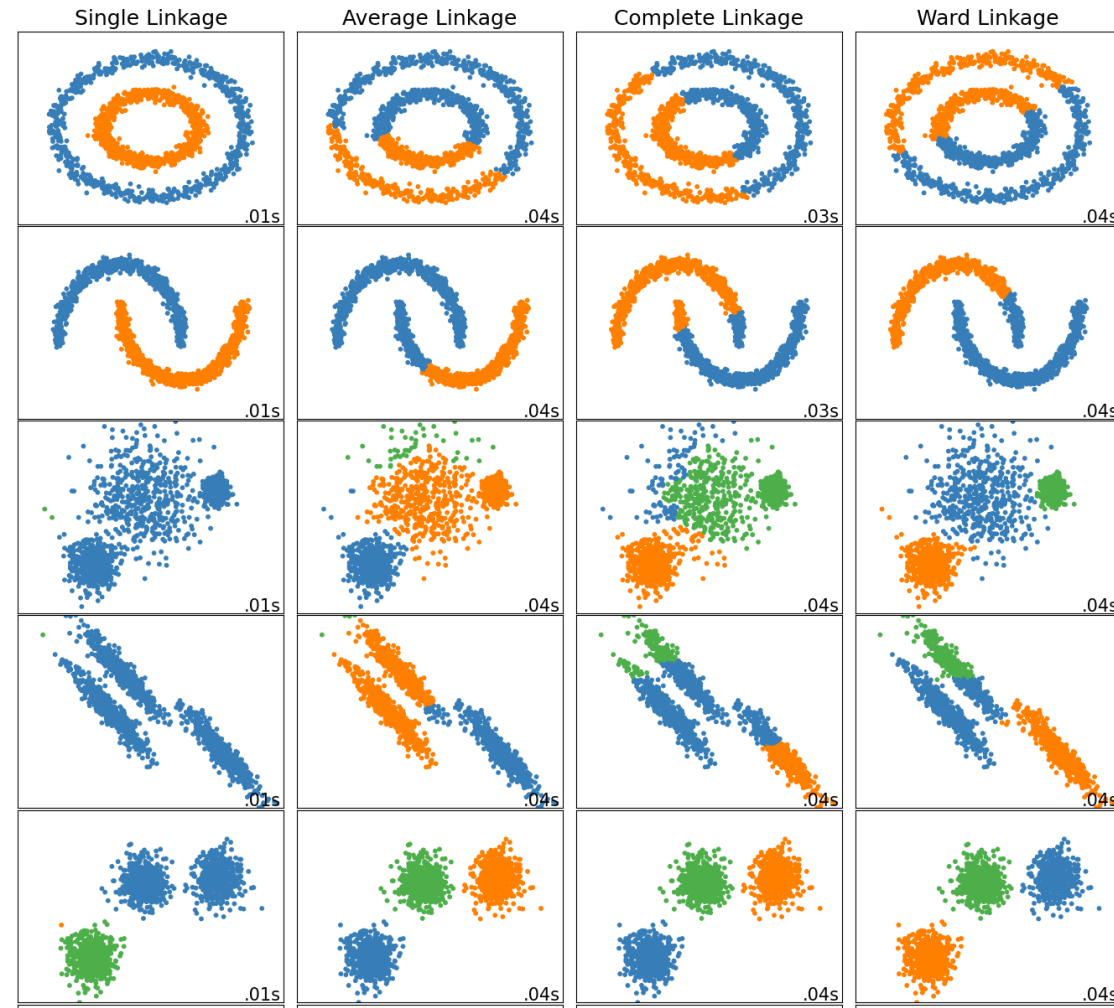
	1	2	3	4	5	6
1	0	.24	.22	.37	.34	.23
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4	.37	.20	.15	0	.29	.22
5	.34	.14	.28	.29	0	.39
6	.23	.25	.11	.22	.39	0

Dendrogram



Clustering > Distance-based > Hierarchical

□ Cluster distance example



Clustering > Distance-based > Hierarchical > BIRCH

❑ Limitation of Hierarchical Approach

- Not reversible merge and split at each level
- Not scalable with the time complexity $O(n^3)$ (i.e., pairwise distances $O(n^2)$ for up to $n-1$ levels)

❑ Efficient multi-phase hierarchical clustering

- *Challenge: calculating various distance statistics repeatedly?*

- Basis distance measures

- Centroid (average of objects)
- Radius (average distance to centroid)
- Diameter (average pair-wise distance)

- Additional distance measures

- D0 = Euclidean distance of two clusters
- D1 = Manhattan distance of two clusters
- D2 = Average inter-cluster distance
- D3 = Average intra-cluster distance
- D4 = Variance increase distance

$$C = \frac{\sum_{i=1}^N \vec{X}_i}{N} \quad R = \sqrt{\frac{\sum_{i=1}^N (\vec{X}_i - \vec{X}_0)^2}{N}} \quad D = \sqrt{\frac{\sum_{i=1}^N \sum_{j=1}^N (\vec{X}_i - \vec{X}_j)^2}{N(N-1)}}$$

$$D_0 = \sqrt{(\vec{X}_{o_1} - \vec{X}_{o_2})^2} \quad D_1 = |\vec{X}_{o_1} - \vec{X}_{o_2}| = \sum_{i=1}^d |\vec{X}_{o_1}[i] - \vec{X}_{o_2}[i]|$$

$$D_2 = \sqrt{\frac{\sum_{i=1}^{N_1} \sum_{j=N_1+1}^{N_1+N_2} (\vec{X}_i - \vec{X}_j)^2}{N_1 N_2}} \quad D_3 = \sqrt{\frac{\sum_{i=1}^{N_1+N_2} \sum_{j=1}^{N_1+N_2} (\vec{X}_i - \vec{X}_j)^2}{(N_1 + N_2)(N_1 + N_2 - 1)}}$$

$$D_4 = \sum_{k=1}^{N_1+N_2} \left(\vec{X}_k - \frac{\sum_{l=1}^{N_1+N_2} \vec{X}_l}{N_1 + N_2} \right)^2 - \sum_{i=1}^{N_1} \left(\vec{X}_i - \frac{\sum_{l=1}^{N_1} \vec{X}_l}{N_1} \right)^2 - \sum_{j=N_1+1}^{N_1+N_2} \left(\vec{X}_j - \frac{\sum_{l=N_1+1}^{N_1+N_2} \vec{X}_l}{N_2} \right)^2$$

Clustering > Distance-based > Hierarchical > BIRCH

□ BIRCH (Balanced Iterative Reducing and Clustering Using Hierarchies)

▪ **Clustering feature (CF)** to summarize a cluster and **CF-tree** to represent a cluster hierarchy

• Clustering feature CF = (N, LS, SS)

– N: Number of data points

– LS: linear sum of N points: $\sum_{i=1}^N \vec{X}_i$

– SS: square sum of N points: $\sum_{i=1}^N \vec{X}_i^2$

• CF-tree

– A height-balanced tree that stores the CFs of clusters (nodes)

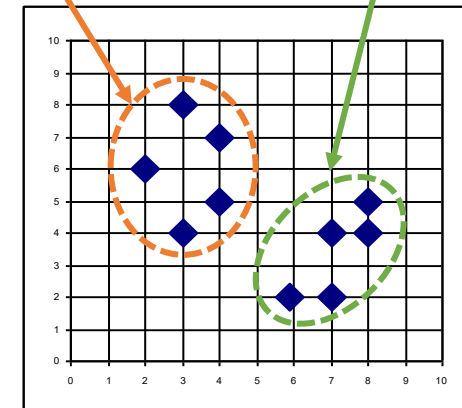
– The non-leaf nodes store the sum of CFs of their child nodes

$$CF = (N, LS = \sum_{i=1}^N \vec{X}_i, SS = \sum_{i=1}^N \vec{X}_i^2)$$

$$C1 = \{(3,4), (2,6), (4,5), (4,7), (3,8)\} \quad C2 = \{(6,2), (7,2), (7,4), (8,4), (8,5)\}$$

$$CF1 = (5, (16,30), 244)$$

$$CF2 = (5, (36,17), 327)$$



Clustering > Distance-based > Hierarchical > BIRCH

□ BIRCH (Balanced Iterative Reducing and Clustering Using Hierarchies)

■ Benefits of CF

- **Compact** than storing all the data objects in a cluster
- **Sufficient** to calculate various distance statistics

$$\text{e.g., Radius } R = \sqrt{\frac{\sum_{i=1}^N (\vec{X}_i - \vec{X}_o)^2}{N}} = \sqrt{\frac{SS}{N} - \left(\frac{LS}{N}\right)^2}$$

(try deriving other measures if you would like)

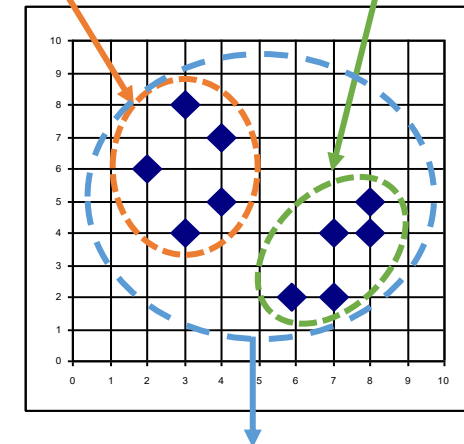
- **Additive** to allow merging clusters incrementally and consistently
 - i.e., $CF1 + CF2 = (N1+N2, LS1+LS2, SS1+SS2)$

$$CF = (N, LS = \sum_{i=1}^N \vec{X}_i, SS = \sum_{i=1}^N \vec{X}_i^2)$$

$$C1 = \{(3,4), (2,6), (4,5), (4,7), (3,8)\} \quad C2 = \{(6,2), (7,2), (7,4), (8,4), (8,5)\}$$

$$CF1 = (5, (16,30), 244)$$

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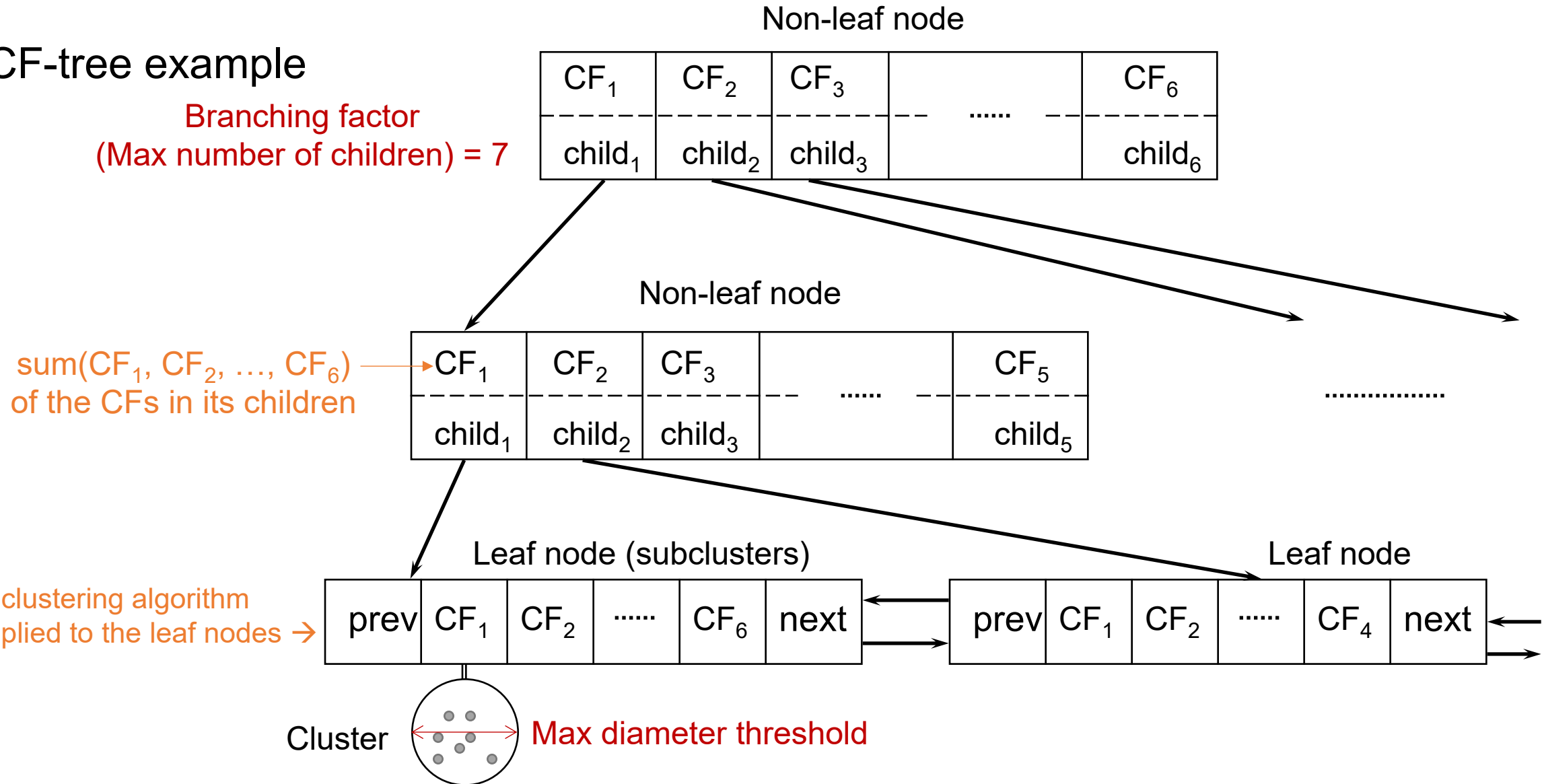
$$C3 = C1 \cup C2$$

$$CF3 = (5+5, (16+36, 30+17), 244+327) = (10, (52, 47), 571)$$

Clustering > Distance-based > Hierarchical > BIRCH

CF-tree example

Branching factor
(Max number of children) = 7



■ Clustering > Distance-based > Hierarchical > BIRCH

□ BIRCH (Balanced Iterative Reducing and Clustering Using Hierarchies)

■ Procedure

- Two parameters

- Branching factor: maximum number of children
- Threshold: maximum diameter of sub-clusters stored at the leaf nodes

1. CF-Tree building (i.e., low-level micro-clustering)

- For each object in the dataset, find the closest leaf entry, add the object to the leaf entry, and update the CF
- If the leaf node is full, split it into two leaf nodes and add one more entry in the parent

2. Global clustering (i.e., high-level macro-clustering)

- Applying a standard clustering method (e.g., k-means) on the leaf nodes of the CF tree

■ Issue

- Sensitive to insertion order of data points
- Since we fix the size of leaf nodes, clusters may not be so natural
- Clusters tend to be spherical given the diameter measures

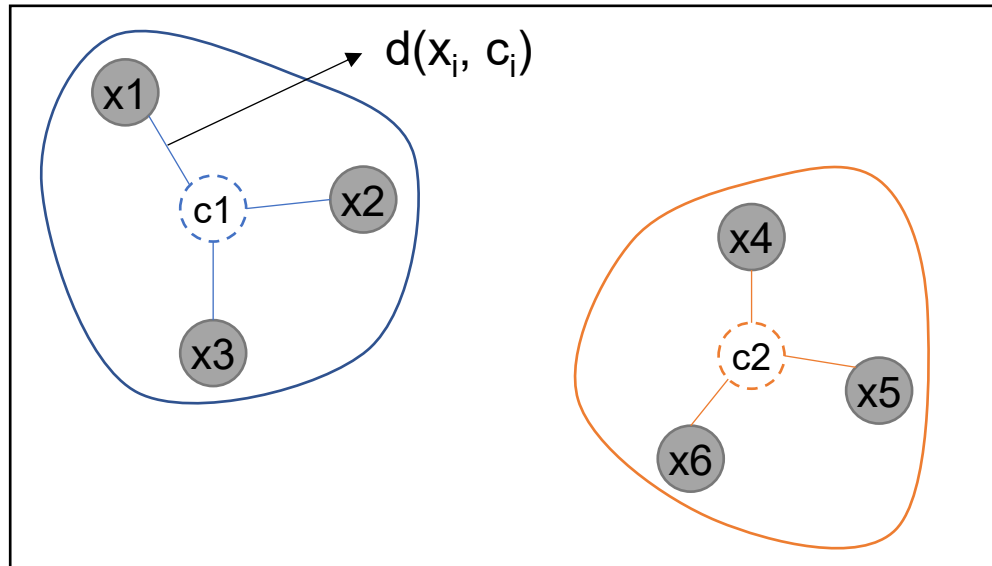
Outline

- ❑ Basic concepts of clustering
- ❑ Distance-based methods
- ❑ **Density-based methods**
 - DBSCAN
- ❑ Probabilistic methods
- ❑ Clustering evaluation

Clustering > Density-based

□ Motivation

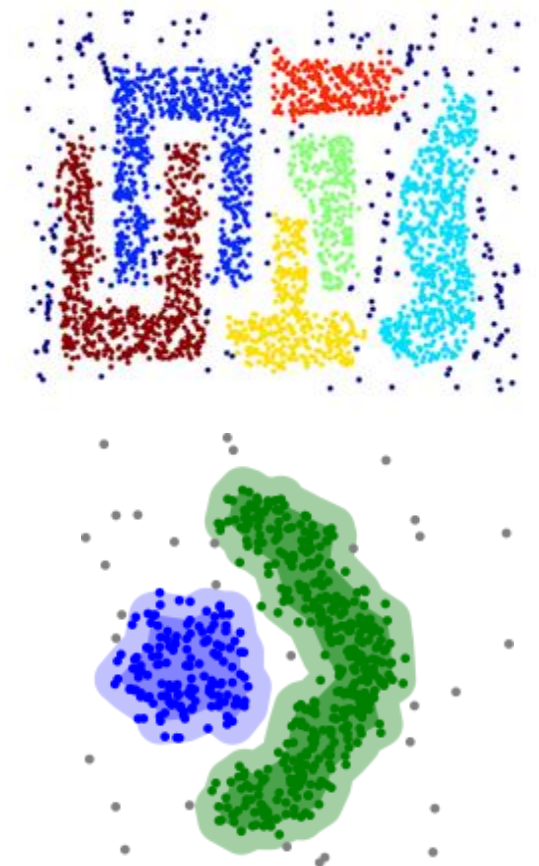
- Most partitioning and hierarchical methods are designed to find spherical-shaped clusters
- What if we want to find arbitrarily-shaped clusters
 - e.g. “S” shape or oval clusters?



Clustering > Density-based

□ Density-based clustering

- Clusters are formed as dense regions in the data space, separated by sparse regions
- Major features
 - Discover clusters of arbitrary shape
 - Robust to noise
 - Single-scan of datasets (only examine the local region to estimate density)
 - No need for the predefined number of clusters
 - Need density parameters for density estimation and termination condition
- Methods
 - **DBSCAN (connected region-based density)**
 - DENCLUE (density estimation function)
 - CLIQUE (grid-based density approximation)

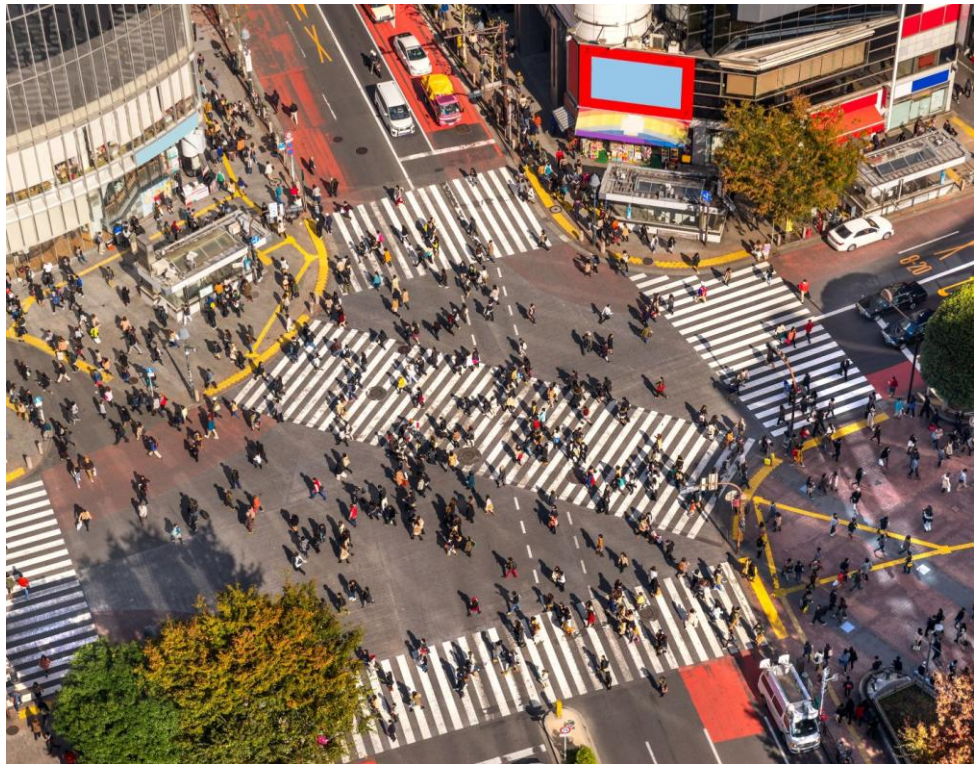


Clustering > Density-based > DBSCAN

□ DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

- *How can we find dense (and sparse) regions?*

→ The density of an object can be measured by the number of other objects close to it

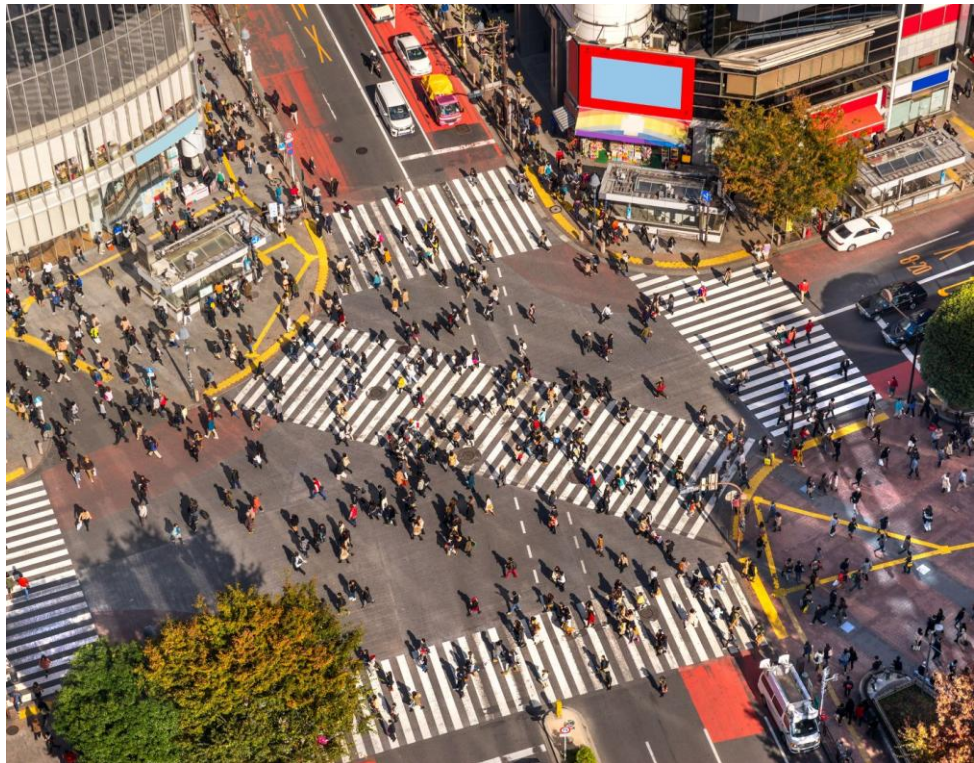


Clustering > Density-based > DBSCAN

□ DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

- *How can we find dense (and sparse) regions?*

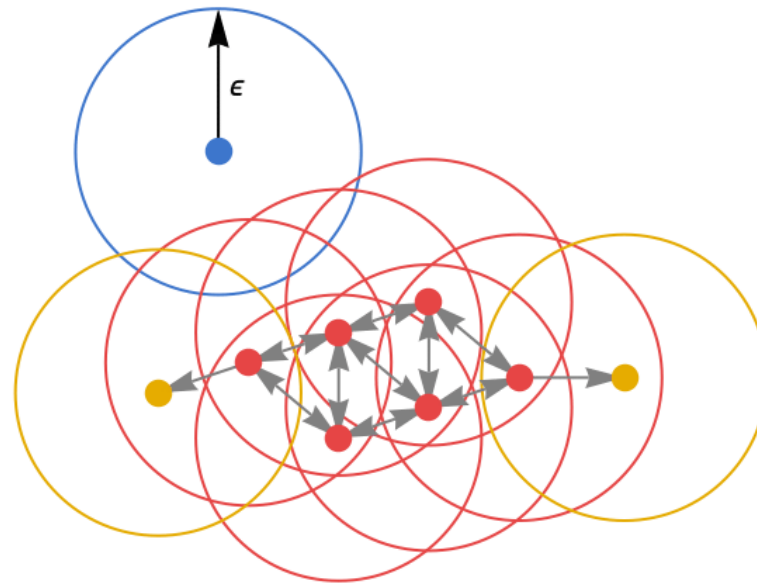
→ The density of an object can be measured by the number of other objects close to it
→ Connect all the objects in the dense regions and group them as clusters



Clustering > Density-based > DBSCAN

□ DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

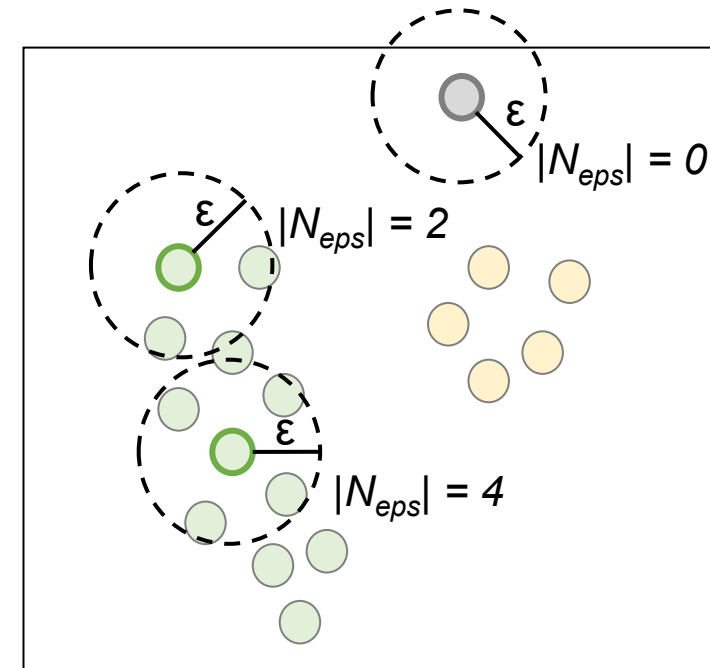
- *How can we find dense (and sparse) regions?*
 - The density of an object can be measured by the number of other objects close to it
 - Connect all the objects in the dense regions and group them as clusters
- A density-based notion of cluster as a maximal set of density-connected points



Clustering > Density-based > DBSCAN

□ DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

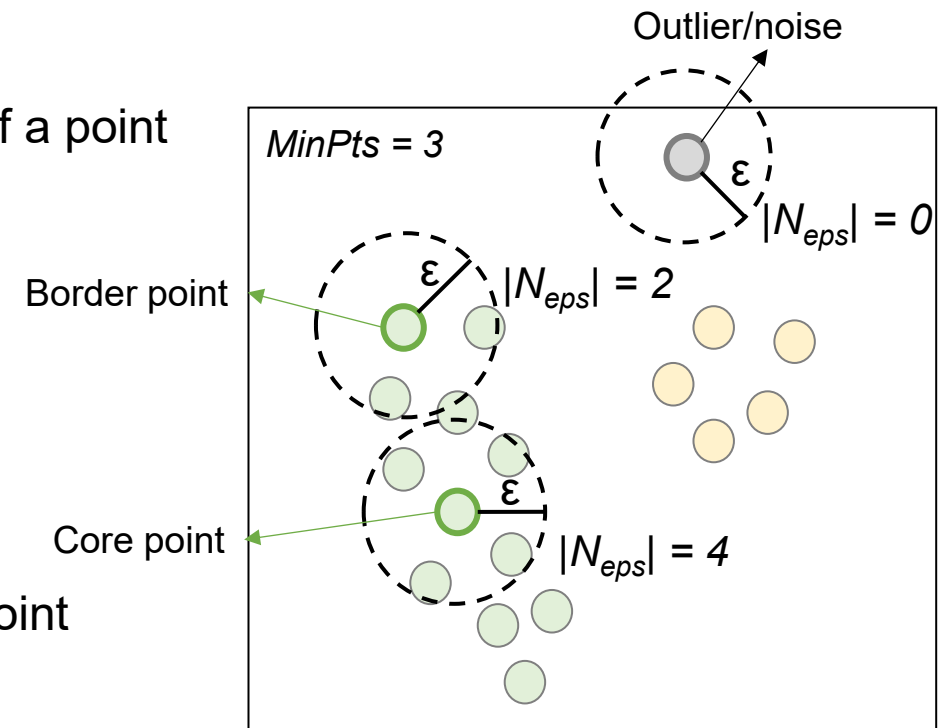
- A density-based notion of cluster as a maximal set of density-connected points
- Main parameters
 - $Eps(\epsilon)$: maximum radius of the neighborhood
 - Eps -neighborhood of a point x : $N_{Eps}(x) = \{p \mid d(p, x) \leq Eps, p \neq x\}$
 - $MinPts$: minimum number of points in the Eps -neighborhood of a point



Clustering > Density-based > DBSCAN

□ DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

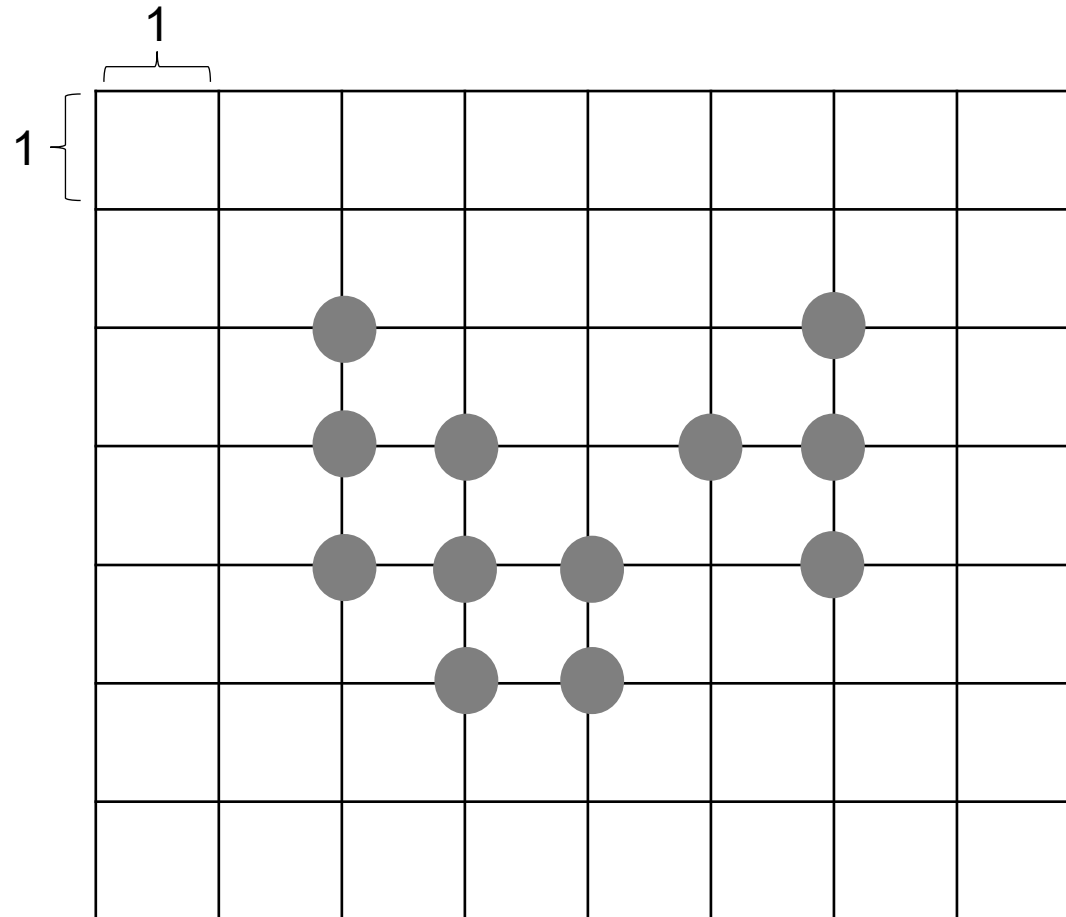
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 - $MinPts$: minimum number of points in the Eps -neighborhood of a point
- Types of points
 - Core point: in a dense neighborhood (i.e., cluster)
 - i.e., $|N_{Eps}(x)| \geq MinPts$
 - Border point: not a core point, but the neighbor of a core point
 - i.e., $|N_{Eps}(x)| < MinPts$, but $x \in N_{Eps}(x')$ for a core point x'
 - Outlier/noise: neither a core point nor the neighbor of a core point



Clustering > Density-based > DBSCAN

Quiz

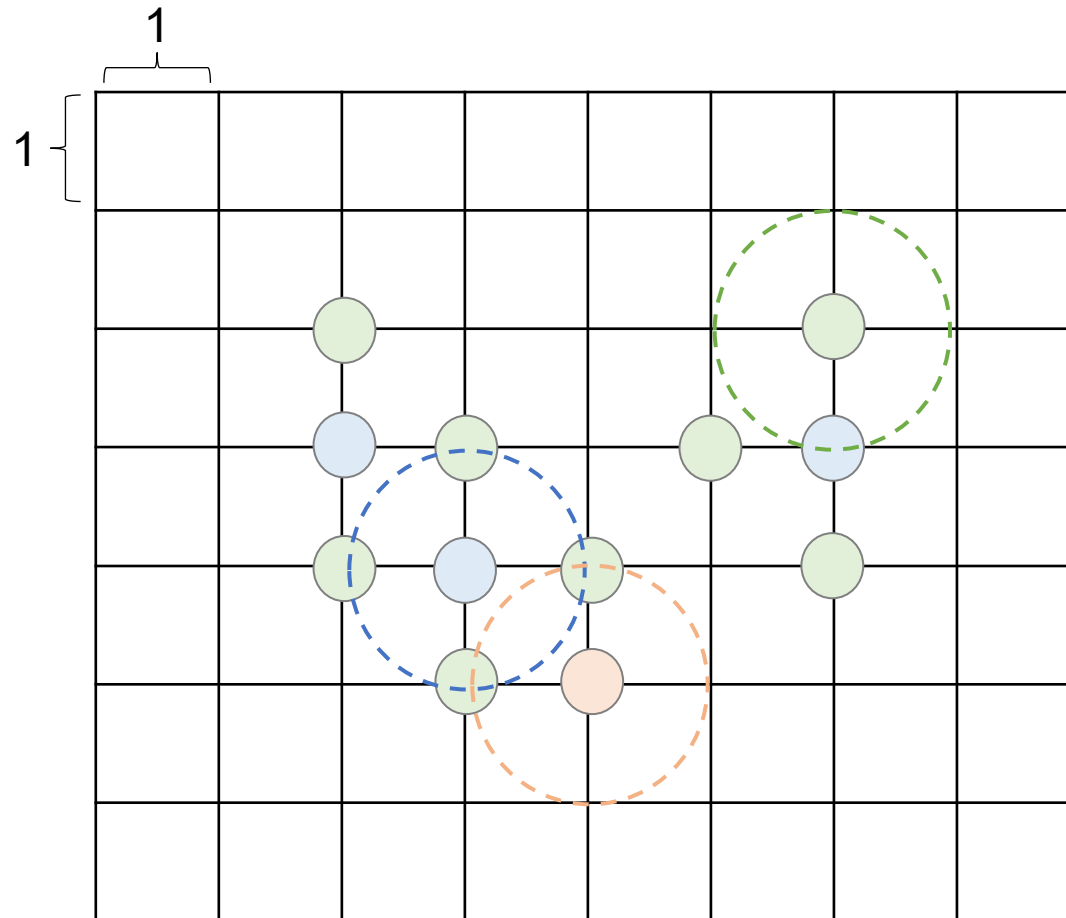
- Given $Eps = 1$ and $MinPts = 3$, find core points, border points, and noises



Clustering > Density-based > DBSCAN

Quiz

- Given $Eps = 1$ and $MinPts = 3$, find core points, border points, and noises



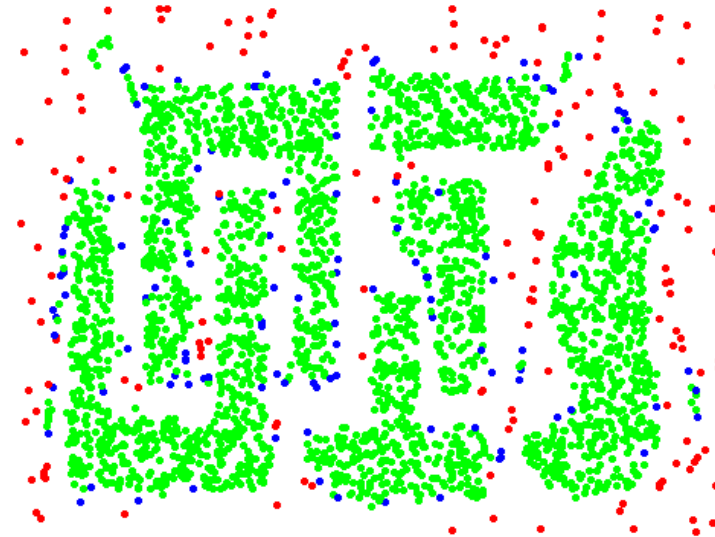
Clustering > Density-based > DBSCAN

□ Example of point types

Eps = 10, MinPts = 4



Original points



Core, border, and noise points

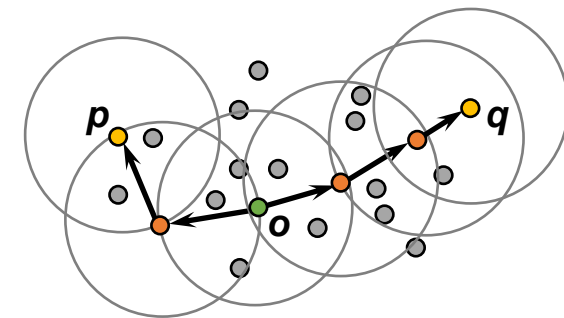
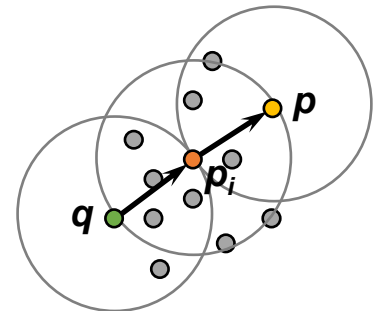
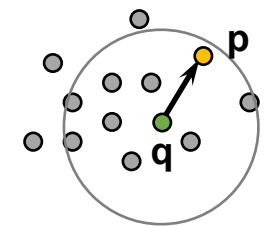
Can you find density-based clusters?

Clustering > Density-based > DBSCAN

□ DBSCAN

- A density-based notion of cluster as a maximal set of density-connected points
- Reachability and Connectivity
 - Directly density-reachable
 - A point p is directly density-reachable from a point q , if p is in the neighborhood of a core point q
 - i.e., $p \in N_{Eps}(q)$ and $|N_{Eps}(q)| \geq MinPts$
 - Density-reachable
 - A point p is density-reachable from a point q , if there is a chain of points where consecutive points are directly density-reachable
 - i.e., For $p_1(=q), \dots, p_{i-1}, p_i, p_{i+1}, \dots, p_n(=p)$, p_{i+1} is *directly density-reachable* from p_i
 - Density-connected
 - A point p is density-connected to a point q , if there is a point o such that both p and q are density-reachable from o
- [Recap] a density-based cluster = a maximal set of density-connected points
 - i.e., every point is density-reachable from any point within the set

Eps = the radius of the circles
 $MinPts = 5$



Clustering > Density-based > DBSCAN

□ Pseudo-code

ALGORITHM 1: Pseudocode of Original Sequential DBSCAN Algorithm

```

Input: DB: Database
Input:  $\epsilon$ : Radius
Input: minPts: Density threshold
Input: dist: Distance function
Data: label: Point labels, initially undefined

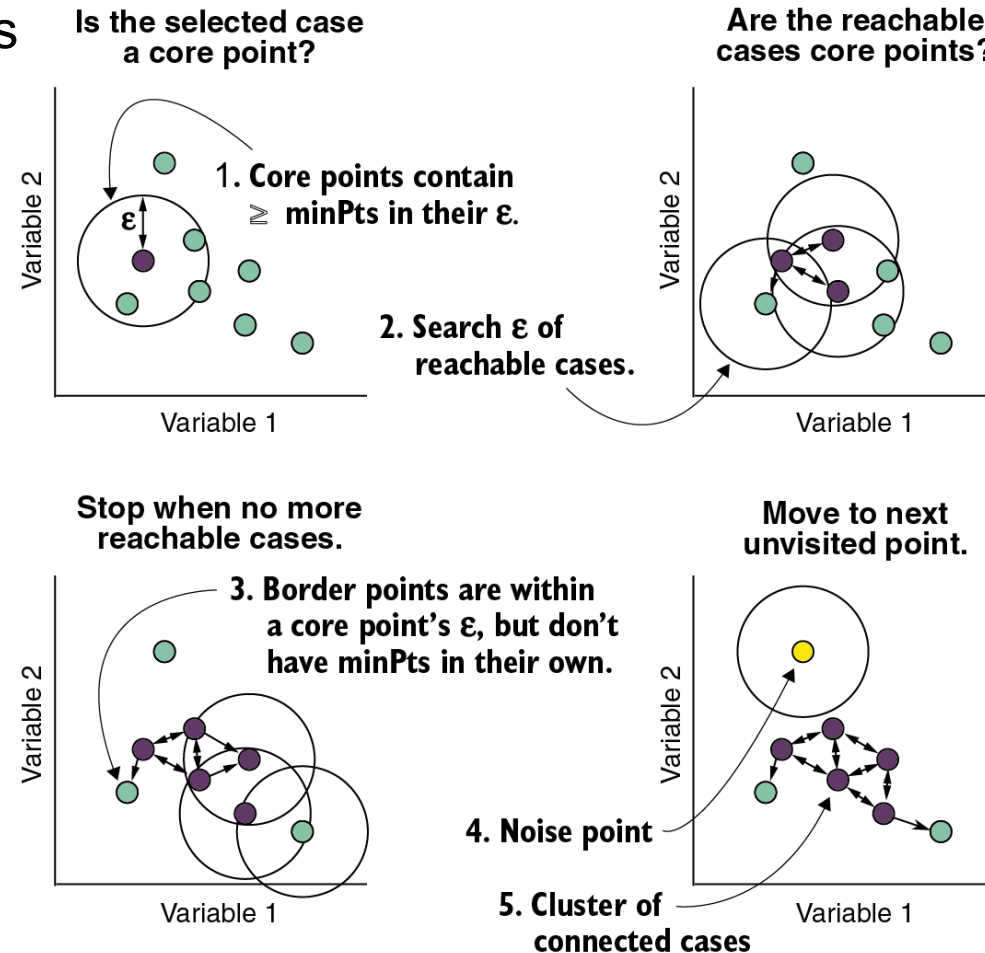
1 foreach point p in database DB do                                // Iterate over every point
2   if label(p)  $\neq$  undefined then continue                        // Skip processed points
3   Neighbors N  $\leftarrow$  RANGEQUERY(DB, dist, p,  $\epsilon$ )           // Find initial neighbors
4   if  $|N| < \textit{minPts}$  then                                          // Non-core points are noise
5     label(p)  $\leftarrow$  Noise
6     continue
7   c  $\leftarrow$  next cluster label                                    // Start a new cluster
8   label(p)  $\leftarrow$  c
9   Seed set S  $\leftarrow N \setminus \{p\}$                              // Expand neighborhood
10  foreach q in S do
11    if label(q) = Noise then label(q)  $\leftarrow$  c
12    if label(q)  $\neq$  undefined then continue
13    Neighbors N  $\leftarrow$  RANGEQUERY(DB, dist, q,  $\epsilon$ )
14    label(q)  $\leftarrow$  c
15    if  $|N| < \textit{minPts}$  then continue                             // Core-point check
16    S  $\leftarrow S \cup N$ 

```

Clustering > Density-based > DBSCAN

□ Example

- Eps = the radius
- MinPts = 2



Clustering > Density-based > DBSCAN

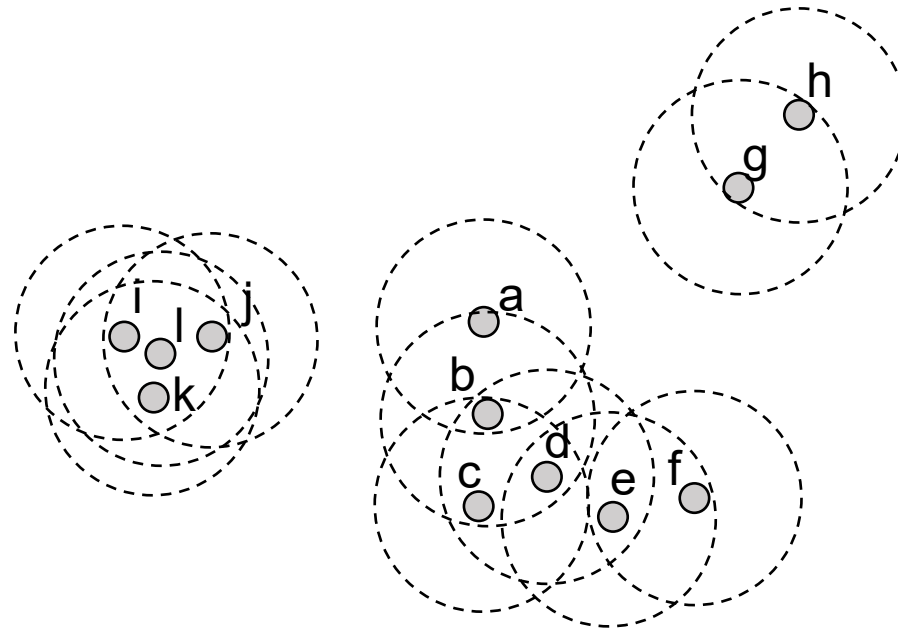
□ Pseudo-code (simplified)

ALGORITHM 2: Abstract DBSCAN Algorithm

1	Compute neighbors of each point and identify core points	// Identify core points
2	Join neighboring core points into clusters	// Assign core points
3	foreach non-core point do	
4	Add to a neighboring core point if possible	// Assign border points
5	Otherwise, add to noise	// Assign noise points

□ Example

- Eps = the radius of the circles
- MinPts = 3
- Clusters?
- Noises (outliers)?



Clustering > Density-based > DBSCAN

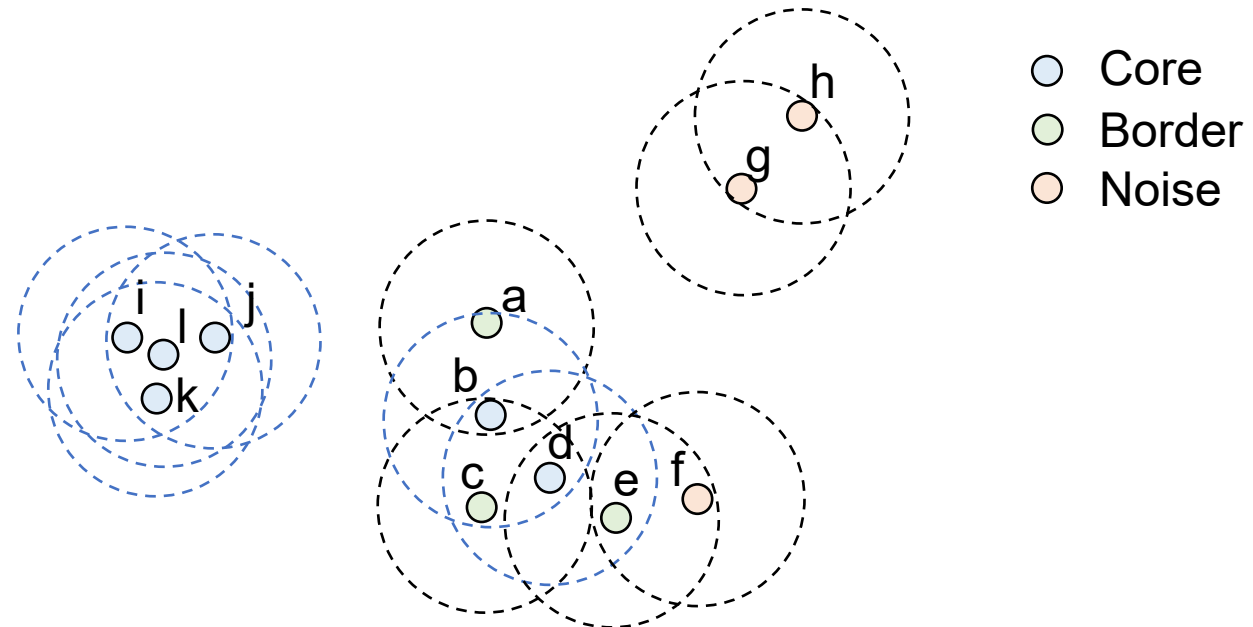
❑ Pseudo-code (simplified)

ALGORITHM 2: Abstract DBSCAN Algorithm

1	Compute neighbors of each point and identify core points	// Identify core points
2	Join neighboring core points into clusters	// Assign core points
3	foreach non-core point do	
4	Add to a neighboring core point if possible	// Assign border points
5	Otherwise, add to noise	// Assign noise points

❑ Example

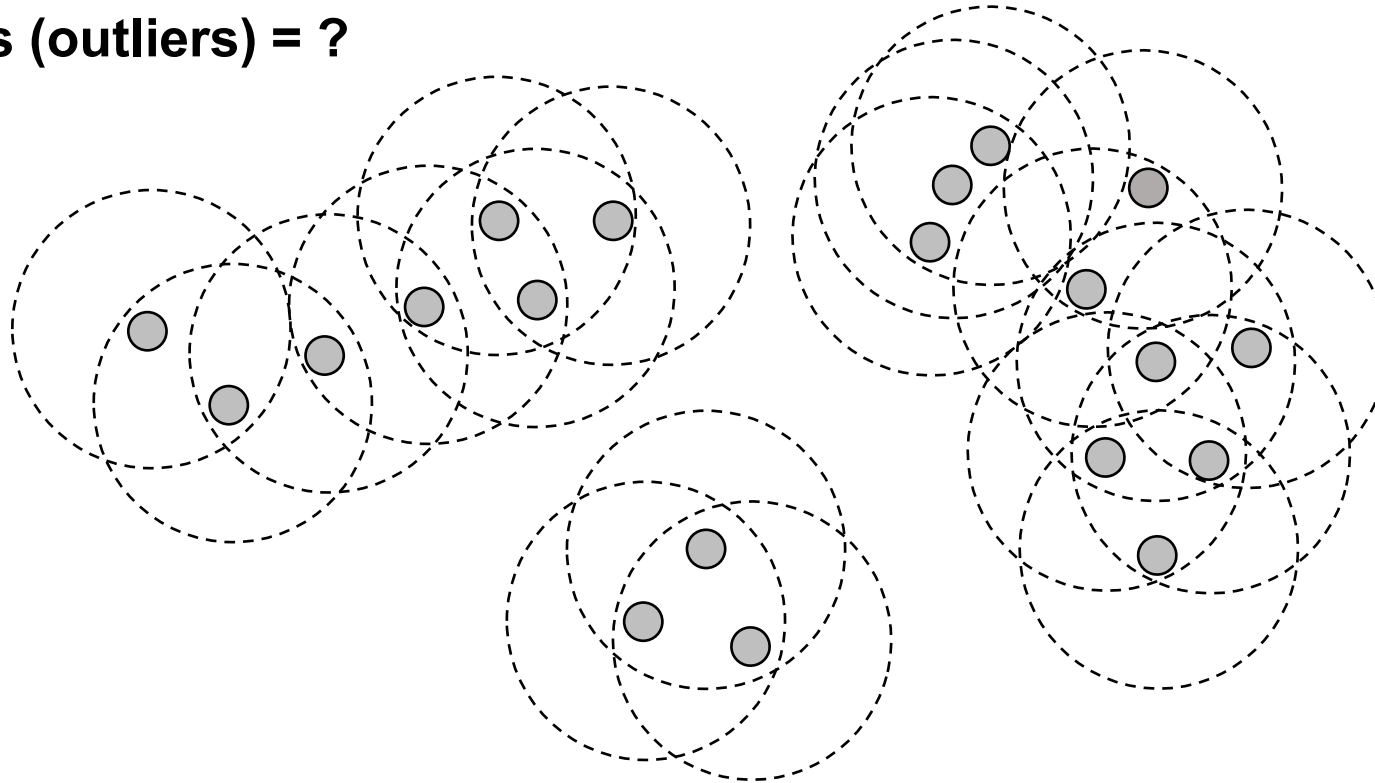
- Eps = the radius of the circles
- MinPts = 3
- 2 Clusters
 - {a, b, c, d, e}, {i, j, k, l}
- 3 noises (outliers)
 - f, g, h



Clustering > Density-based > DBSCAN

□ Quiz

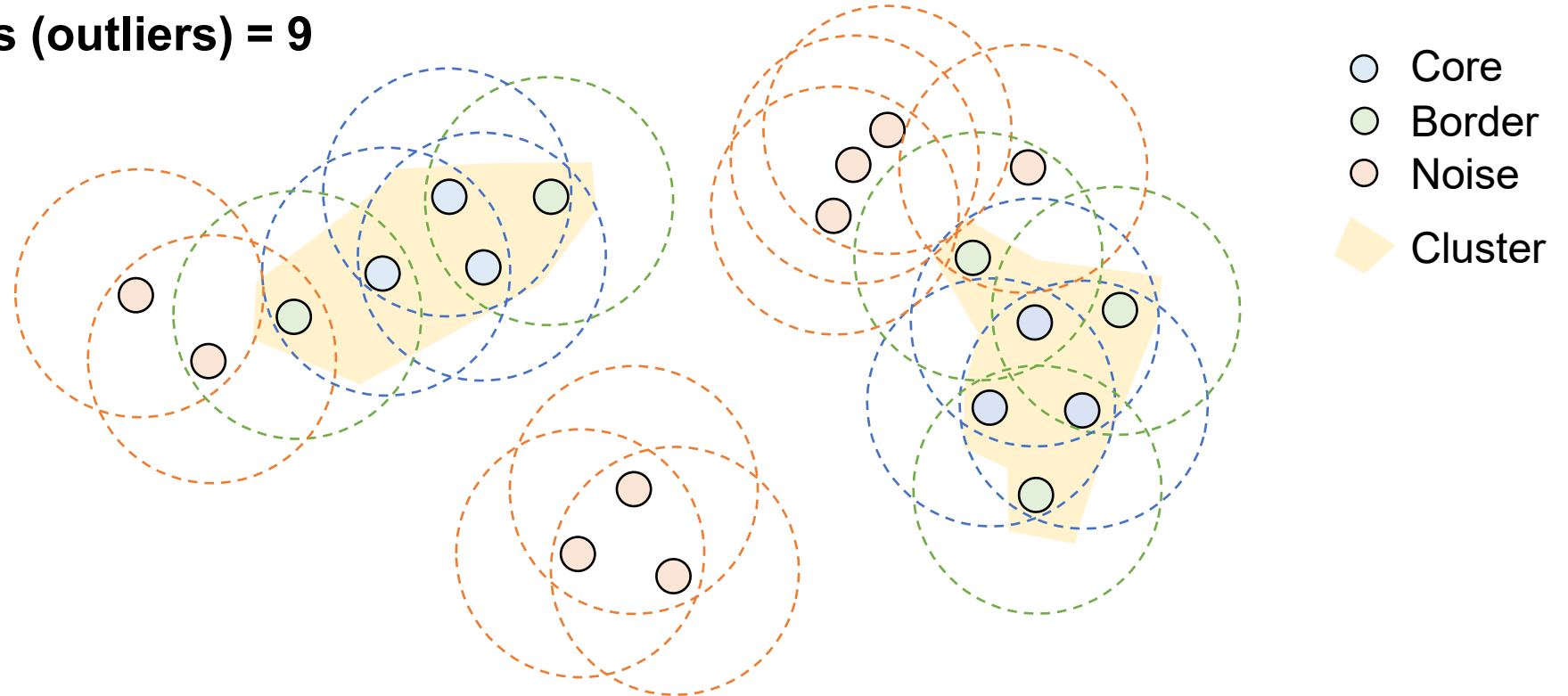
- Given Eps = the radius of the circle and $MinPts$ = 3, conduct DBSCAN to find clusters
- **Number of clusters = ?**
- **Number of noises (outliers) = ?**



Clustering > Density-based > DBSCAN

□ Quiz – Answer

- Given Eps = the radius of the circle and $MinPts$ = 3, conduct DBSCAN to find clusters
- **Number of clusters = 2**
- **Number of noises (outliers) = 9**



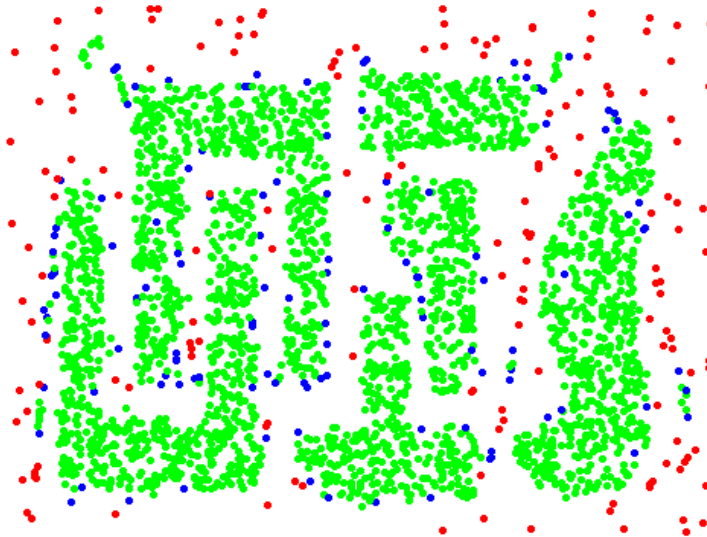
Clustering > Density-based > DBSCAN

□ Example of point types and clusters by DBSCAN

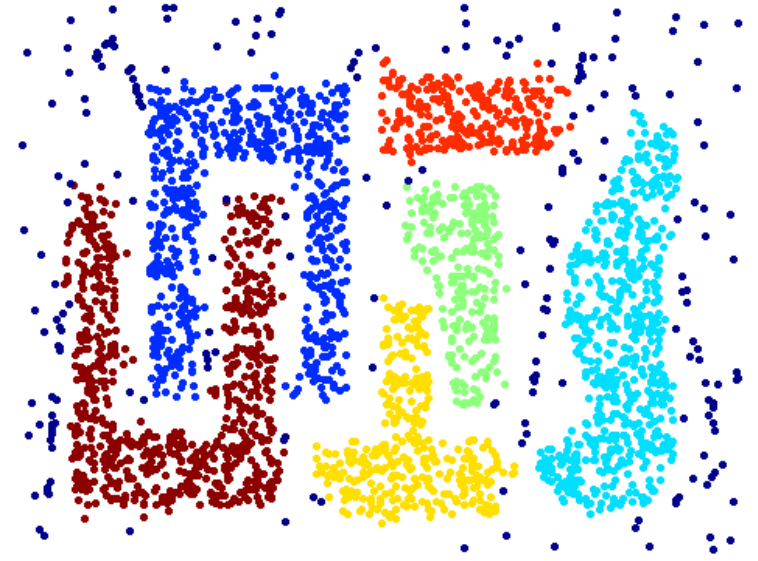
Eps = 10, MinPts = 4



Original points



Core, border, and noise points

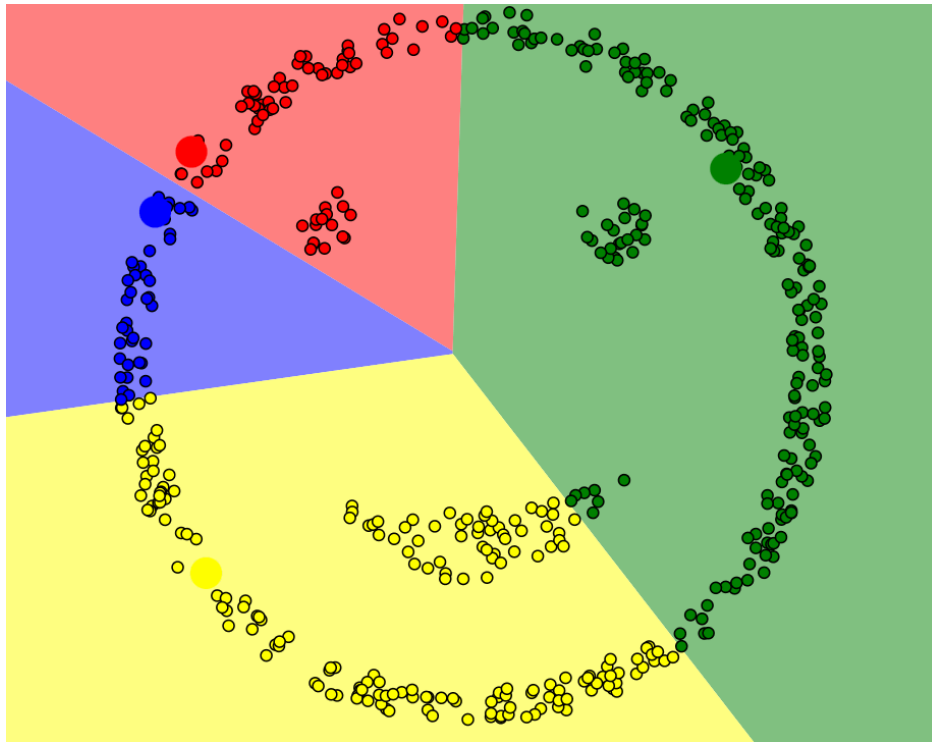


Clusters with noises
(a minimum cluster size applied)

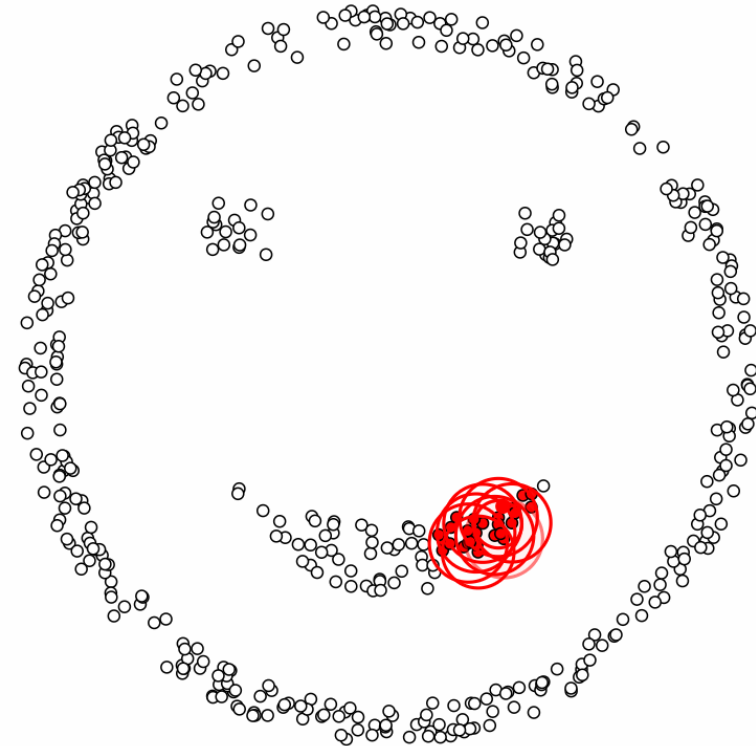
Clustering > Density-based > DBSCAN

□ Comparison of distance-based and density-based clustering

k-Means



DBSCAN ([Demo](#))

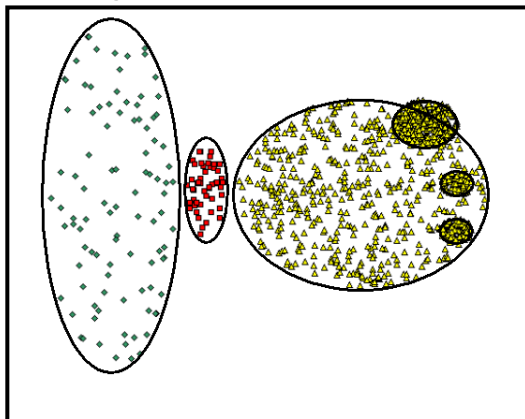


Clustering > Density-based > DBSCAN

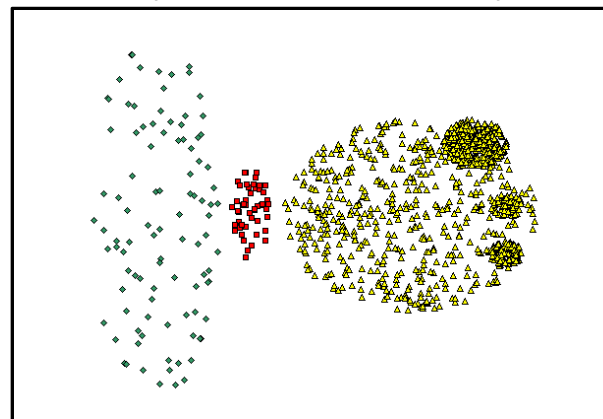
□ Discussion

- Complexity: $O(n^2)$ or $O(n \log n)$ if spatial index is used
- Advantages
 - Does not need to know the number of clusters
 - Can find arbitrarily shaped clusters
 - Has a notion of noise – robust to outliers
- Disadvantages
 - Hard to determine the proper parameter values (*Eps* and *MinPts*)
 - Ineffective to data sets with large density difference
 - Example

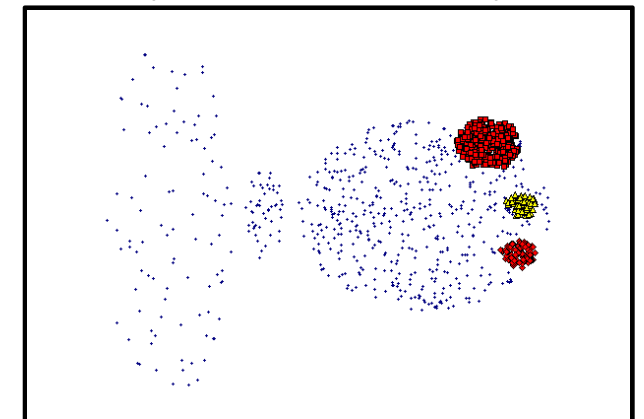
Original data and clusters



(MinPts=4, Eps=9.92)



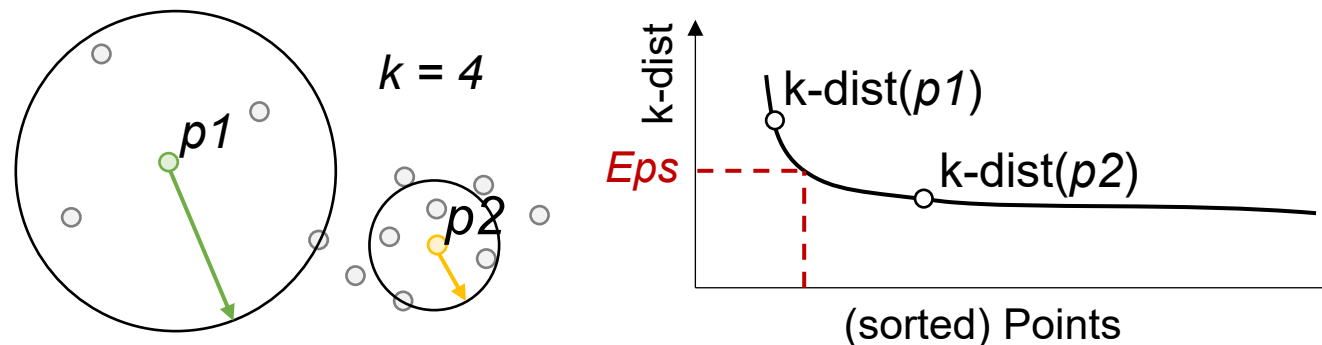
(MinPts=4, Eps=9.75)



Clustering > Density-based > DBSCAN

□ Heuristics for determining *Eps* and *MinPts*

- Main idea
 - For points in a cluster, their k -th nearest neighbors are at closer distance
 - For noise points, their k -th nearest neighbor at farther distance
- *MinPts*: a rule of thumb
 - E.g., $2 \cdot \text{dim} = 4$ for two-dimensional data or $\ln(n)$ for n points
- *Eps*: the first valley of the sorted k (=*MinPts*)-dist graph
 - The slope of the curve becomes noticeably more horizontal (i.e., the elbow method)
 - $k\text{-dist}(p)$: return the distance from a point p to its k -th nearest neighbor
 - Sorted k -dist graph: sorting the points by their k -dist values



Outline

- ❑ Basic concepts of clustering
- ❑ Distance-based methods
- ❑ Density-based methods
- ❑ **Probabilistic methods**
 - Expectation Maximization
- ❑ Clustering evaluation

Clustering > Probabilistic

□ Motivation

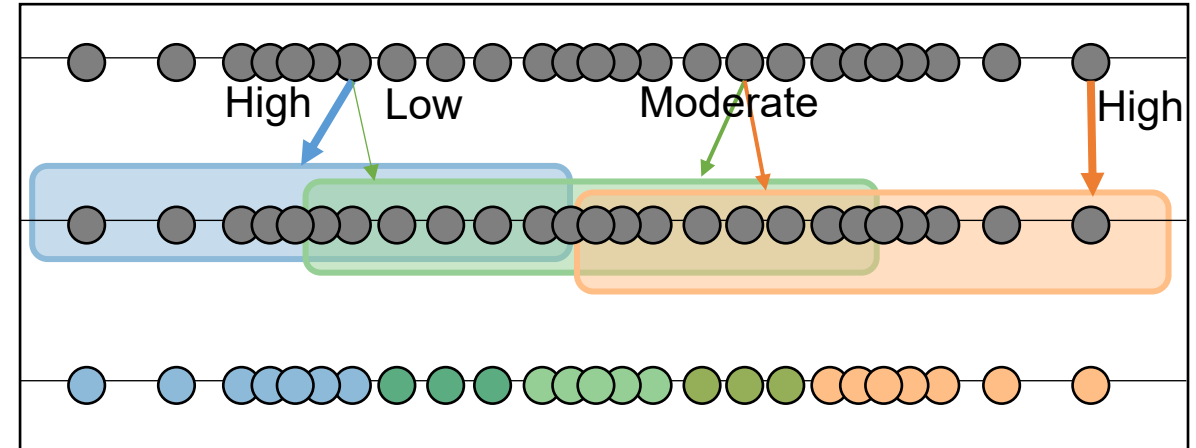
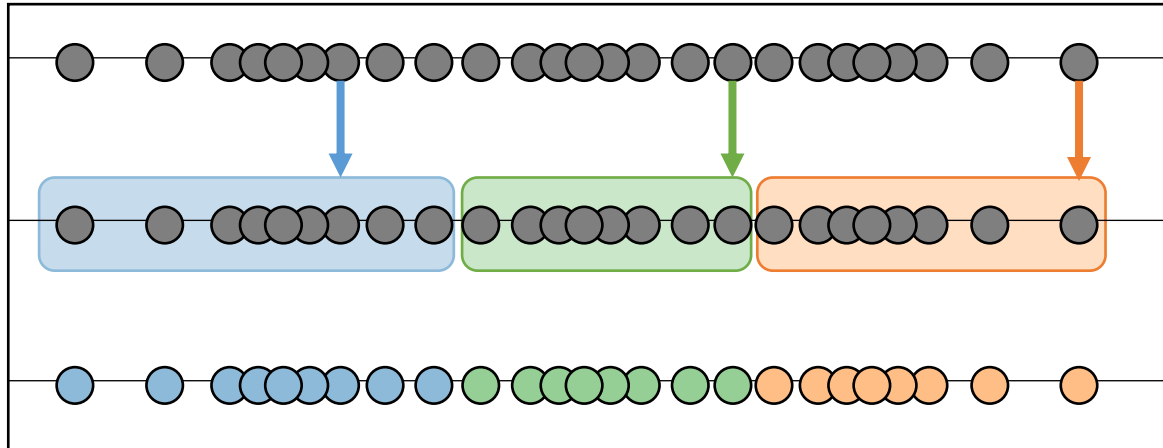
- Traditional clustering
 - Assign each data point exclusively to one cluster with clear boundaries between clusters
 - i.e., hard clustering
- Probabilistic clustering
 - Assign each data object over multiple clusters with the probability of it belonging to each cluster
 - i.e., soft clustering



Clustering > Probabilistic

□ Motivation

- Traditional clustering
 - Assign each data point exclusively to one cluster with clear boundaries between clusters
 - i.e., hard clustering
- Probabilistic clustering
 - Assign each data object over multiple clusters with the probability of it belonging to each cluster
 - i.e., soft clustering



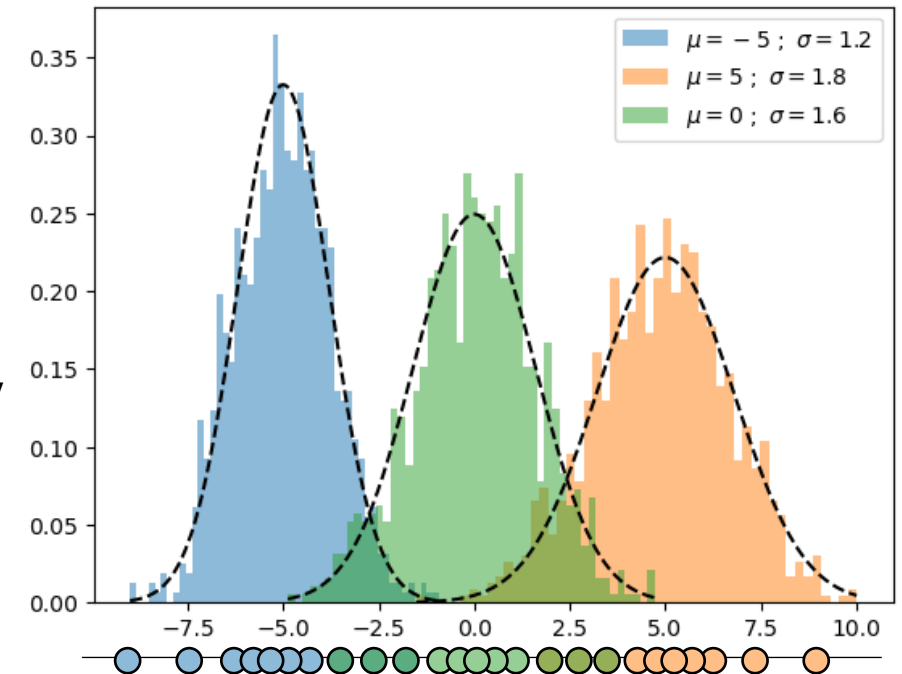
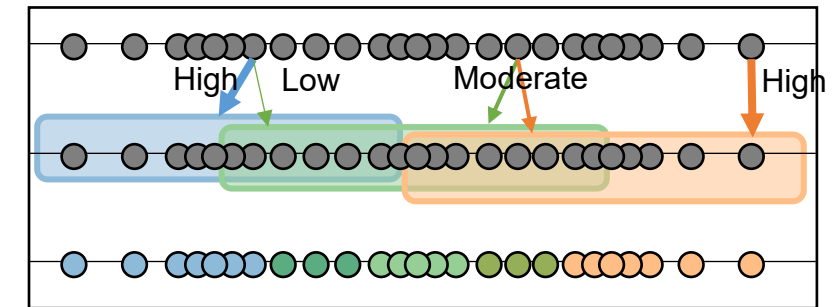
Clustering > Probabilistic

□ Motivation

- Assign each data object over multiple clusters with the probability of it belonging to each cluster

□ Idea

- A probabilistic cluster is a distribution over the data space
 - Can be represented as a probability density function
- A set of observed objects is a mixture of instances from multiple probabilistic clusters
 - Assume that each observed object is generated independently
- Infer a set of k probabilistic clusters that are mostly likely to generate the data set



Clustering > Probabilistic

□ Probabilistic model-based clustering

- A set C of k probabilistic clusters $c_1, \dots, c_k$ with probability density functions (PDF) $f_1, \dots, f_k$, respectively, and their weights $w_1, \dots, w_k$
- Probability of an object o generated by cluster c_j is

$$P(o|c_j) = w_j f_j(o)$$

- Probability of an object o generated by the set of clusters $C = \{c_1, \dots, c_k\}$ is

$$P(o|C) = \sum_{j=1}^k w_j f_j(o)$$

- Since objects are assumed to be generated independently, for a data set $D = \{o_1, \dots, o_n\}$,

$$P(D|C) = \prod_{i=1}^n P(o_i|C) = \prod_{i=1}^n \sum_{j=1}^k w_j f_j(o_i)$$

□ How to find a set C of k probabilistic clusters such that $P(D|C)$ is maximized?

- Maximizing $P(D|C)$ is often intractable since the PDF of a cluster can take an arbitrary form
- To make it computationally feasible, assume the PDFs are some parameterized distributions

Clustering > Probabilistic

□ Probabilistic mixture model

- For $D = \{o_1, \dots, o_n\}$ (n observed objects) and $\theta = \{\theta_1, \dots, \theta_k\}$ (parameters of the k distributions), $P_j(o_i|\theta_j)$ is the probability that o_i is generated from the j-th distribution using parameter θ_j

$$P(o_i|\theta) = \sum_{j=1}^k w_j P_j(o_i|\theta_j) \rightarrow P(D|\theta) = \prod_{i=1}^n \sum_{j=1}^k w_j P_j(o_i|\theta_j)$$

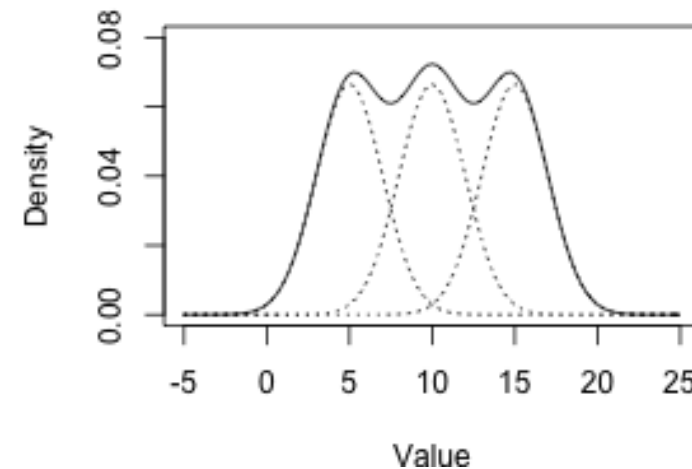
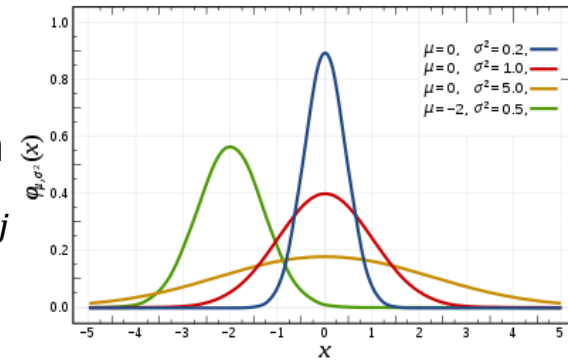
□ Example: univariate Gaussian mixture model

- Assume the probability density functions of k clusters follow a 1-d Gaussian distribution
- The probability density function of a cluster j is centered at μ_j with standard deviation σ_j
 - i.e., $\theta_j = (\mu_j, \sigma_j)$
- Each cluster has the same weights $w_k = 1/k$

$$P(o_i|\theta_j) = \frac{1}{\sqrt{2\pi}\sigma_j} e^{-\frac{(o_i - \mu_j)^2}{2\sigma_j^2}}$$

$$P(o_i|\theta) = \sum_{j=1}^k \frac{1}{k} \frac{1}{\sqrt{2\pi}\sigma_j} e^{-\frac{(o_i - \mu_j)^2}{2\sigma_j^2}}$$

$$P(D|\theta) = \prod_{i=1}^n \sum_{j=1}^k \frac{1}{k} \frac{1}{\sqrt{2\pi}\sigma_j} e^{-\frac{(o_i - \mu_j)^2}{2\sigma_j^2}}$$



■ Clustering > Probabilistic > EM

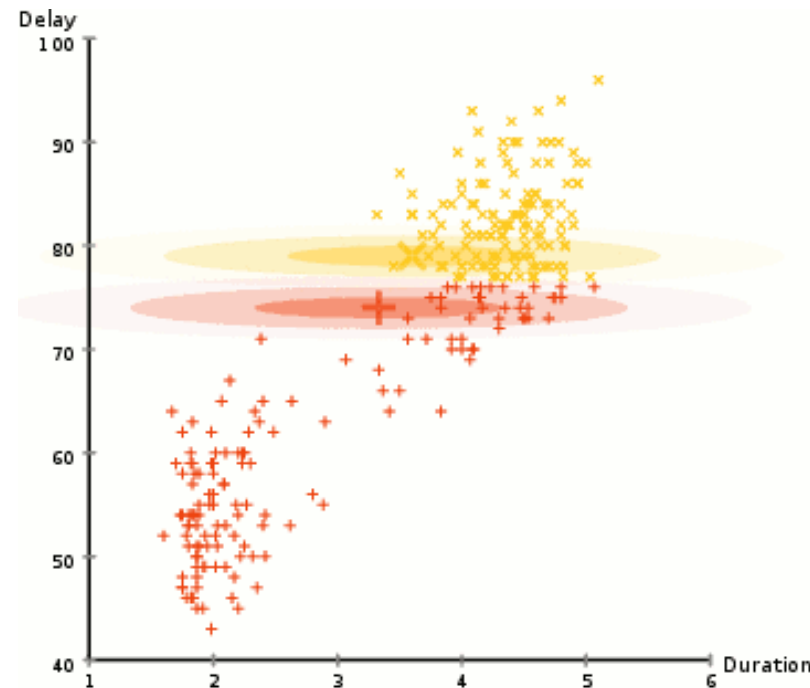
□ Expectation-Maximization

- An iterative method to find the maximum likelihood of parameters in statistical models, where the model depends on unobserved latent variables
- Two repeated main steps
 - Expectation step: computes the expected likelihood using the current estimate for the parameters
 - Maximization step: computes parameters maximizing the expected likelihood found on the E step→ Repeat the E-M steps until the parameters converge
- Example
 - In the context of clustering
 - Expectation: assign each object to the cluster with a center that is closest to the object
 - Maximization: adjust the centers so that the similarity of objects in a cluster to its center is maximized→ Repeat the E-M steps until the clustering results converge
 - *Anything come to mind?*

■ Clustering > Probabilistic > EM

□ Computing probabilistic mixture models with EM

- Given n objects $D = \{x_1, \dots, x_n\}$, find a set of parameters $\theta = \{\theta_1, \dots, \theta_k\}$ to maximize $P(D|\theta)$
 - e.g., $\theta_j = (\mu_j, \sigma_j)$ are the mean and standard deviation of the j -th univariate Gaussian distribution
- Initially assign random values to θ_j , then iteratively conduct the E/M steps until convergence



Clustering > Probabilistic > EM

□ Computing probabilistic mixture models with EM

- Given n objects $D = \{x_1, \dots, x_n\}$, find a set of parameters $\theta = \{\theta_1, \dots, \theta_k\}$ to maximize $P(D|\theta)$
 - e.g., $\theta_j = (\mu_j, \sigma_j)$ are the mean and standard deviation of the j -th univariate Gaussian distribution
- Initially assign random values to θ_j , then iteratively conduct the E/M steps until convergence

□ Example with Gaussian mixture model

- Expectation-step
 - For each object x_i , calculate the probability w_{ij}^{t+1} that x_i belongs to each distribution j with θ_j

$$w_{ij}^{t+1} = \frac{w_j^t P(x_i | \mu_j^t, \sigma_j^t)}{\sum_k w_k^t P(x_i | \mu_k^t, \sigma_k^t)} \quad P(x_i | \mu_j^t, \sigma_j^t) = \frac{1}{\sqrt{2\pi}\sigma_j^t} e^{-\frac{(x_i - \mu_j^t)^2}{2(\sigma_j^t)^2}}$$

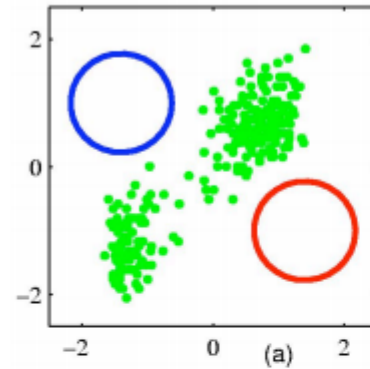
- Maximization-step
 - Adjust the parameters $\theta_j = (\mu_j, \sigma_j)$ so that the expected log likelihood of $P(D|\theta) = \prod_{i=1}^n \sum_{j=1}^k w_j P_j(x_i | \theta_j)$ is maximized

$$\mu_j^{t+1} = \frac{\sum_i w_{ij}^{t+1} x_i}{\sum_i w_{ij}^{t+1}} \quad (\sigma_j^2)^{t+1} = \frac{\sum_i w_{ij}^{t+1} (x_i - \mu_j^{t+1})^2}{\sum_i w_{ij}^{t+1}} \quad w_j^{t+1} = \frac{\sum_i w_{ij}^{t+1}}{n}$$

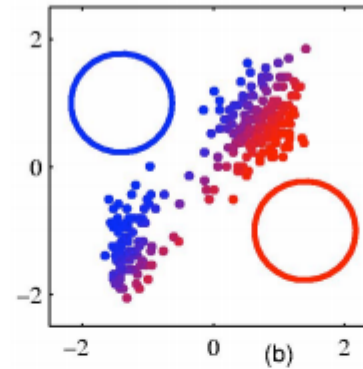
Clustering > Probabilistic > EM

□ Example (2D)

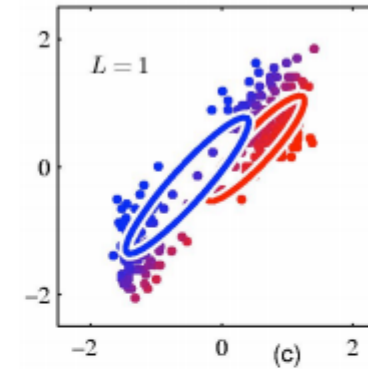
Data points and
Initial mixture model



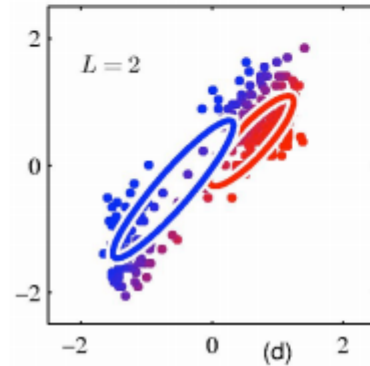
Initial E step



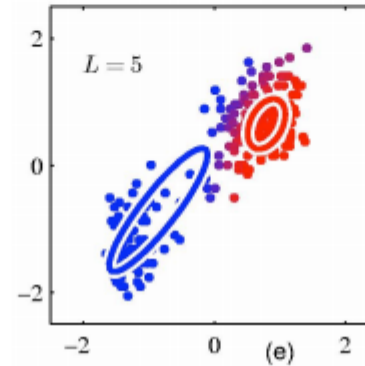
After first M step



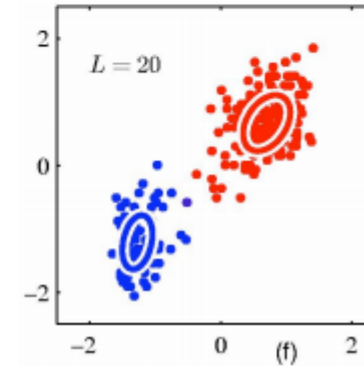
After 2 cycles



After 5 cycles



After 20 cycles



Clustering > Probabilistic > EM

❑ Comparison with k-Means

- k-Means as a special case of the EM algorithm with hard assignment

	k-Means	EM with GMM
Cluster representation	Mean (Centroid point)	Distribution parameters (e.g., Mean, Variance, Weight)
Cluster initialization	Initial k centers	Initial k distributions (e.g., Gaussian)
Expectation	Hard-assign each point to closest center	Soft-assign each point to each distribution
Maximization	Compute new centers of current clusters	Compute new parameters of each distribution
Membership	Exclusive	Non-exclusive

Outline

- ❑ Basic concepts of clustering
- ❑ Distance-based methods
- ❑ Density-based methods
- ❑ Probabilistic methods
- ❑ **Clustering evaluation**
 - Clustering tendency
 - Number of clusters
 - Clustering quality

Clustering > Evaluation

□ *“Clusters are in the eye of the beholder”*

- For (supervised) classification, we have a variety of measures to evaluate how good our model is
 - e.g., accuracy, precision, recall, F1
- For (unsupervised) clustering analysis, how can we evaluate the “goodness” of the clusters?

□ Major issues in clustering analysis

- Clustering tendency
 - Assess the suitability of clustering - any inherent grouping structure?
- The number of clusters
 - Determine the right number of clusters for a given data set and algorithm
- Clustering quality
 - Evaluate the quality of the clustering results in extrinsic or intrinsic ways

Clustering > Evaluation > Clustering Tendency

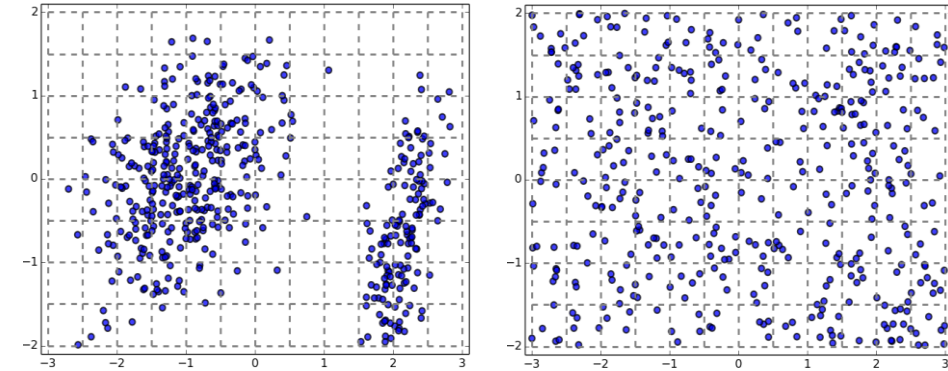
❑ Clustering tendency (i.e., clusterability)

- Evaluate whether the dataset has any “inherent grouping structure”
 - Non-random structure that may lead to meaningful clusters
- Measures
 - Spatial histogram (qualitative)
 - Contrast the histogram of the data with that generated from random samples
 - Distribution distance (quantitative)
 - Compare the empirical probability mass function from the data with those from the randomly generated samples
 - e.g., Kullback-Leibler (KL) divergence
 - Hopkins statistic
 - A statistic for the spatial randomness of a variable as distributed in a space.
 - Given a d-dimensional space S and a data set D in S ,

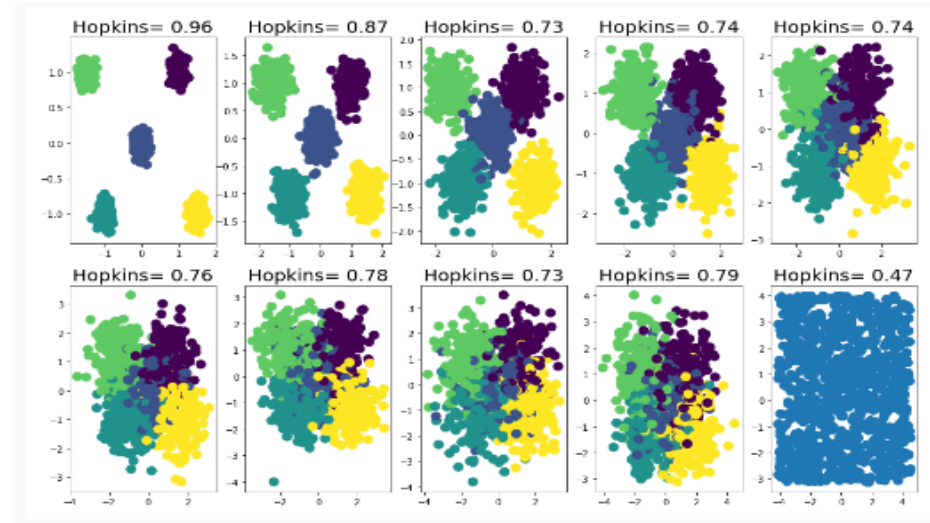
$$H = \frac{\sum_{i=1}^n x_i^d}{\sum_{i=1}^n x_i^d + \sum_{i=1}^n y_i^d}$$

x_i = distance from a randomly sampled data in S to its nearest neighbor in D

y_i = distance from a randomly sampled data in D to its nearest neighbor in D

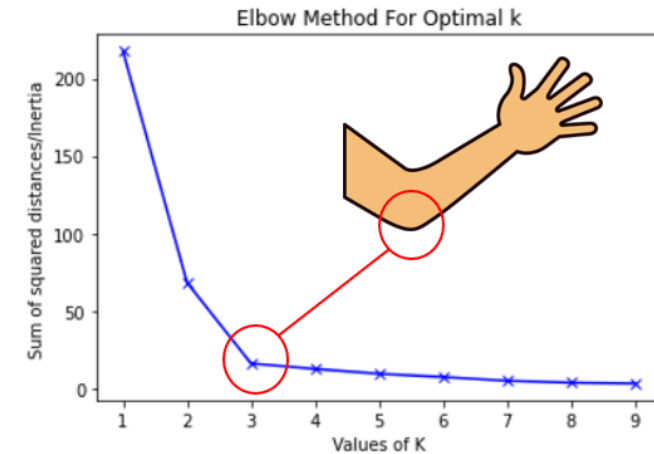


$$D_{KL}(p(x)||q(x)) = \sum_{x \in X} p(x) \ln \frac{p(x)}{q(x)}$$



Clustering > Evaluation > Number of Clusters

- ❑ The number of clusters controls the proper granularity of cluster analysis
 - Finding a good balance between compressibility and accuracy in cluster analysis
 - Depending on the distribution's shape and scale in the data set
 - Depending on the clustering resolution required by the user
 - Two undesirable extremes
 - The whole data set is one cluster: no value of clustering $\leftrightarrow$ Each point as a cluster: no data summarization
- ❑ Methods for determining the number of clusters k
 - Empirical approach
 - Rule of thumb: $k \approx \sqrt{n/2}$ for a dataset of n points (e.g., $n = 200 \rightarrow k = 10$)
 - Each cluster is expected to have about $\sqrt{2n}$ points
 - Elbow method: the turning point in the curve of the cluster variance with k
 - The cluster variance: the sum of squared distances of data to its centroid, a.k.a. inertia
 - Statistical approach
 - Find the optimal number of clusters that results the best statistics
 - e.g., Dunn index, Silhouette coefficient, etc.



■ Clustering > Evaluation > Clustering Quality

□ Unsupervised measure (i.e., intrinsic index)

- Examine *how compact* clusters are and *how well clusters are separated* with distance metrics
- Measures
 - Dunn index
 - Silhouette coefficient

□ Supervised measure (i.e., extrinsic index)

- Examine the extent to which cluster labels match *externally given ground truth cluster labels*
- Measures
 - Information-based: Normalized mutual information (NMI)
 - Pairwise comparison-based: Jaccard index, Rand index

Clustering > Evaluation > Clustering Quality > Dunn Index

□ Unsupervised measure (i.e., intrinsic index)

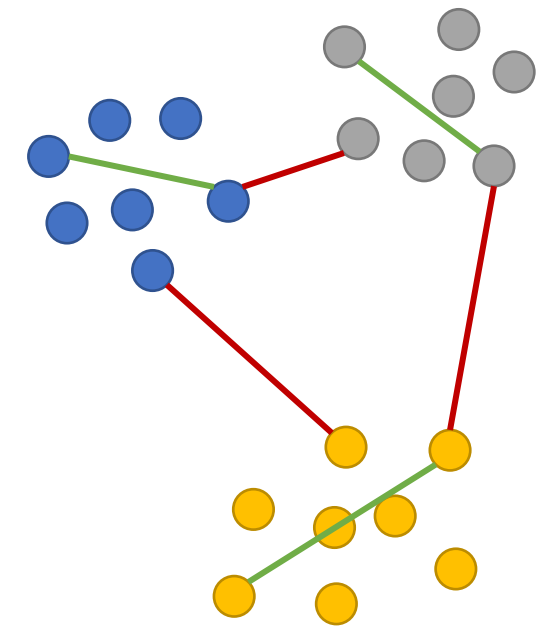
▪ Criteria

- Cluster cohesion: how closely related are objects in a cluster?
- Cluster separation: how distinct a cluster is from other clusters?

▪ Dunn index (extreme situations)

- The intra-cluster compactness $\Delta = \max_{C(o_i)=C(o_j)}\{d(o_i, o_j)\}$
 - The maximum distance between two points that belong to the same cluster
- The inter-cluster separation $\delta = \min_{C(o_i) \neq C(o_j)}\{d(o_i, o_j)\}$
 - The minimum distance between two points that belong to different clusters
- The Dunn index $DI = \frac{\delta}{\Delta}$
 - The larger the ratio, the farther away the clusters are separated comparing to their compactness
- *Using the extreme distances can be affected by outliers*

— Intra-cluster distance
— Inter-cluster distance



Clustering > Evaluation > Clustering Quality > Silhouette Coefficient

□ Unsupervised measure (i.e., intrinsic index)

▪ Silhouette coefficient (average situations)

- The average intra-cluster distance of an object

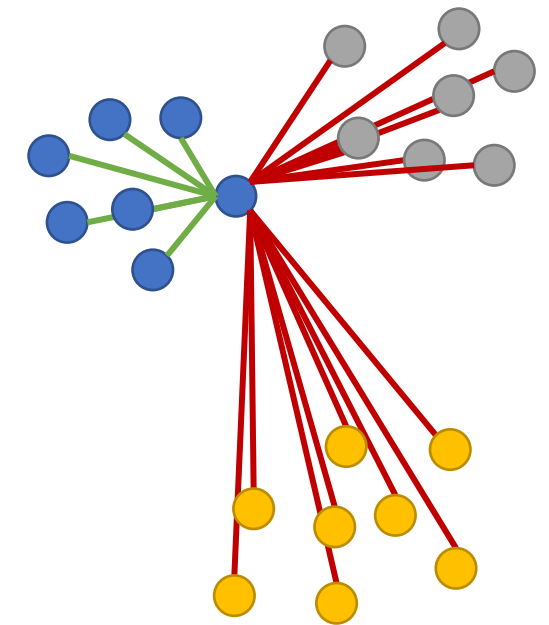
$$- a(o) = \sum_{o' \in C_i, o' \neq o} d(o, o') / (|C_i| - 1)$$

- The minimum average inter-cluster distance of an object

$$- b(o) = \min_{C_j: 1 \leq j \leq k, j \neq i} \left\{ \sum_{o' \in C_j} \text{dist}(o, o') / |C_j| \right\}$$

- Silhouette coefficient of an object $s(o) = \frac{b(o) - a(o)}{\max\{a(o), b(o)\}} \sim [-1, 1]$
 - $s(o) \sim 1$: the cluster containing o is compact and o is far away from other clusters (good)
 - $s(o) < 0$ (i.e., $b(o) < a(o)$): o is closer to the objects in another cluster than to the objects in the same cluster (bad)
- Final silhouette coefficient of the clustering results $= \sum_{o \in D} \frac{1}{n} s(o)$

— Intra-cluster distance
— Inter-cluster distance



Clustering > Evaluation > Clustering Quality > Information-based

□ Supervised measure (i.e., extrinsic index)

- Dataset $D = \{o_1, \dots, o_n\}$, Ground truth $G = \{G_1, \dots, G_l\}$, Clustering result $C = \{C_1, \dots, C_m\}$,
- Information-based
 - Entropy = a measure of “information”, “surprise”, or “uncertainty” in the variable’s possible outcomes
 - Information: $I(x_i) = \log_2 \left(\frac{1}{P(X=x_i)} \right) = -\log_2(P(X = x_i))$
 - Entropy: $H(X = \{x_1, x_2, \dots, x_m\}) = E[I(X)] = \sum_{i=1}^m p_i I(x_i) = -\sum_{i=1}^m p_i \log_2(p_i)$, where $p_i = P(X = x_i)$



X : a dog snack

$$P(X = \text{strawberry}) = 1$$

$$I(\text{strawberry}) = -\log_2(1) = 0$$

$$E(X) = -\log_2(1) = 0$$



X : a dog snack

$$P(X = \text{cup}_i) = 1/3$$

$$I(\text{cup}_1) = -\log_2(1/3) = 1.58$$

$$I(\text{cup}_2) = -\log_2(1/3) = 1.58$$

$$I(\text{cup}_3) = -\log_2(1/3) = 1.58$$

$$E(X) = \sum_{i=1}^3 1/3 \log_2(1/3) = 1.58$$

Clustering > Evaluation > Clustering Quality > Information-based

□ Supervised measure (i.e., extrinsic index)

▪ Dataset $D = \{o_1, \dots, o_n\}$, Ground truth $G = \{G_1, \dots, G_l\}$, Clustering result $C = \{C_1, \dots, C_m\}$,

▪ Information-based

• Entropy of clustering results

$$- H(C) = - \sum_{i=1}^m \frac{|C_i|}{n} \log \frac{|C_i|}{n}, \quad H(G) = - \sum_{i=1}^l \frac{|G_i|}{n} \log \frac{|G_i|}{n}$$

$$- H(G|C) = - \sum_{i=1}^m \sum_{j=1}^l \frac{|C_i \cap G_j|}{n} \log \frac{|C_i \cap G_j|}{|C_i|} \quad (= \sum_{i=1}^m \frac{|C_i|}{n} H(G|C_i) = - \sum_{i=1}^m \frac{|C_i|}{n} \sum_{j=1}^l \frac{|C_i \cap G_j|}{|C_i|} \log \frac{|C_i \cap G_j|}{|C_i|})$$

• Mutual information

$$- I(C; G) = H(G) - H(G|C) (= H(C) - H(C|G))$$

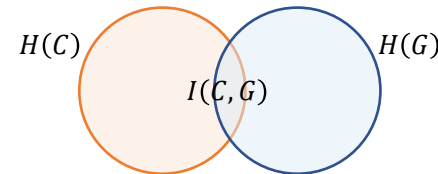
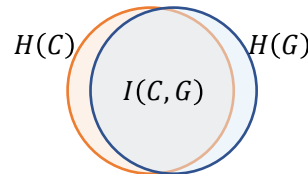
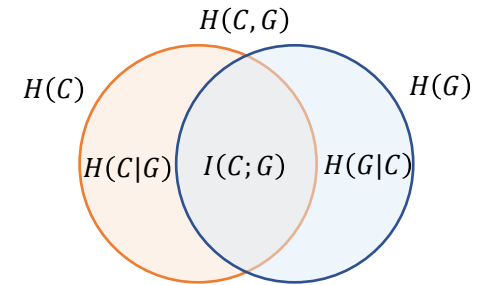
– Quantify the amount of shared info between the clustering C and the ground-truth partitioning G

• Normalized mutual information (NMI)

$$- NMI(C, G) = \frac{I(C; G)}{(H(C) + H(G))/2} \sim [0, 1]$$

– Close to 1 indicates a good clustering

• Cf. advanced metrics: adjusted mutual information (AMI)



■ Clustering > Evaluation > Clustering Quality > Comparison-based

□ Supervised measure (i.e., extrinsic index)

- Dataset $D = \{o_1, \dots, o_n\}$, Ground truth $G = \{G_1, \dots, G_l\}$, Clustering result $C = \{C_1, \dots, C_m\}$,
- Pairwise comparison-based
 - Treat each cluster in the ground truth as a class
 - For each pair of objects (o_i, o_j) in D ,
if they are assigned to the same cluster, the assignment is regarded as positive; otherwise, negative
 - Depending on assignments, we have four possible cases for the total number of pairs (i.e., $\binom{n}{2}$)
 - Basic metrics
 - Jaccard index = $TP / (TP + FP + FN)$
 - Rand index = $(TP + TN) / (TP + TN + FP + FN)$
- Cf. advanced metrics: adjusted rand index (ARI), B-cubed score

	$C(o_i) = C(o_j)$	$C(o_i) \neq C(o_j)$
$G(o_i) = G(o_j)$	true positive	false negative
$G(o_i) \neq G(o_j)$	false positive	true negative

Clustering > Evaluation > Clustering Quality > Comparison-based

□ Example

- Dataset $D = \{a, b, c, d, e\}$
- Ground truth $G = [G1=\{a,b,c\}, G2=\{d,e\}]$
- Clustering result $C = [C1= \{a,b\}, C2 = \{c,d,e\}]$

- All possible pairs $= \binom{5}{2} = 10$
 - $(a,b), (a,c), (a,d), (a,e), (b,c), (b,d), (b,e), (c,d), (c,e), (d,e)$
- Counting each case
 - TP = 2: $(a,b), (d,e)$
 - FP = 2: $(c,d), (c,e)$
 - FN = 2: $(a,c), (b,c)$
 - TN = 4: $(a,d), (a,e), (b,d), (b,e)$

	$C(o_i) = C(o_j)$	$C(o_i) \neq C(o_j)$
$G(o_i) = G(o_j)$	true positive	false negative
$G(o_i) \neq G(o_j)$	false positive	true negative

- Rand index $= (TP + TN) / (TP + TN + FP + FN) = 2 + 4 / (2 + 2 + 2 + 4) = 0.6$

Tutorial References

❑ Library and tutorial

- [Scikit-learn clustering and evaluation](#)
 - [k-Means/k-Means++](#)
 - [DBSCAN](#)
 - [Evaluation](#)
- [B-cubed](#)

❑ Examples

- [K-means](#)
- [Hierarchical clustering](#)
- [DBSCAN](#) / [HDBSCAN](#)
- [Expectation Maximization](#)
- [Elbow method](#)

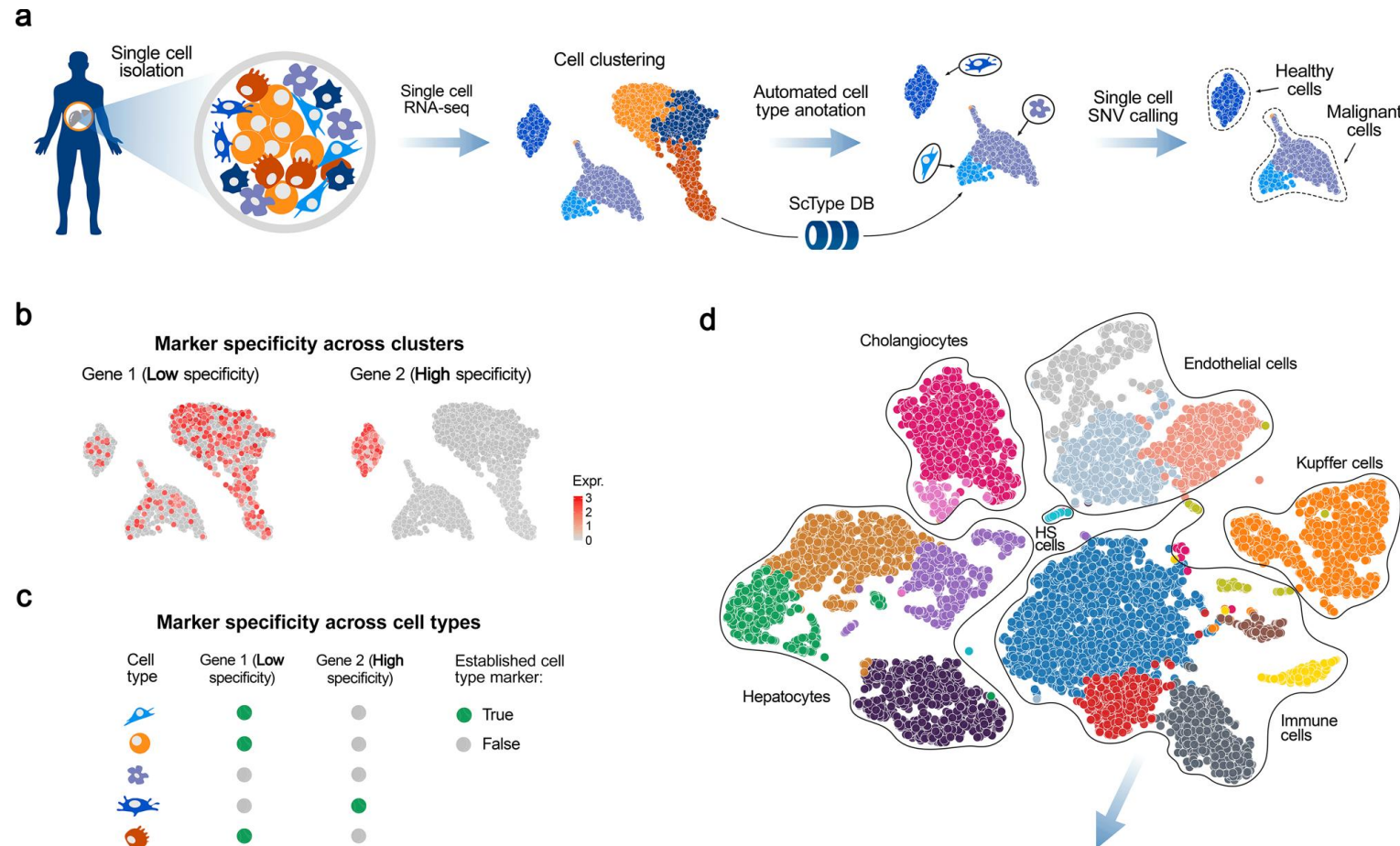
❑ Case study

- [Comprehensive study](#)
- [Customer segmentation](#)

[Cookie Slides] Density-based Clustering in Practice

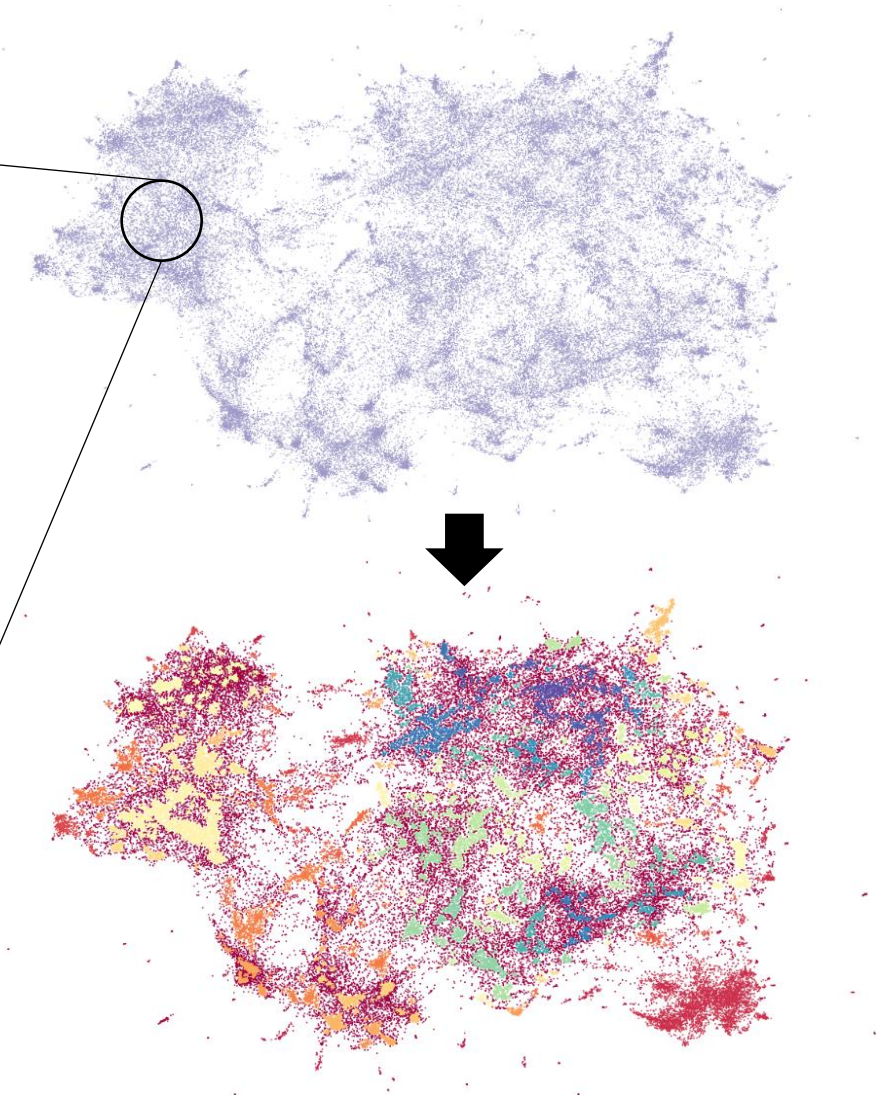
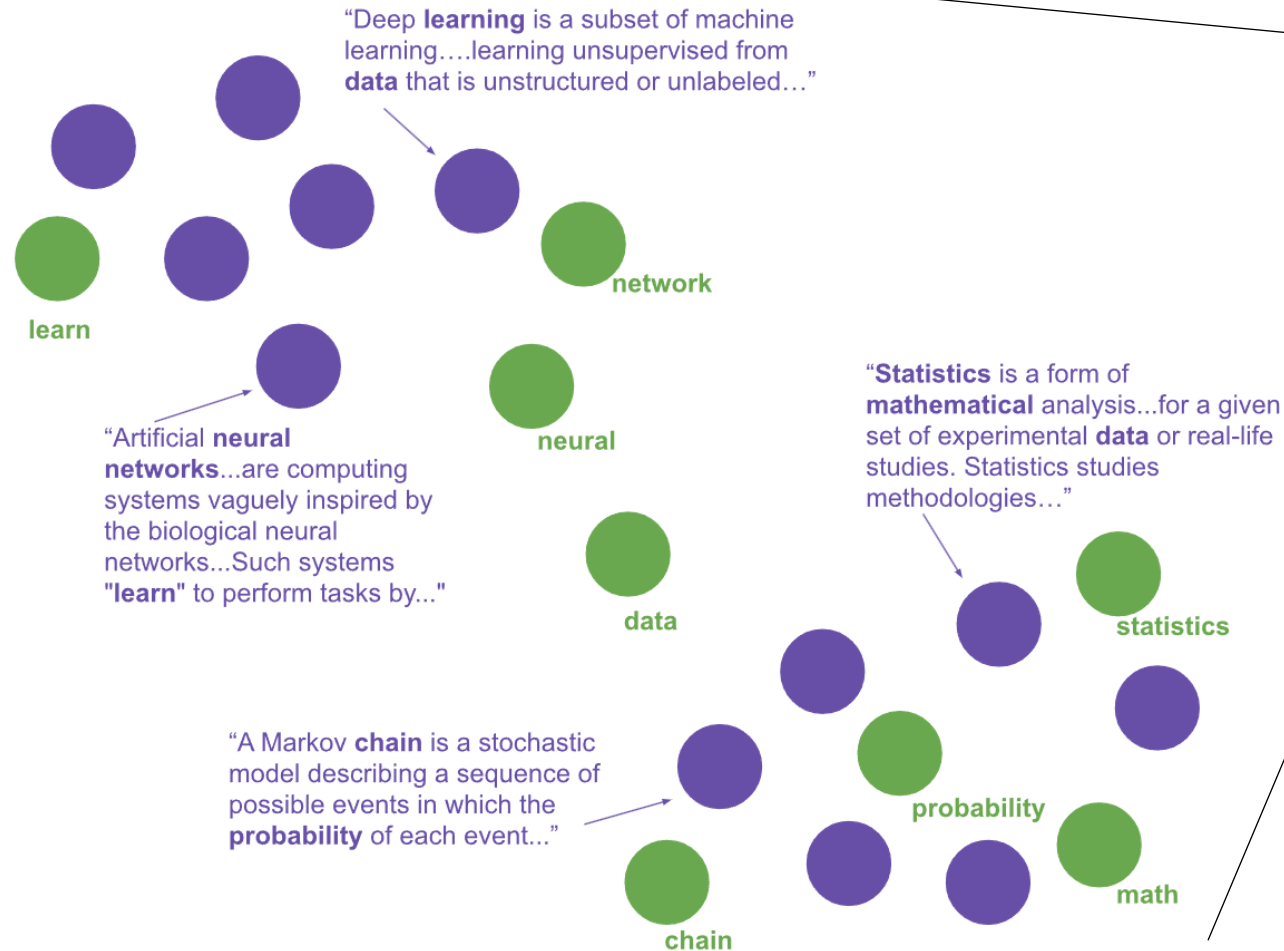
□ Example – cell type clustering by RNA-Seq features

- Used for new biological hypotheses about cell type identities and relationships



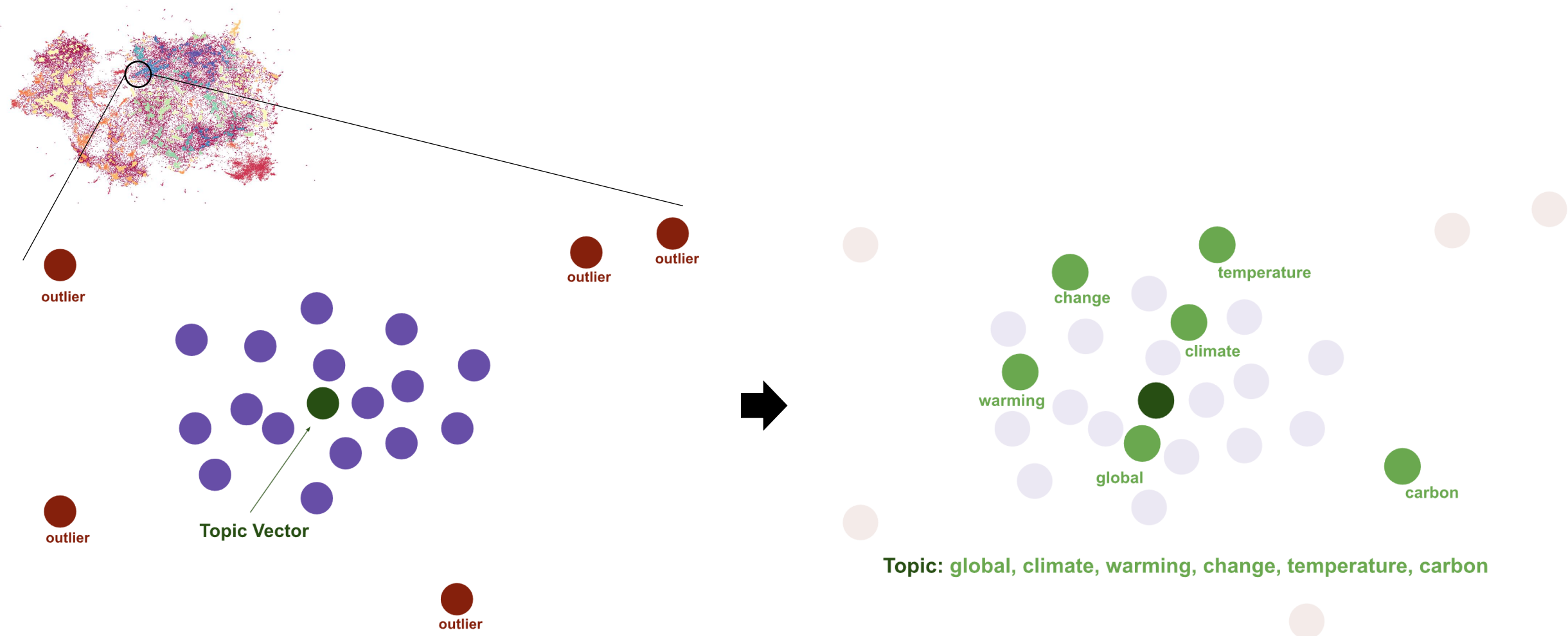
[Cookie Slides] Density-based Clustering in Practice

□ Example – topic modeling of documents



[Cookie Slides] Density-based Clustering in Practice

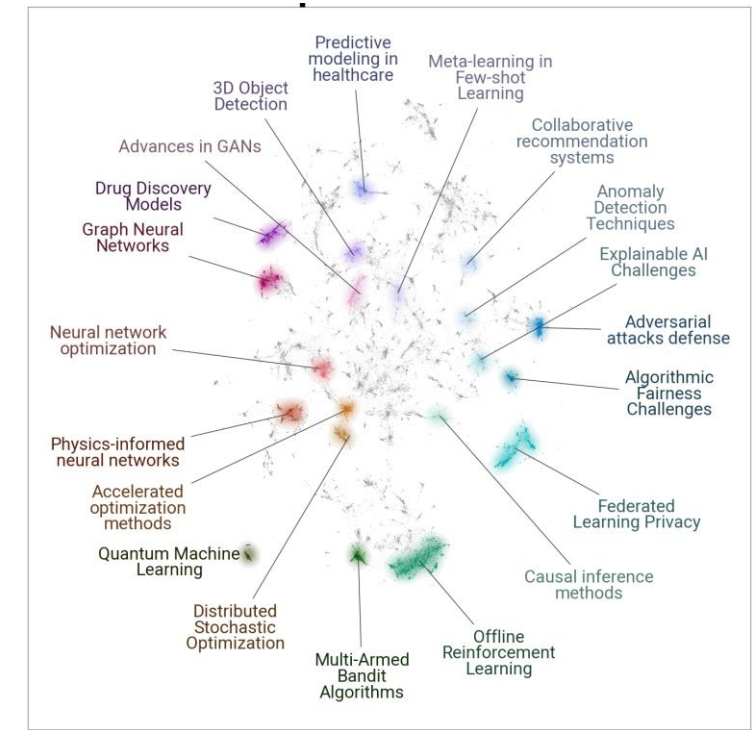
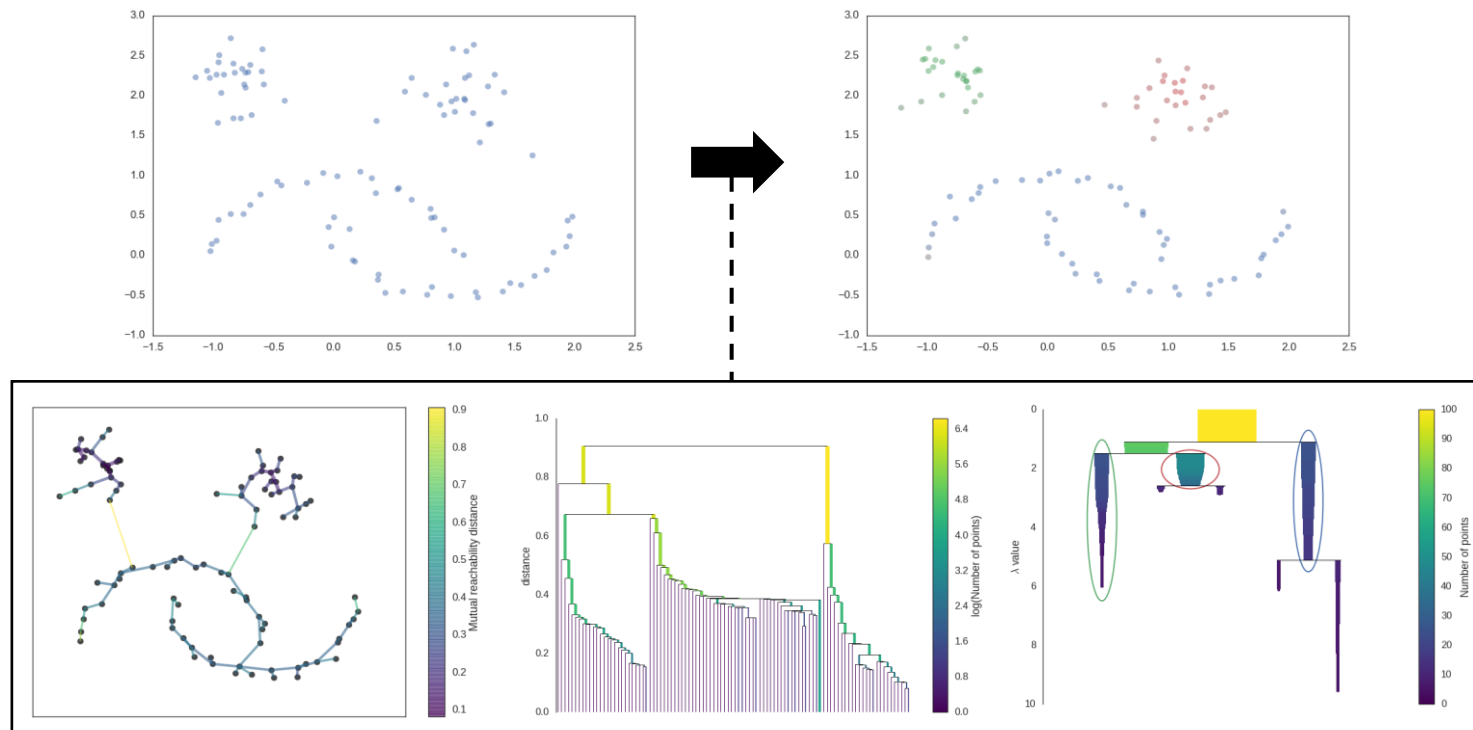
□ Example – topic modeling of documents



[Cookie Slides] HDBSCAN (Hierarchical DBSCAN)

❑ Illustration of HDBSCAN

- Example usage: [BERTopic](#)



[Cookie Slides] Self-supervised Continual Clustering

WWW 2023 Web Mining

SCStory: Self-supervised and Continual Online Story Discovery

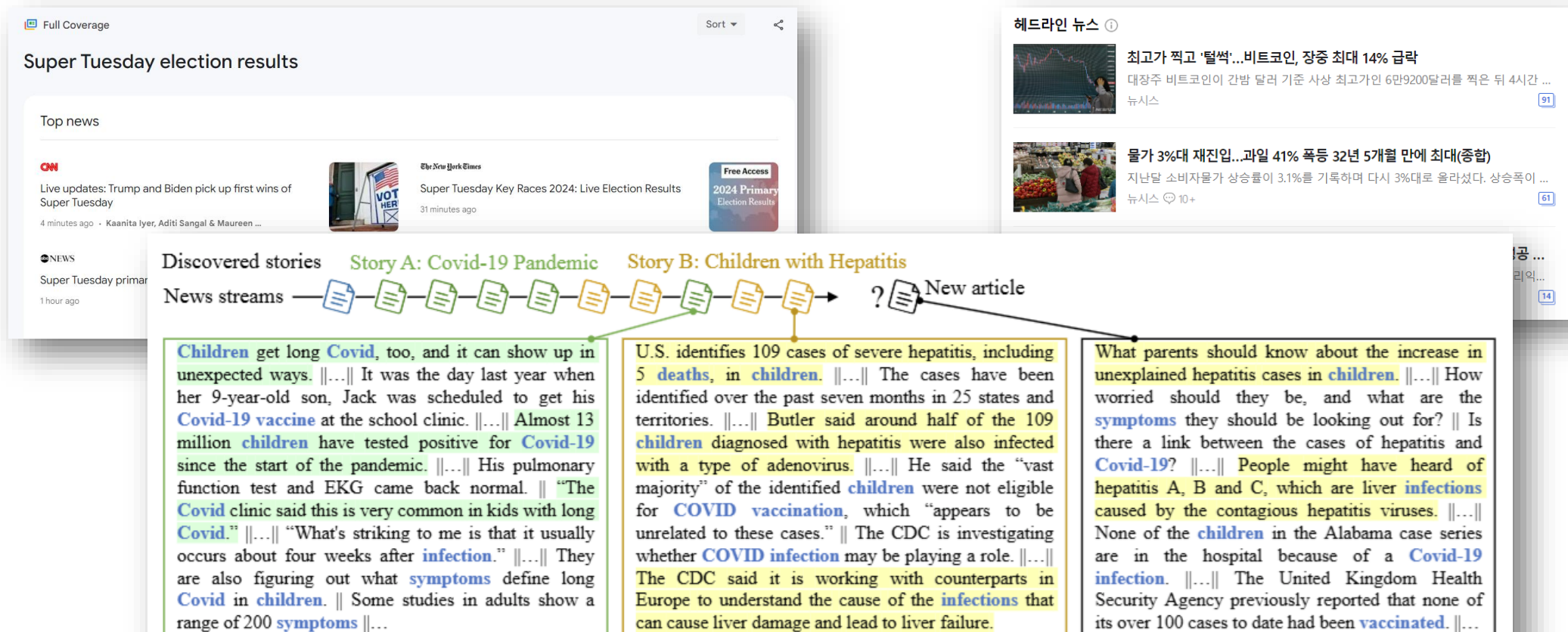


Figure 1: The articles contain common keywords (marked in blue) and too similar/specific semantics. The story-indicative parts (highlighted in green for Story A and in yellow for Story B) are helpful to clearly distinguish articles in different stories.

[Cookie Slides] Self-supervised Continual Clustering

WWW 2023 Web Mining

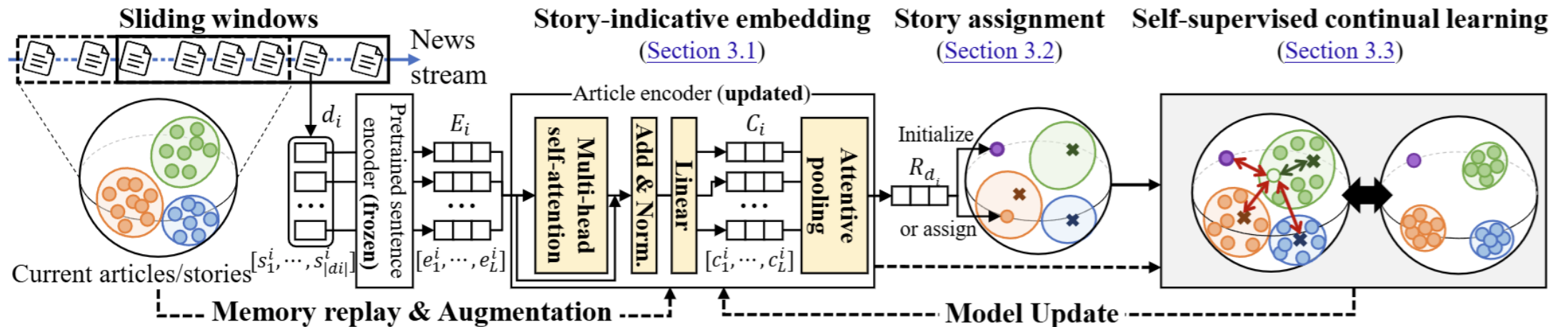


Figure 2: In every sliding window of an article stream, first, SCStory learns story-indicative article representations by modeling the correlations between initial sentence representations from a PSE. Second, SCStory assigns each new article to the most confident current story or forms a new story. Lastly, SCStory updates the story-indicative embedding module based on the contrastive loss with confidence-based memory replay and prioritized-augmentation (the base PSE is frozen).

[Cookie Slides] Self-supervised Continual Clustering

WWW 2023 Web Mining

SCStory: Self-supervised and Continual Online Story Discovery

Data type	Data structure	Data state	Target knowledge	Application
Sequence/ Temporal	Unstructured	Streaming	Story (cluster)	News outlet

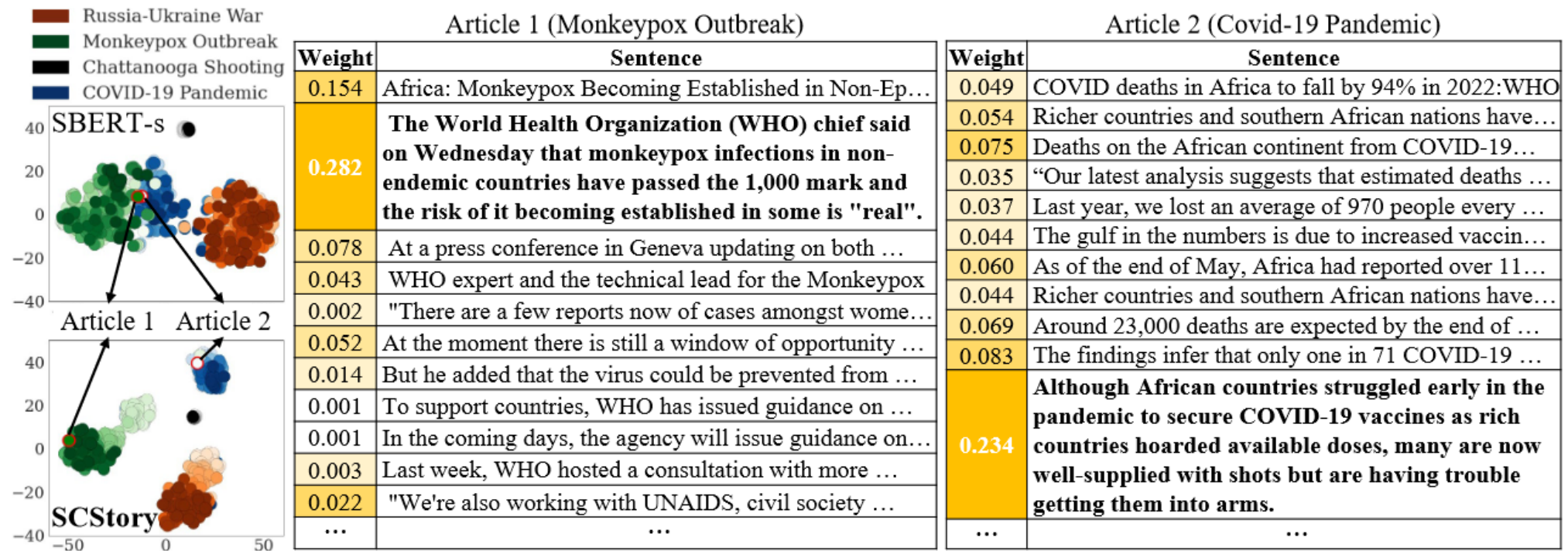


Figure 6: [Left] the article representations in CaseStudy by SBERT-s and SCStory (darker markers indicate newer articles). [Right] the two example boundary articles and their sentences with attentive pooling weights learned by SCStory.